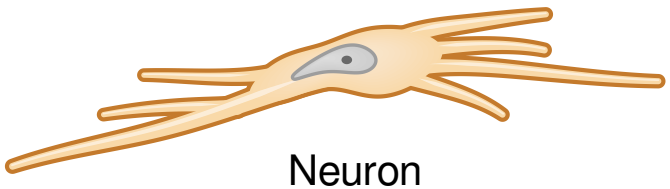
On the precision of dynamic gene regulation

Tarek Tohme

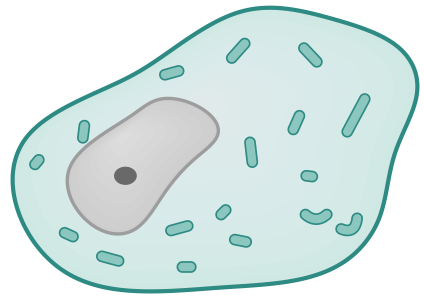
Under the supervision of

Aleksandra Walczak and Thierry Mora

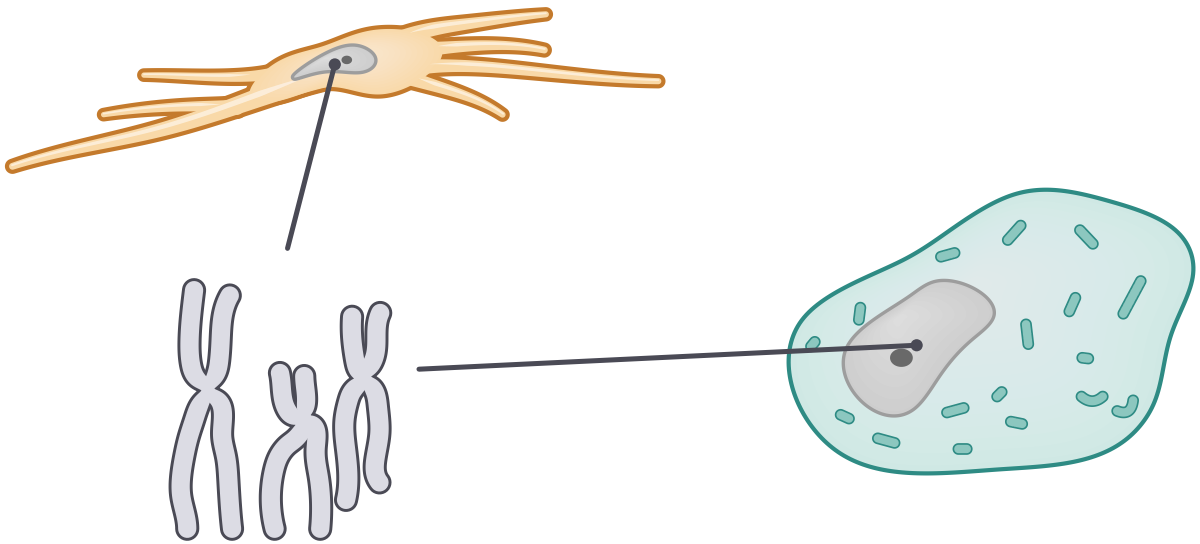
November 9, 2026



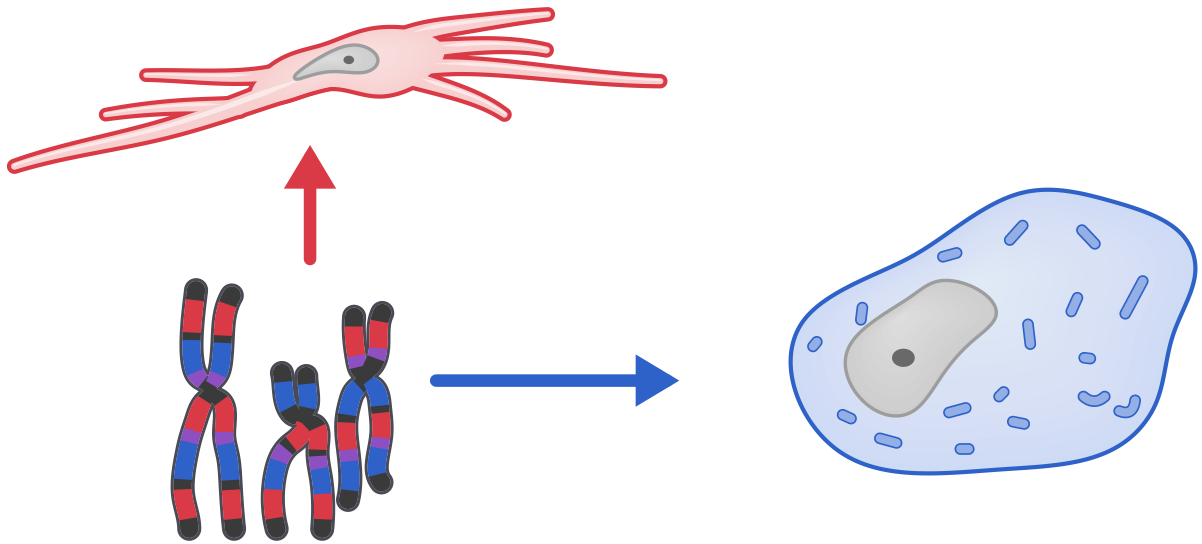
Neuron



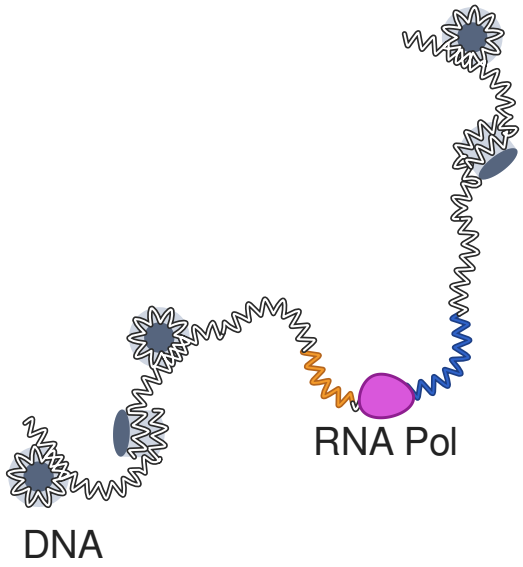
Kidney cell



Gene expression is regulated



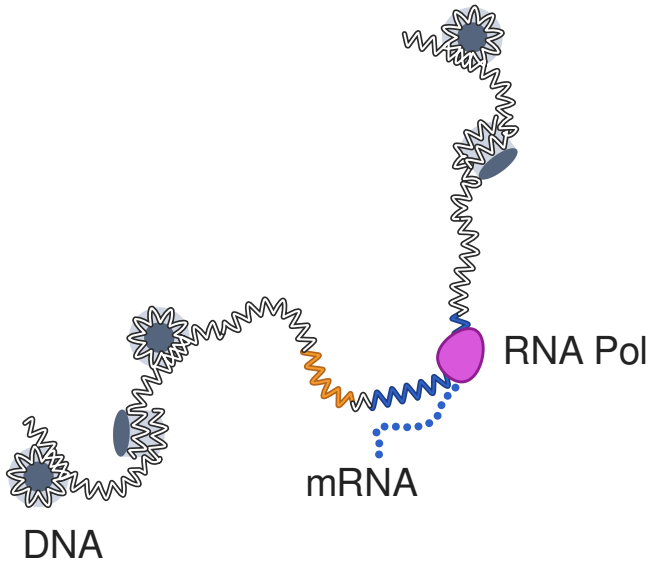
Overview of gene regulation



Transcription

DNA
Promoter
Gene
RNA Polymerase

Overview of gene regulation



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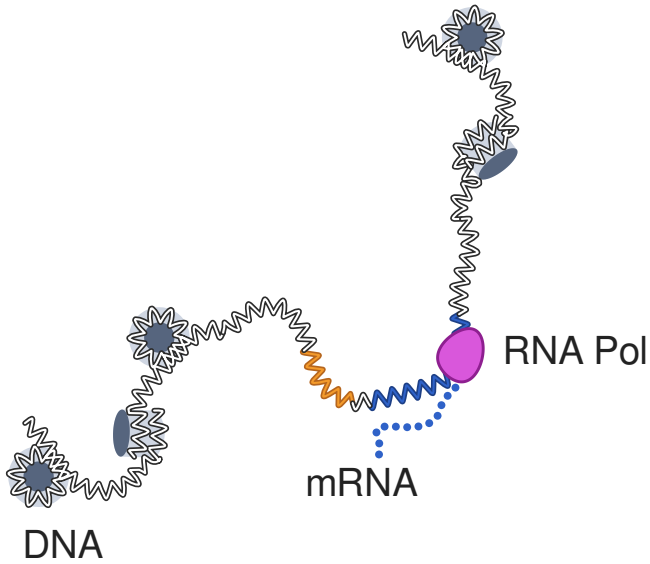

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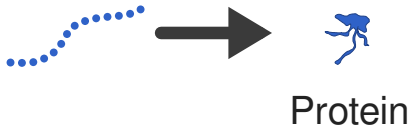

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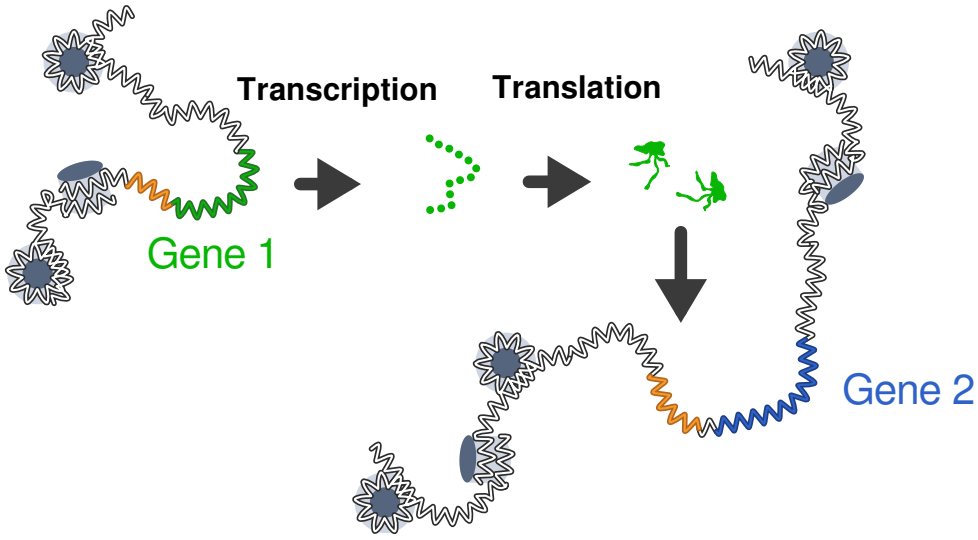

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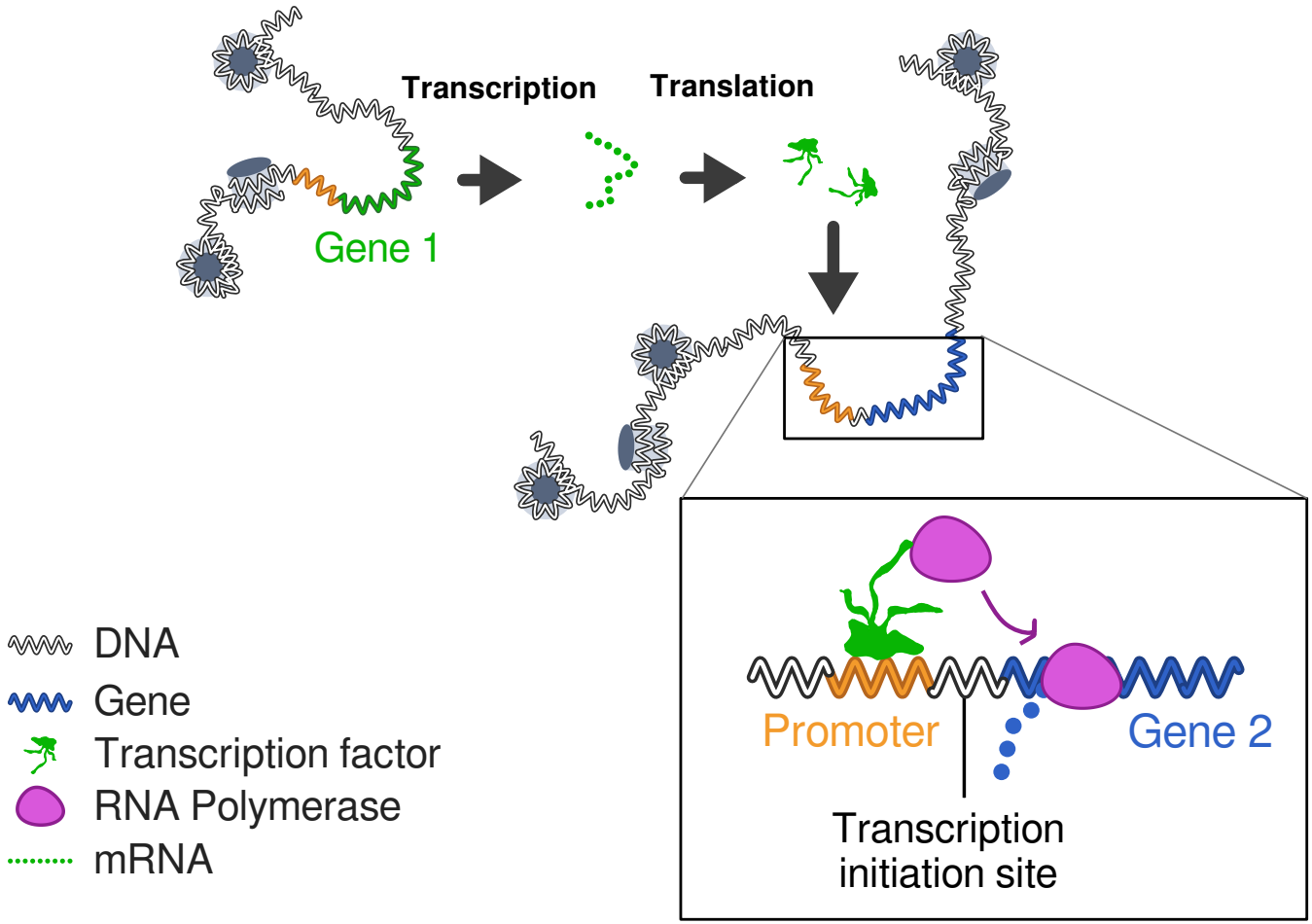
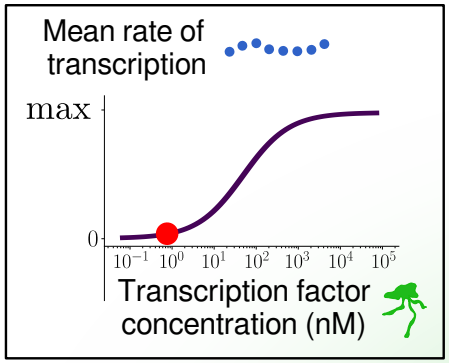
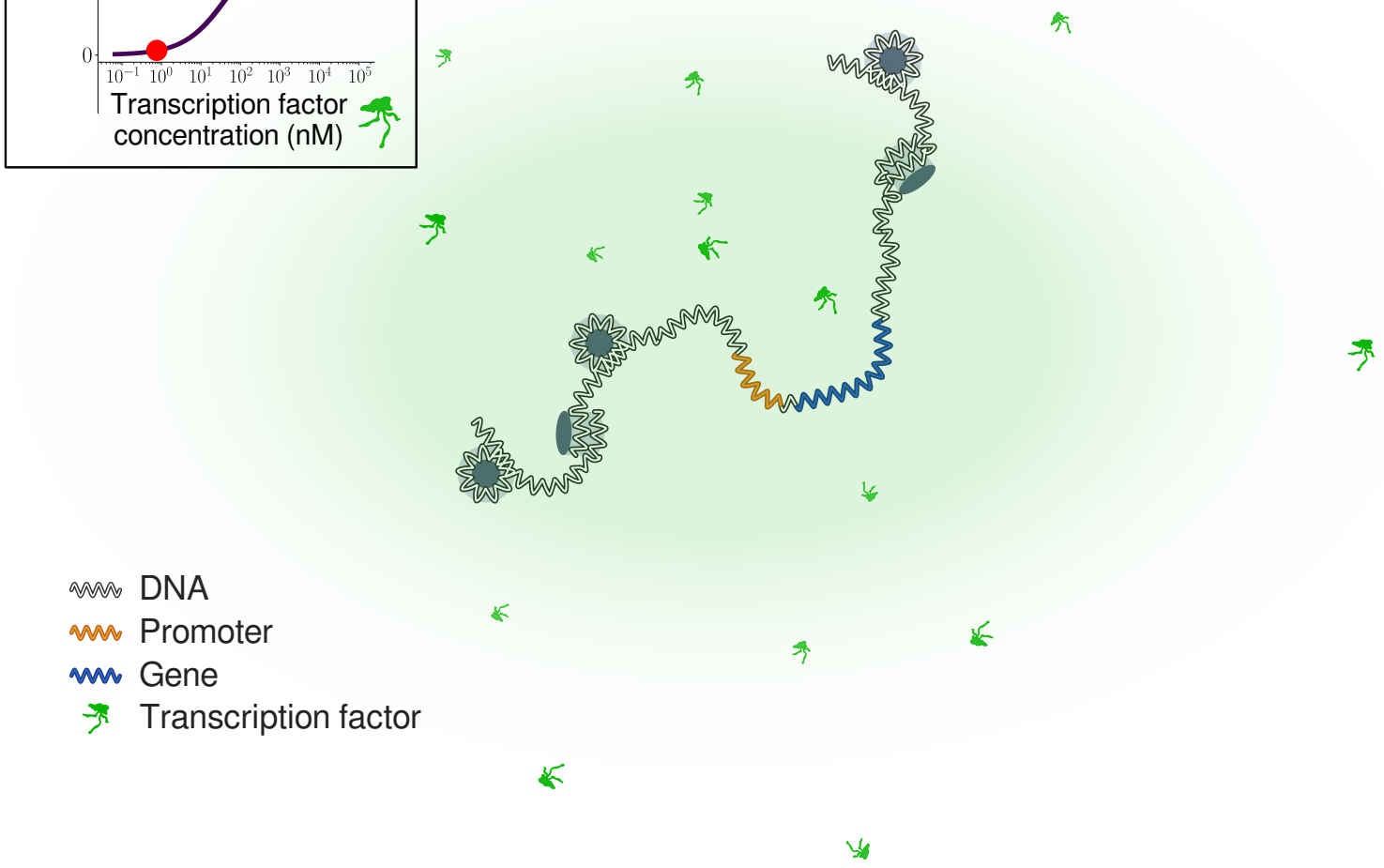


Expression levels are continuous ...





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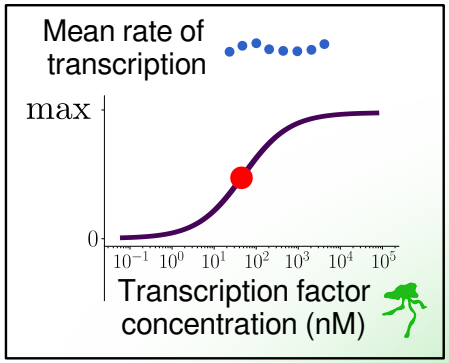

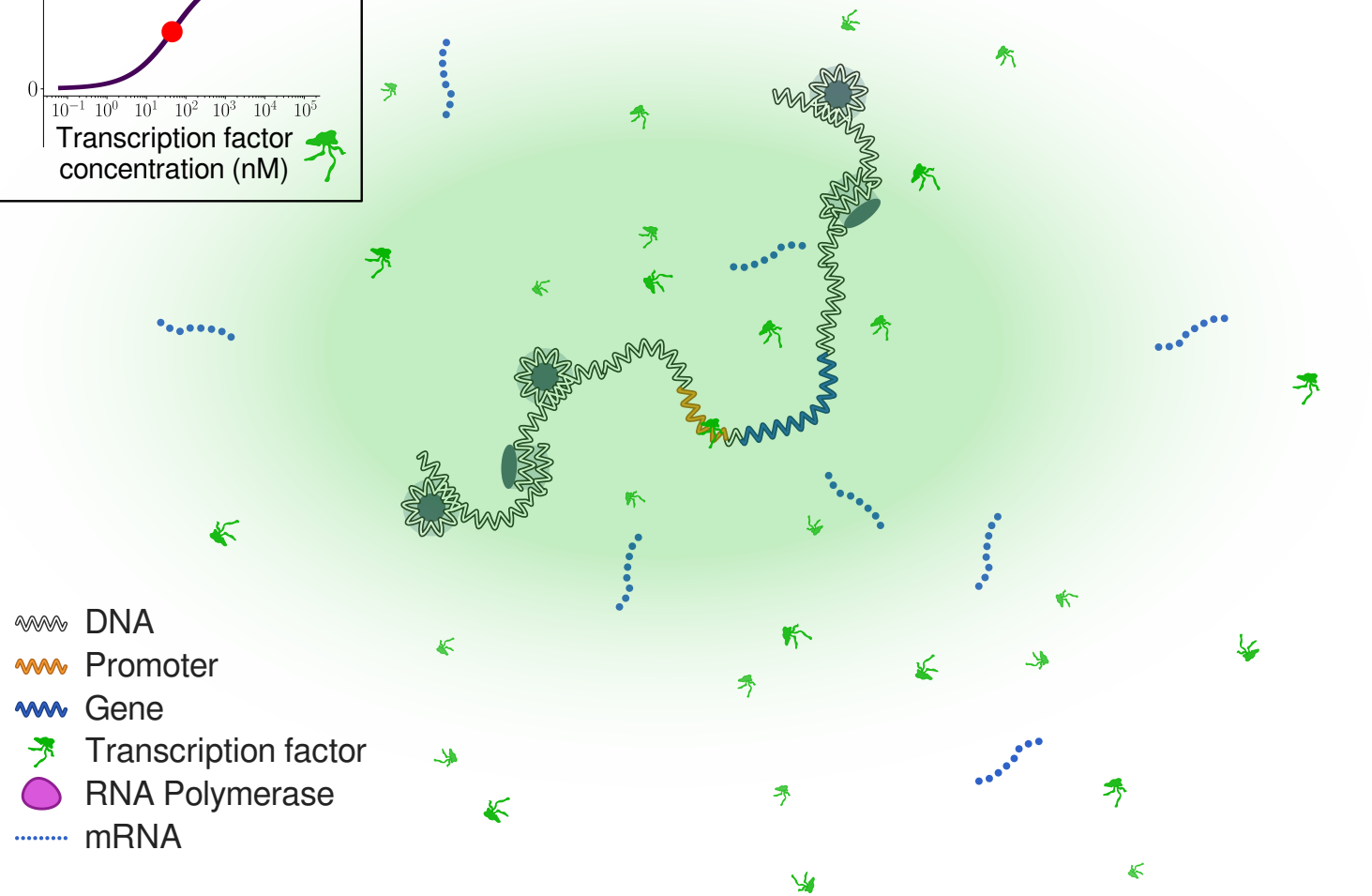

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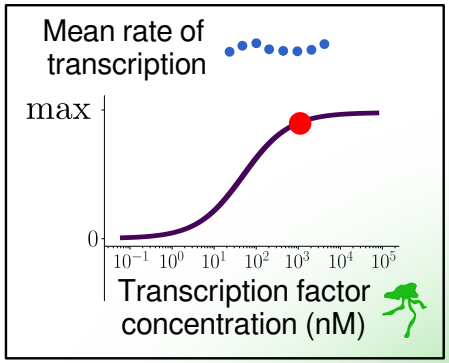
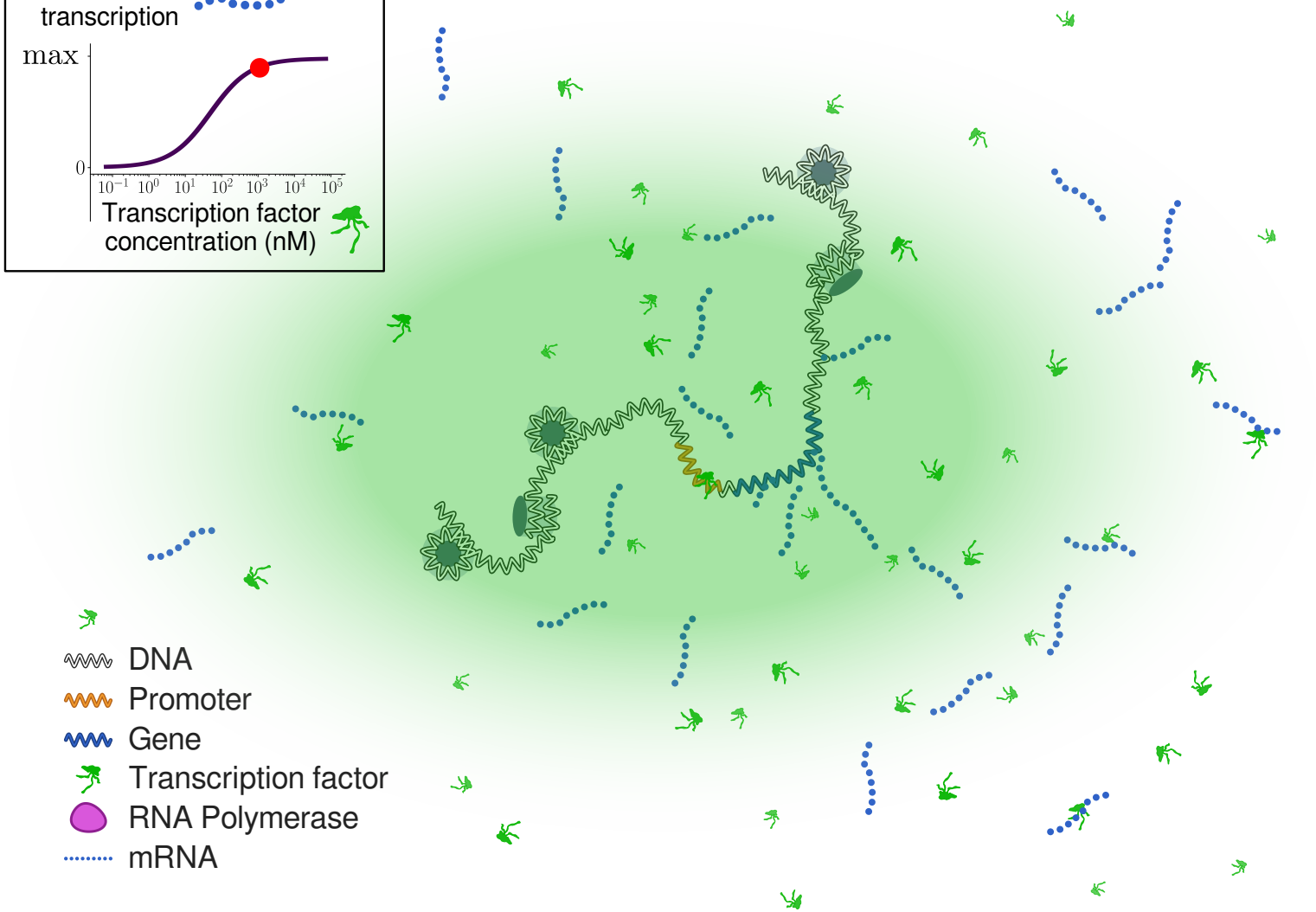
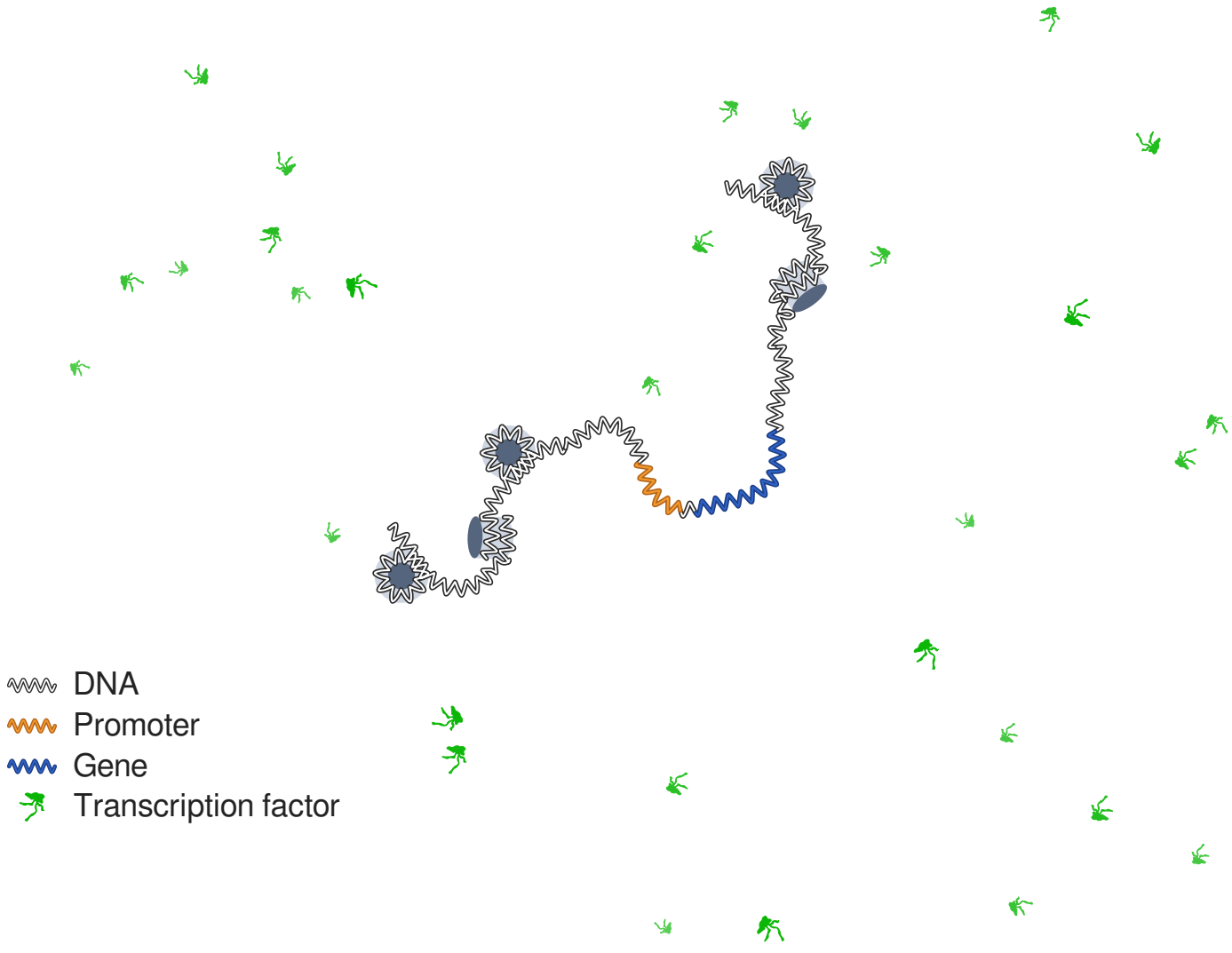




... but transcriptional control is noisy



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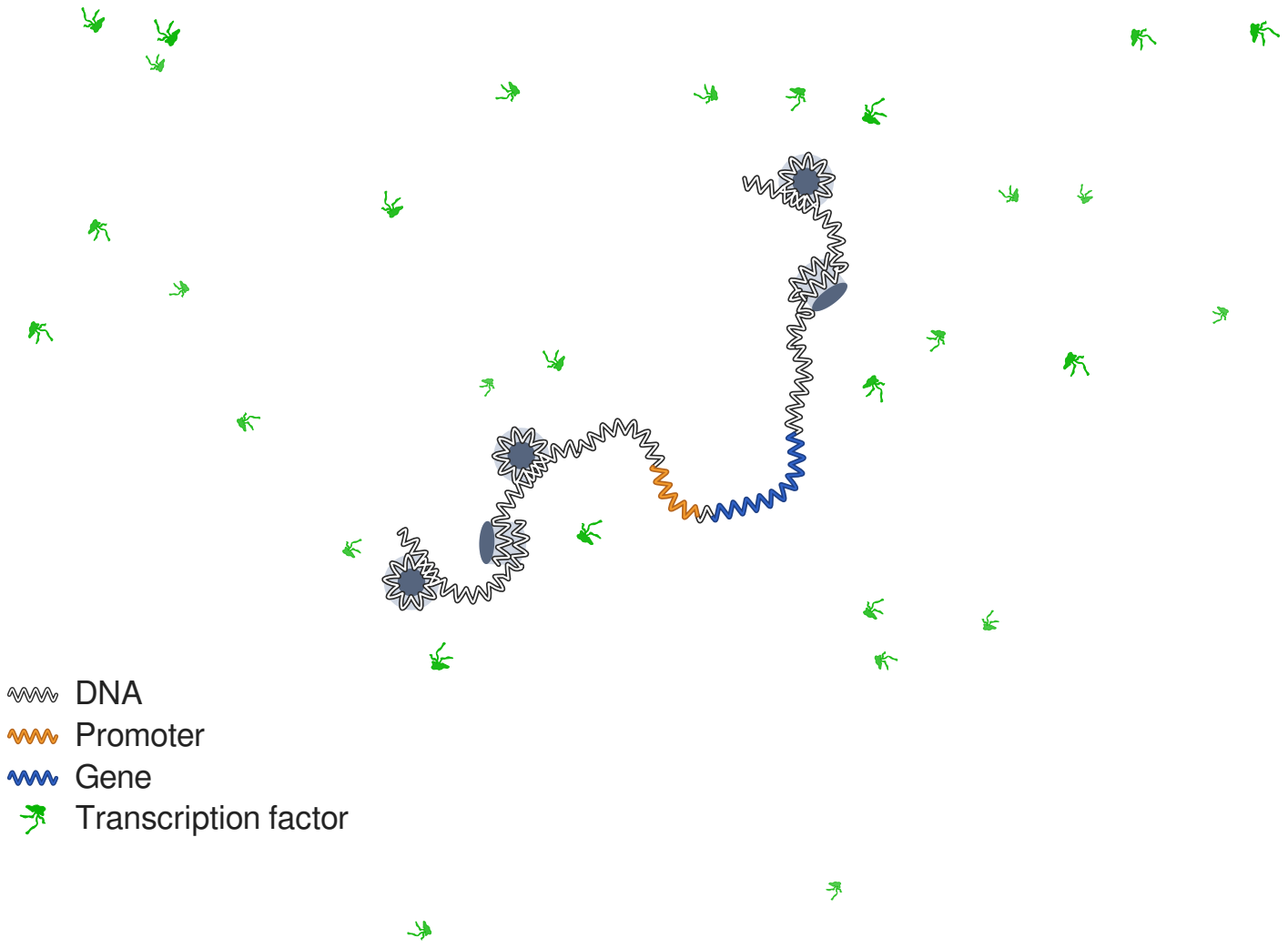

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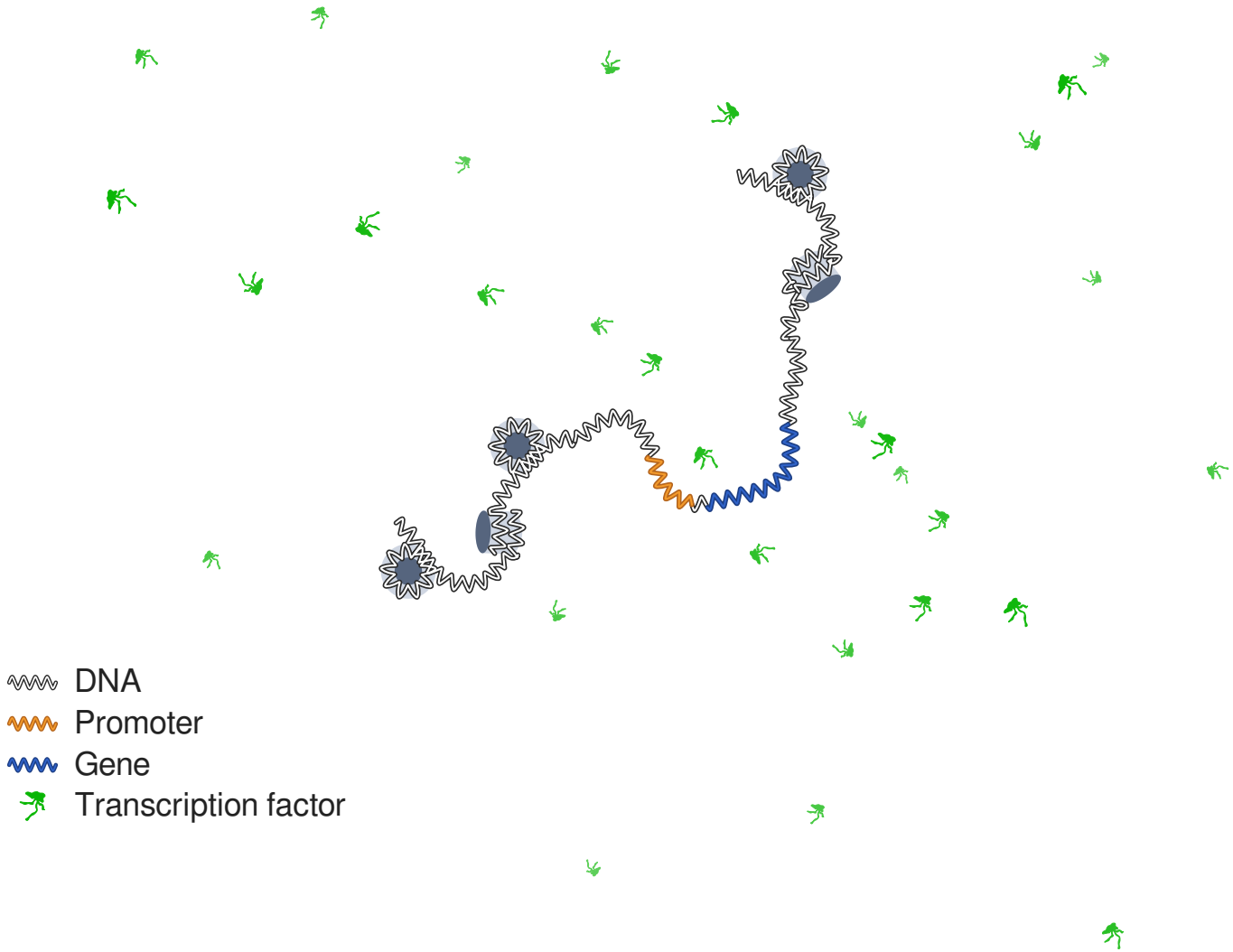

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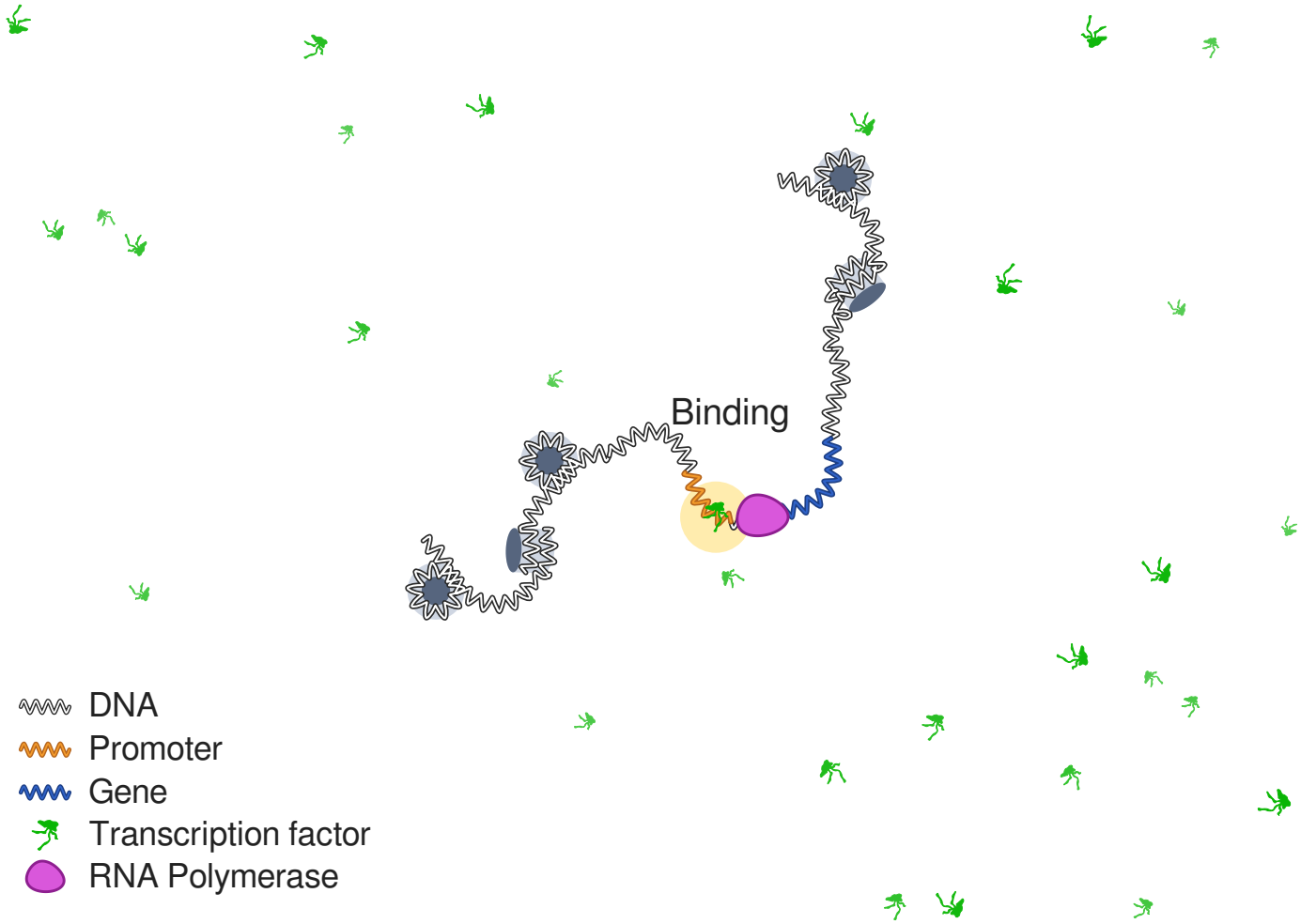

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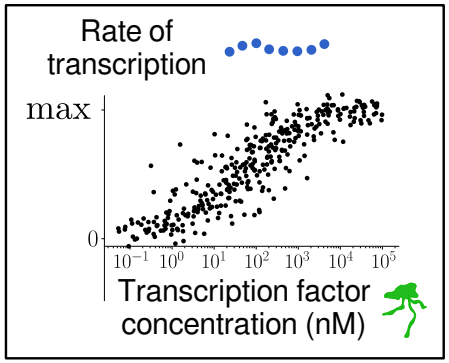
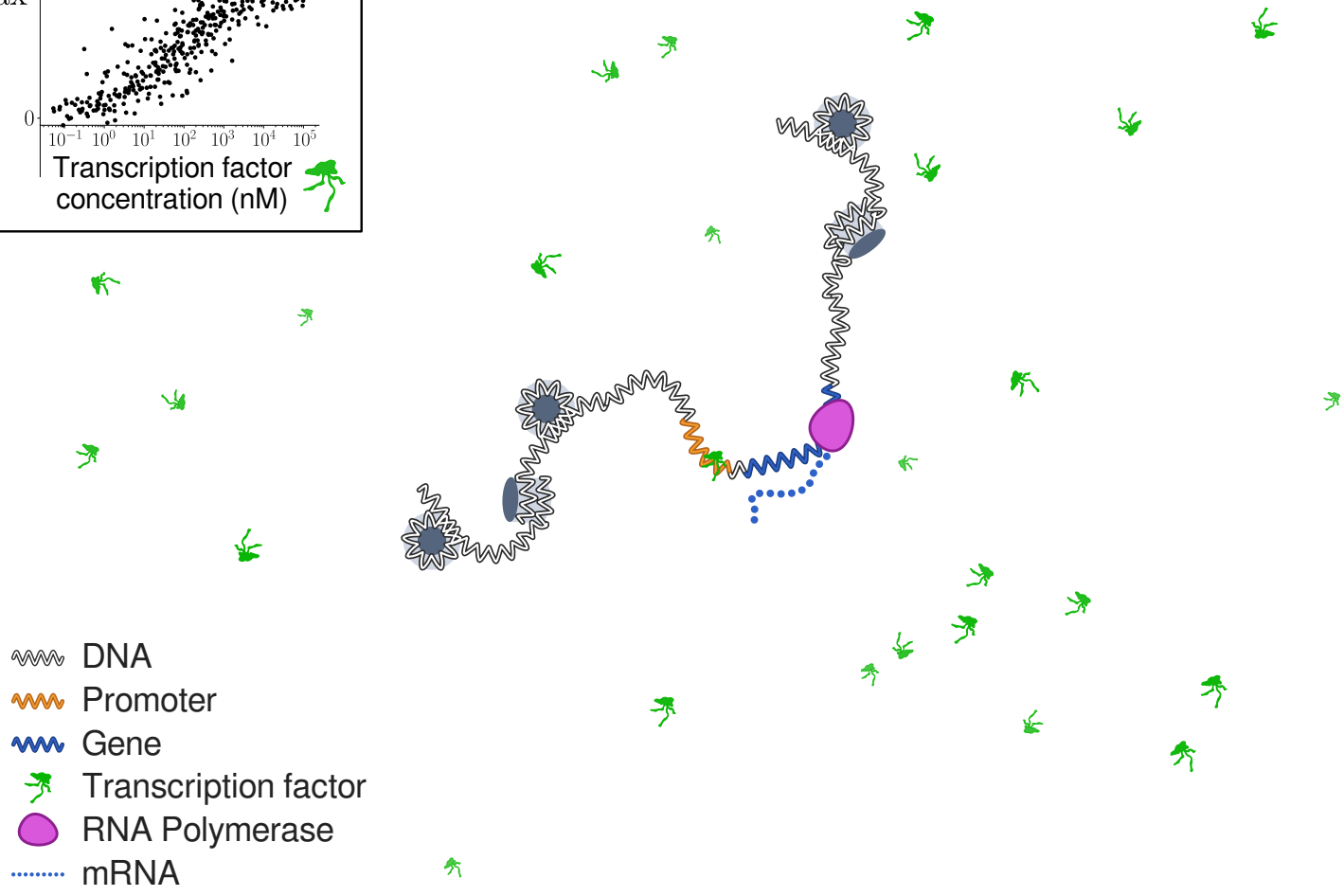
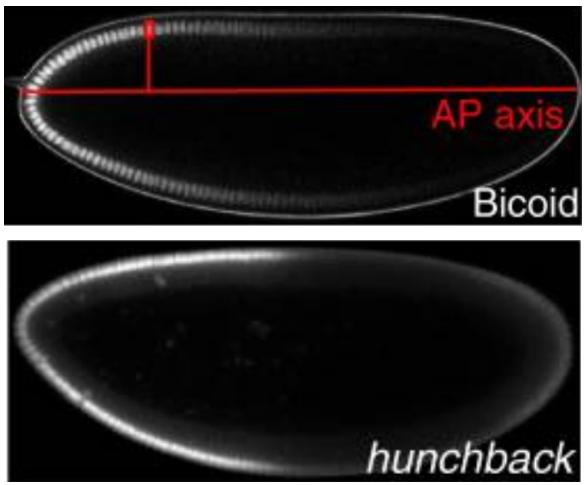
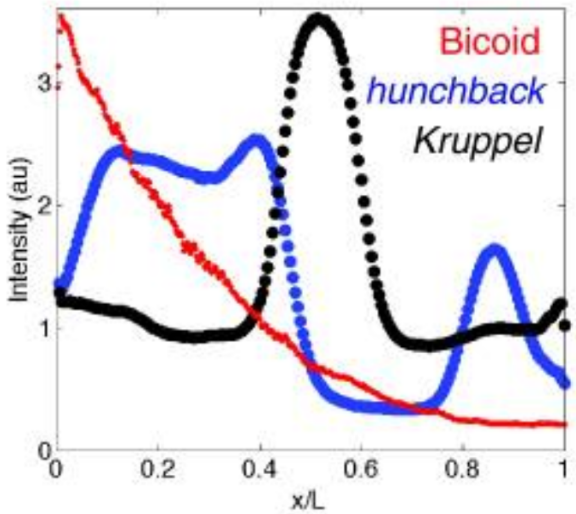




Sensitive response to shallow concentration gradients

Early Drosophila embryo

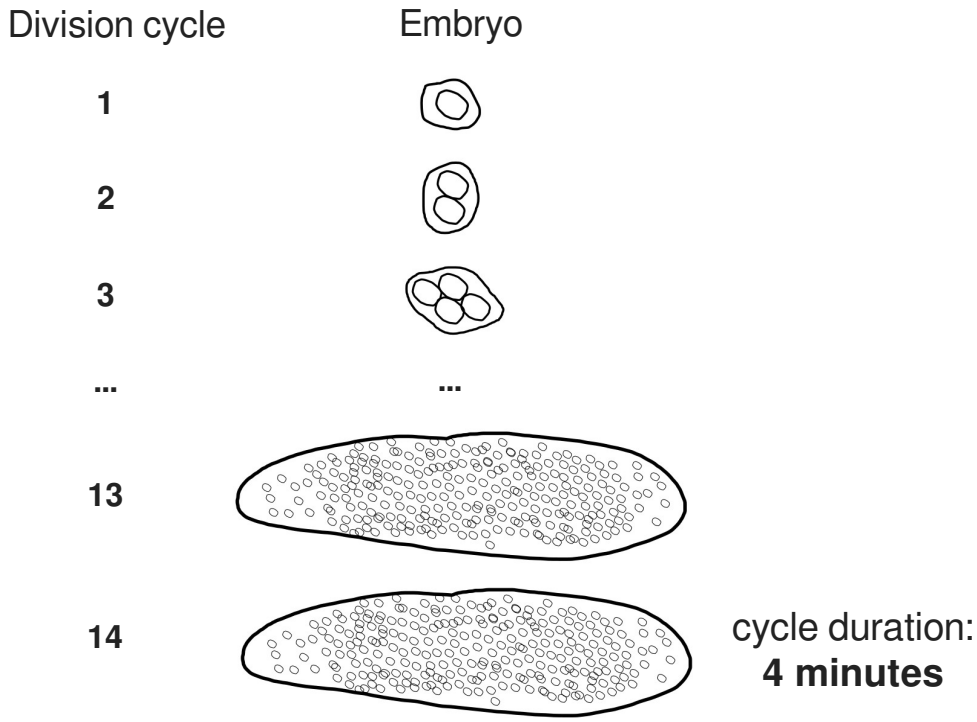




Grimm et al., *Development,* 2010

Positional information of the bicoid gradient: Gregor et al., *Cell* 2007

Precise sensing on a tight schedule



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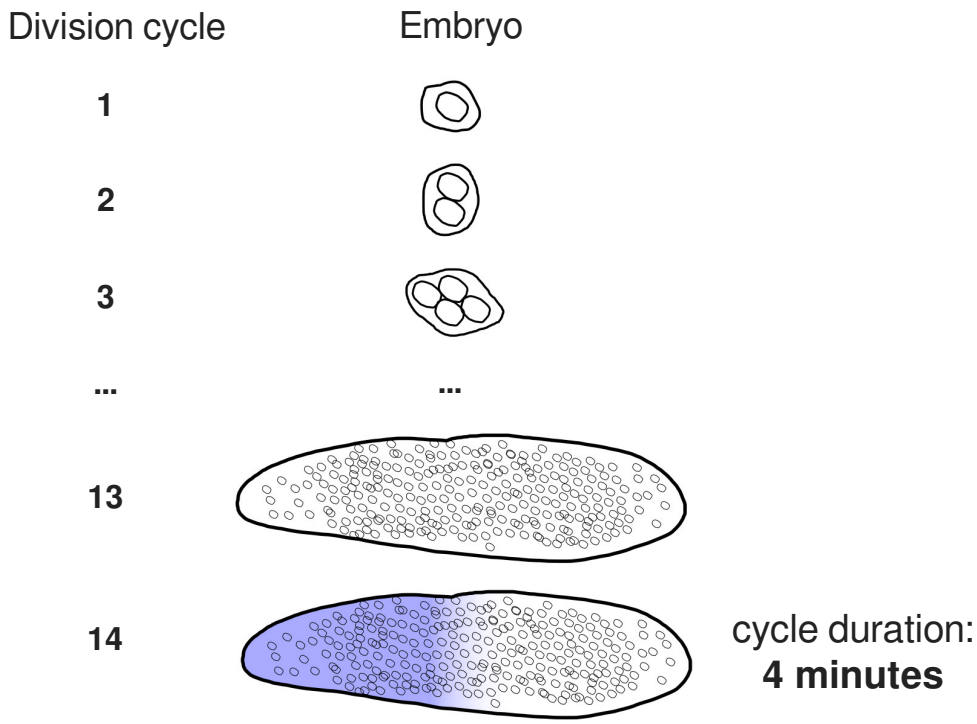

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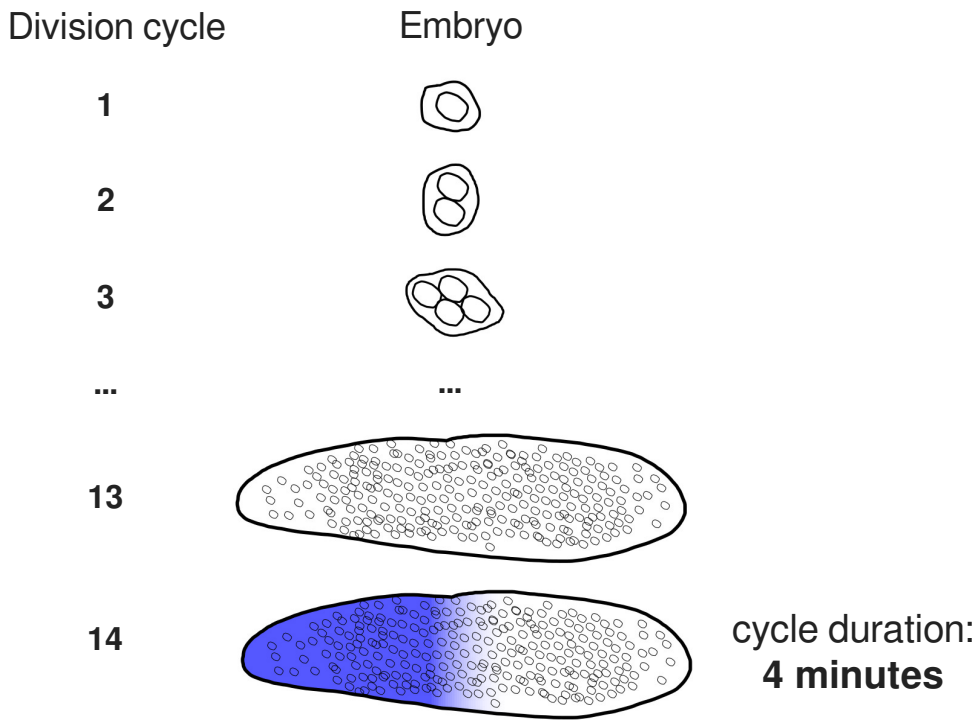

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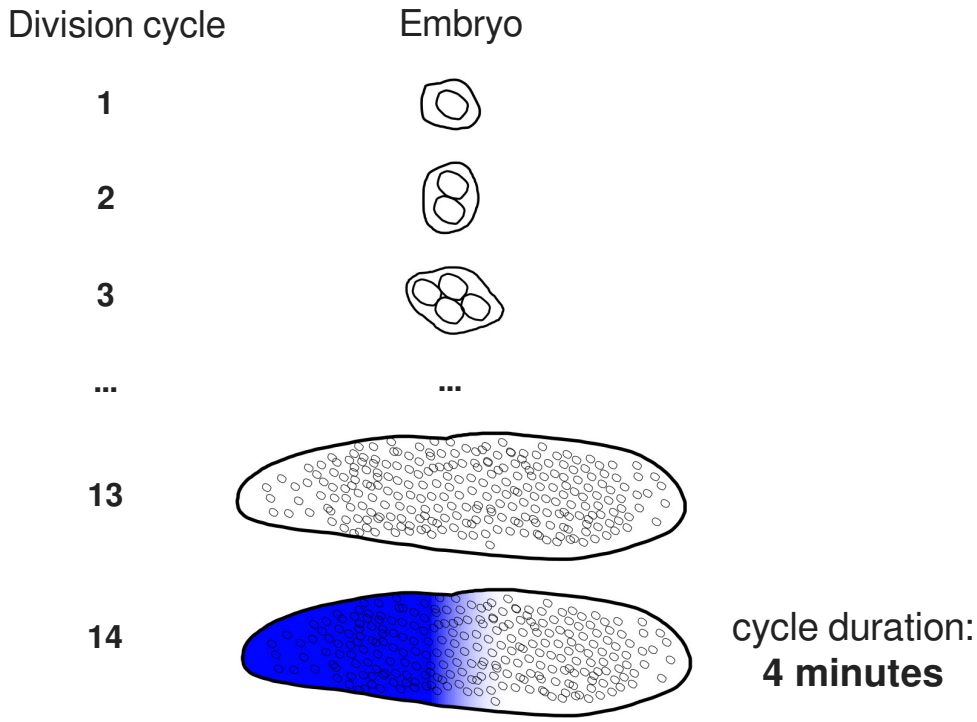

Understanding the concentration sensitivity of promoters

Questions:

1. How to quantify sensitivity, to evaluate/compare promoter models?

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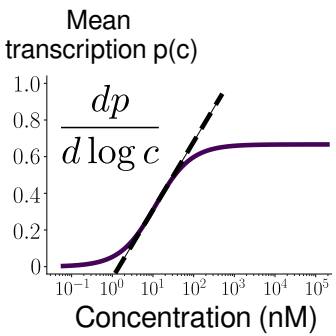

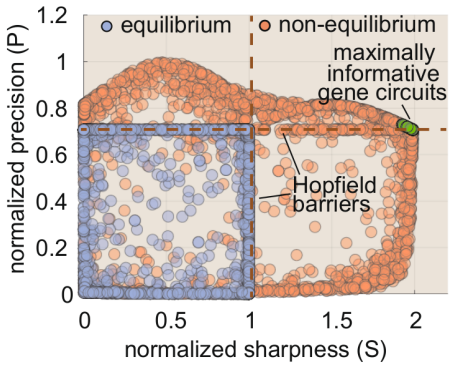

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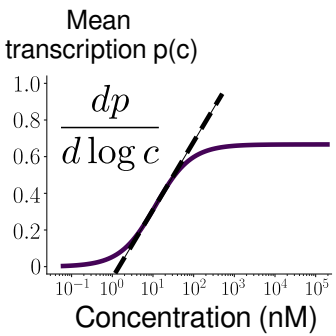

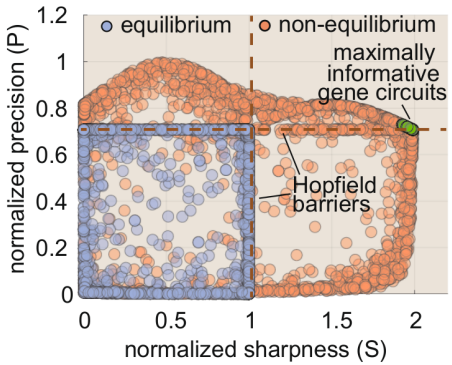

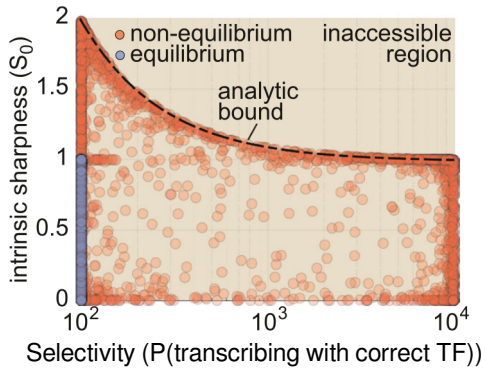

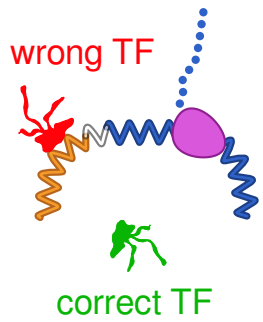

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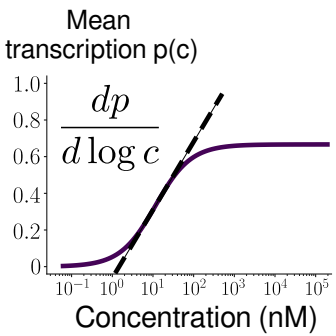

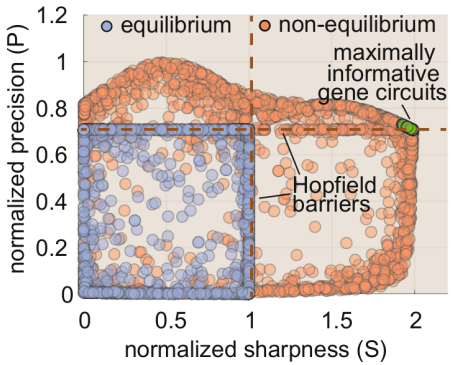

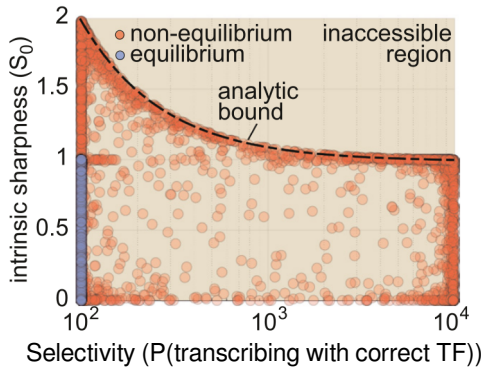

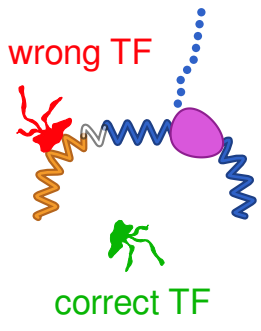

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Issues: 1. No accounting for physics-imposed rate constraints

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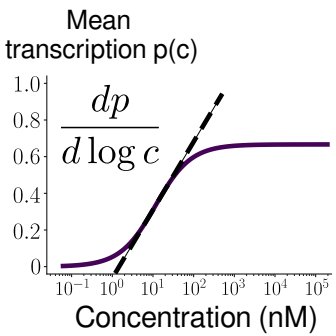

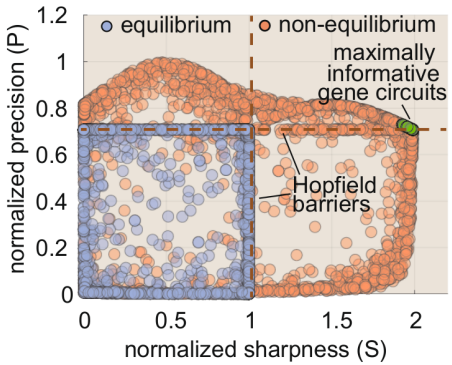

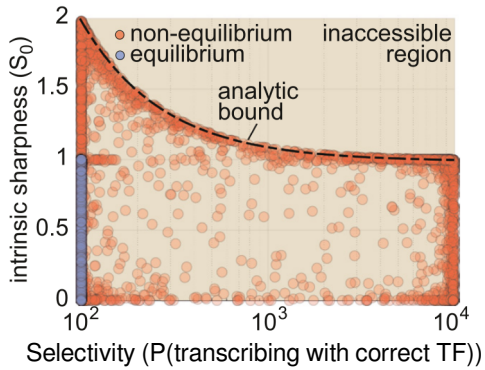

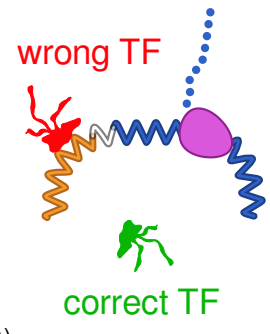

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Issues:
1. No accounting for physics-imposed rate constraints
2. Measure sensitivity in bits/burst

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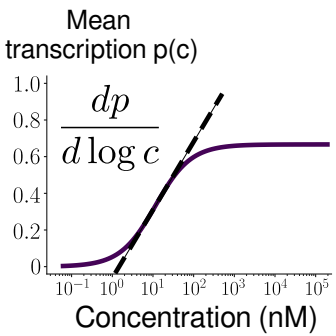

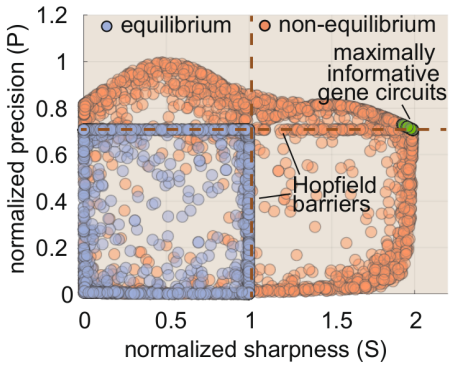

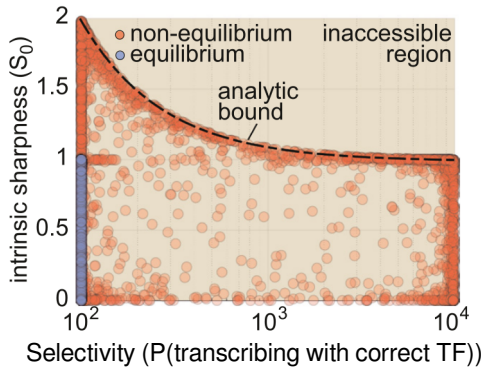

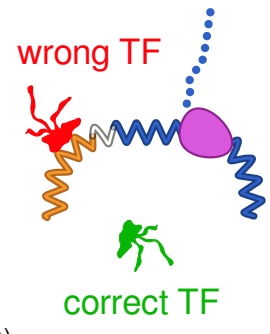

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Issues:

1. No accounting for physics-imposed rate constraints
2. Measure sensitivity in bits/burst
3. No accounting for transcriptional dynamics

Understanding the concentration sensitivity of promoters

Approach:

1. Defining a measure of precision that accounts for **dynamics**

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Approach:

1. Defining a measure of precision that accounts for **dynamics**

2. Identifying physics-imposed **constraints** on promoter architecture

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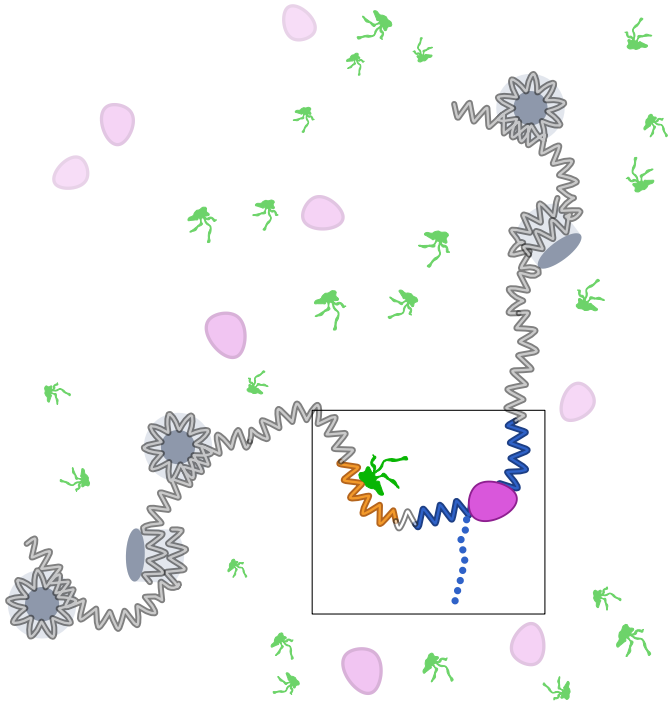
3. Constrained **optimization** of precision

A simple model of transcriptional control



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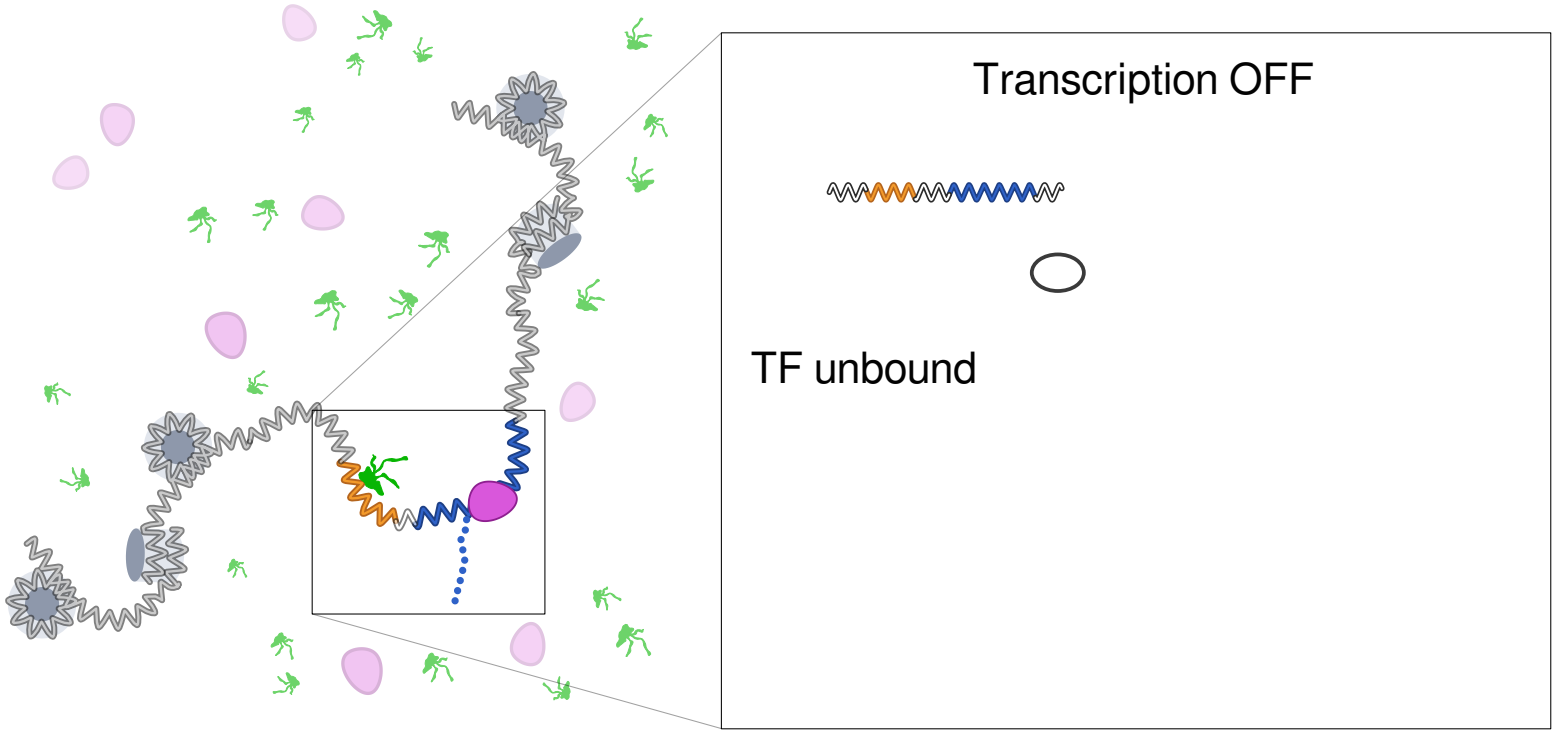

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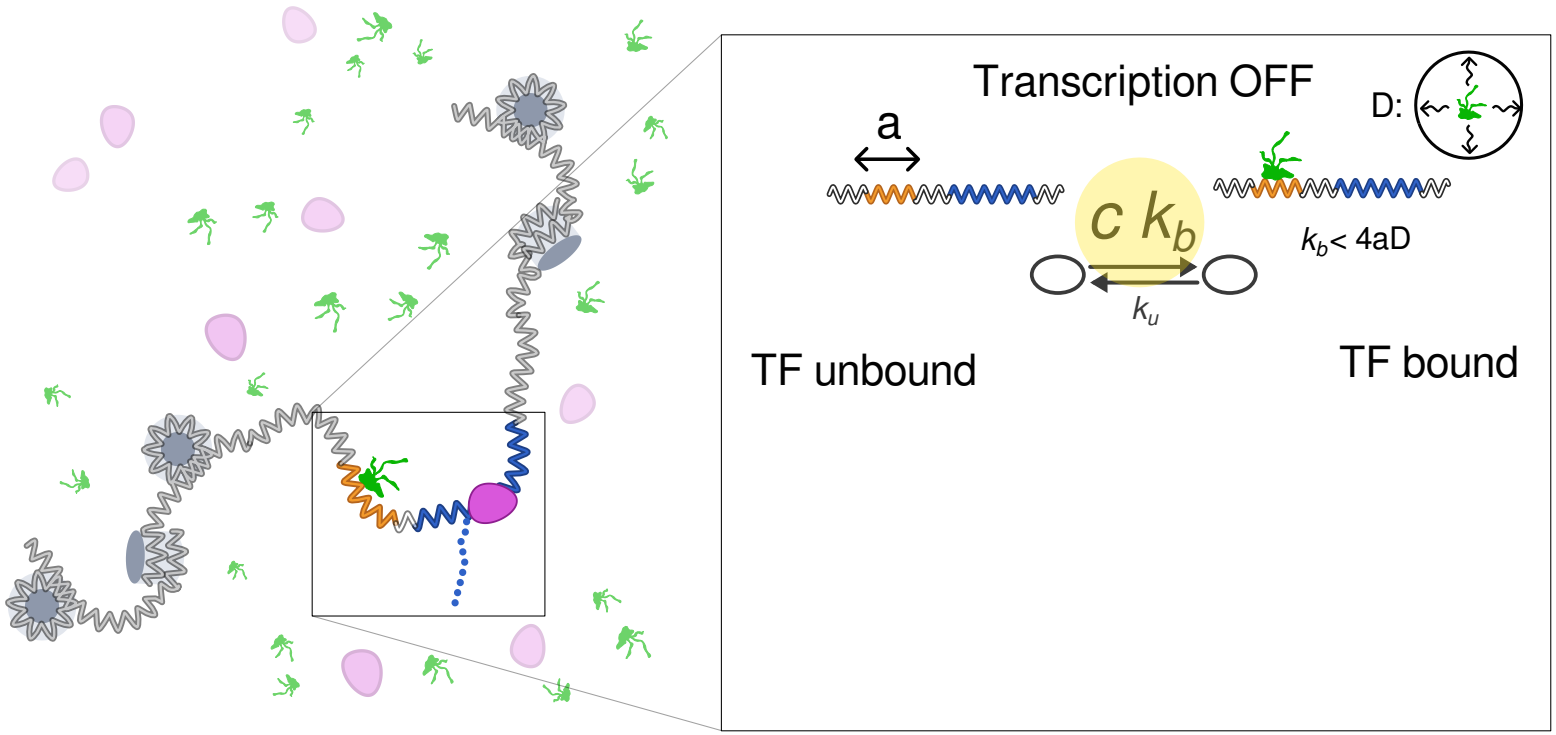

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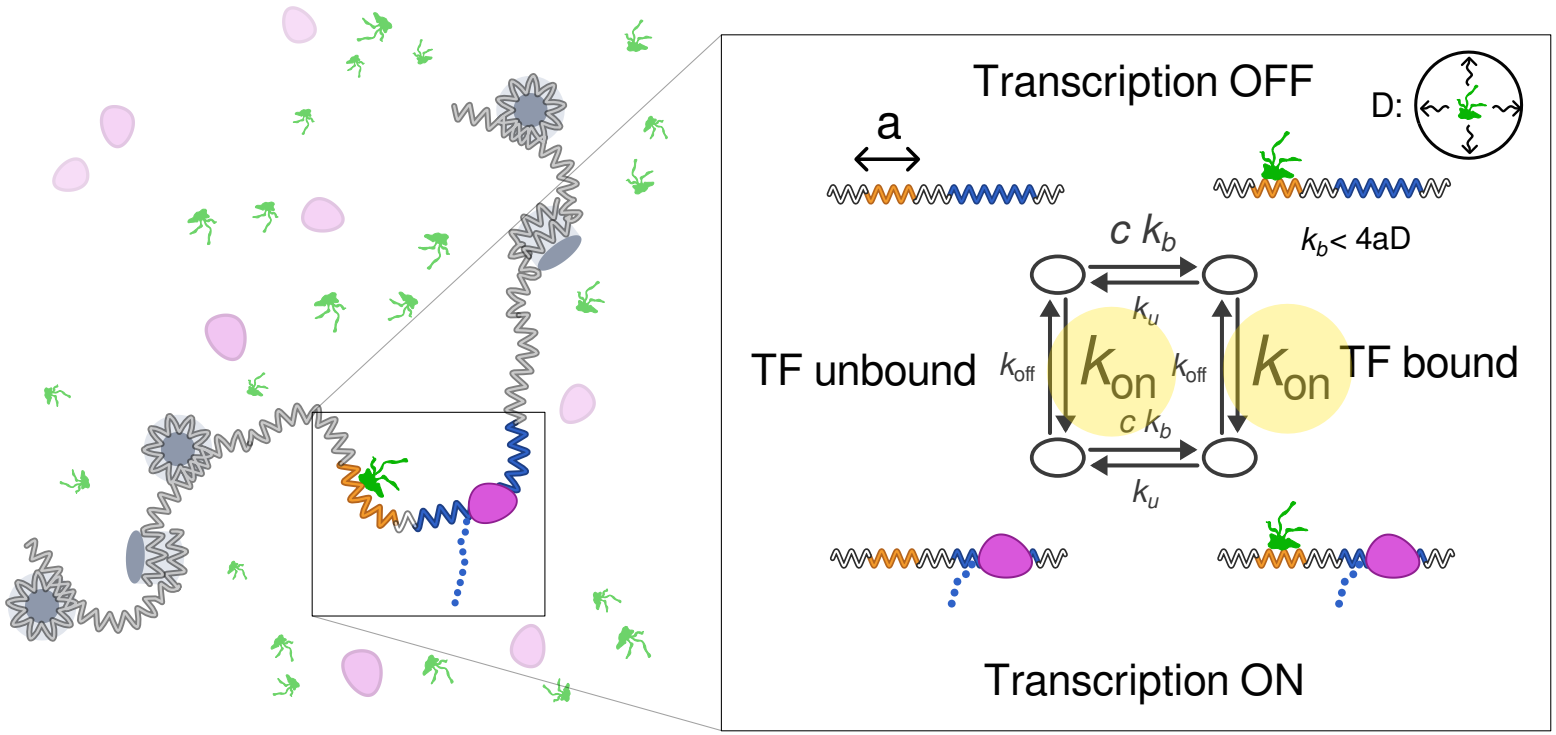

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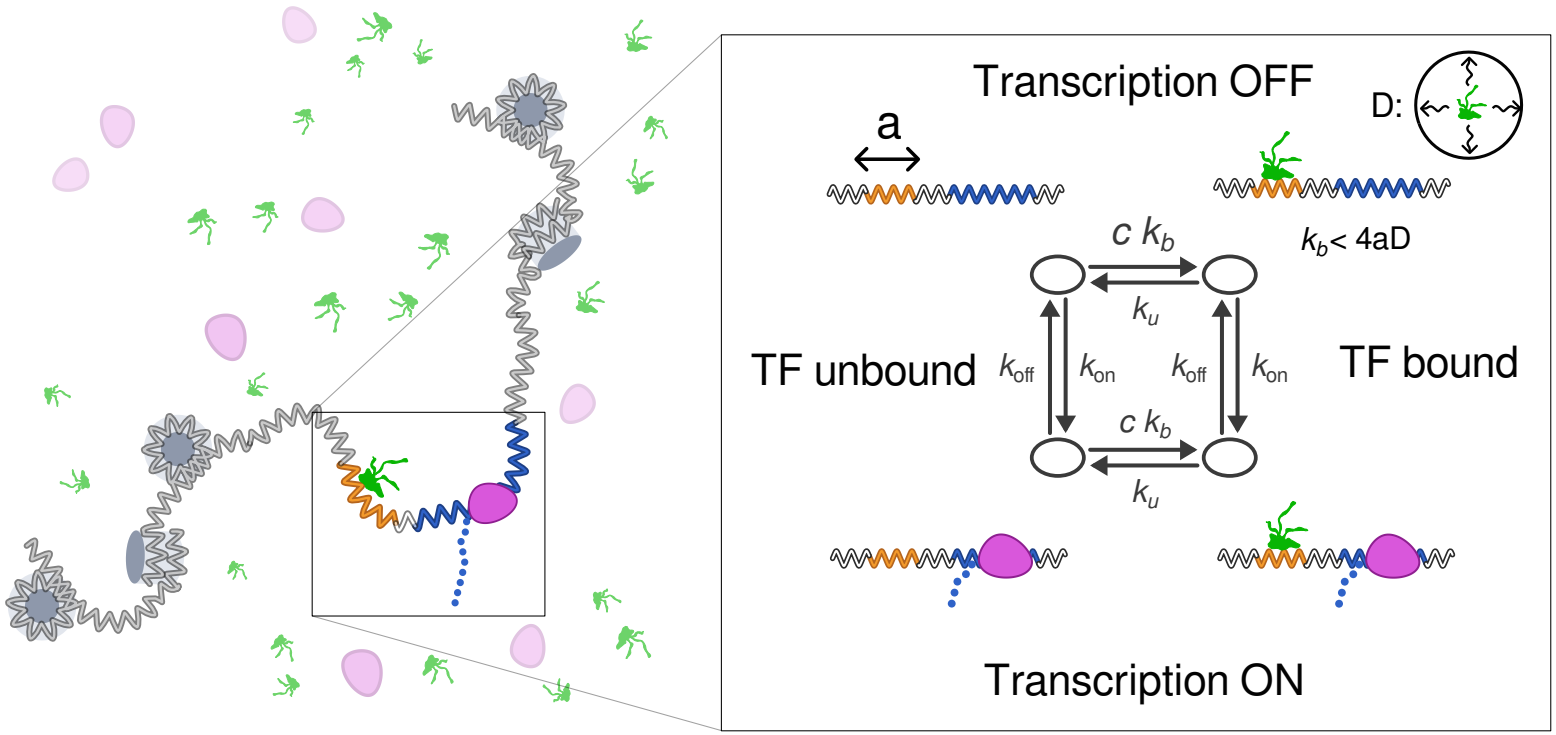

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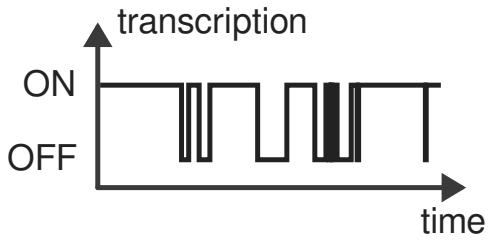
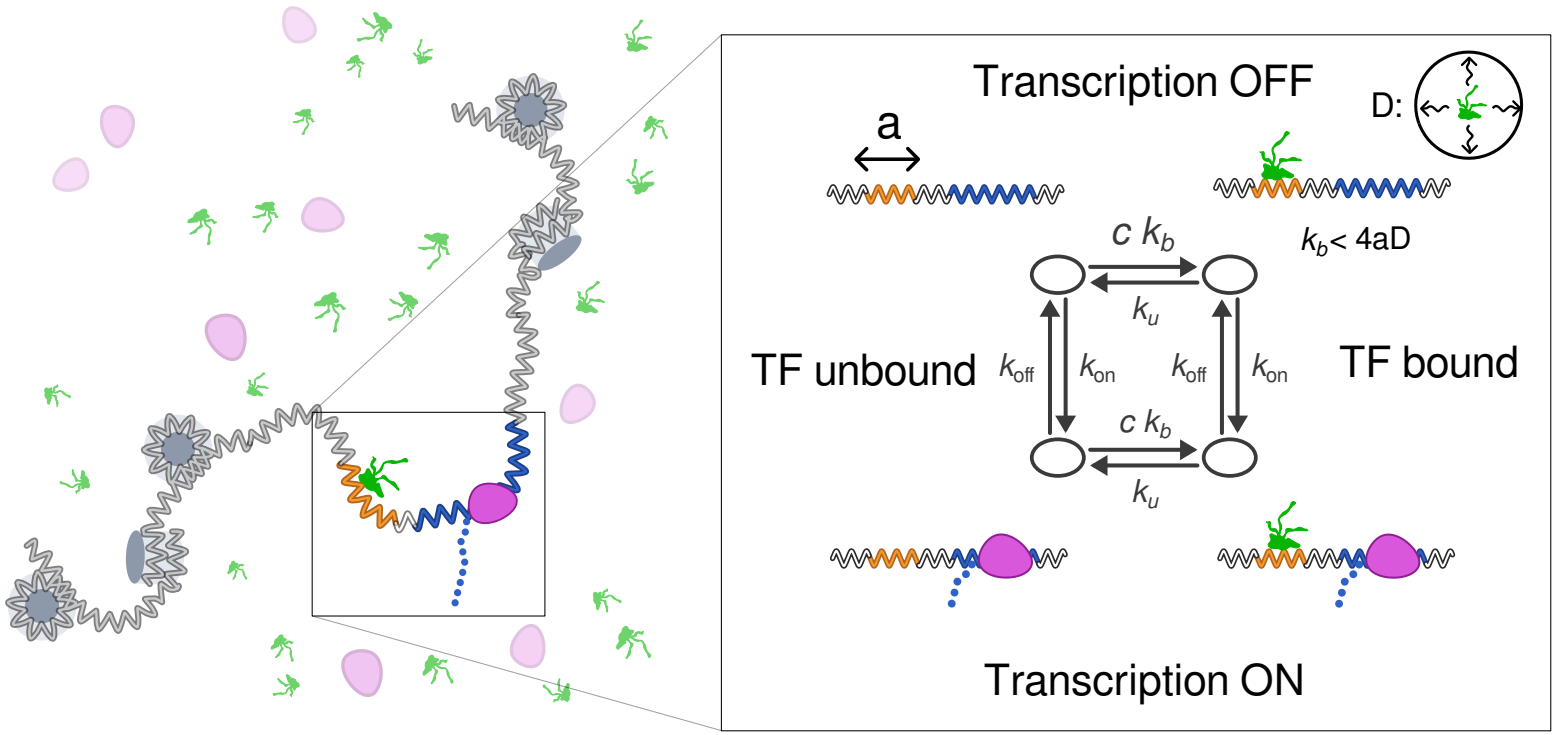
mRNA

Concentration: high

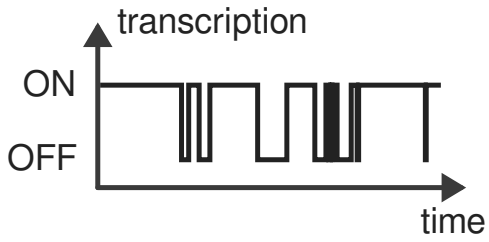


A simple model of transcriptional control



DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

Concentration: high



low

Sequential hypothesis testing

$P($ $\mid c_{\text{low}})$ $\qquad$ $P($ $\mid c_{\text{high}})$

Sequential hypothesis testing

$P($ $| c_{\text{low}})$ $P($ $| c_{\text{high}})$

$$\text{R(t)= log } \frac{P(\text{ } \mid c_{\text{low}})}{P(\text{ } \mid c_{\text{high}})}$$

Sequential hypothesis testing

$P(\text{ } \mid c_{\text{low}})$ $P(\text{ } \mid c_{\text{high}})$

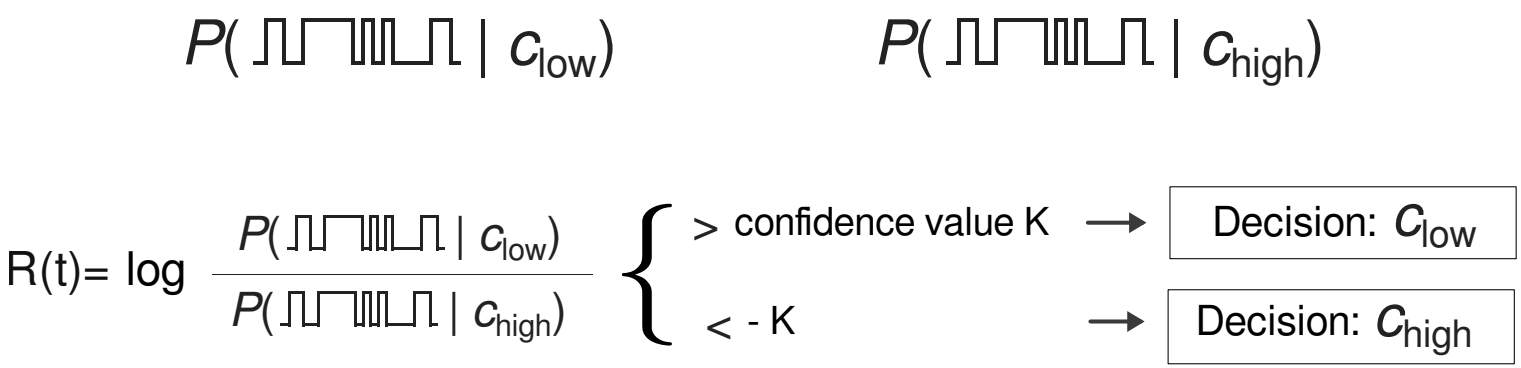
$$R(t) = \log \frac{P(\text{ } \mid c_{\text{low}})}{P(\text{ } \mid c_{\text{high}})} \begin{cases} > \text{confidence value K} \longrightarrow \boxed{\text{Decision: } c_{\text{low}}} \end{cases}$$

Sequential hypothesis testing

$P($ $| c_{low})$ $P($ $| c_{high})$

$$R(t) = \log \frac{P(\text{ } | c_{low})}{P(\text{ } | c_{high})} \begin{cases} > \text{confidence value K} \rightarrow \text{Decision: } c_{low} \\ < -K \rightarrow \text{Decision: } c_{high} \end{cases}$$

Sequential hypothesis testing

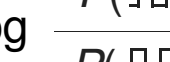


Sequential Probability Ratio Test (SPRT)

Wald 1949 (conception)
Siggia, Vergassola 2012 (application to biological receptors)
Desponds, Vergassola, Walczak 2022 (optimization)
Lammers, Flamholz, Garcia, 2024 (nonequilibrium receptors)

Sequential hypothesis testing

$P(\text{ } \mid c_{\text{low}})$ $P(\text{ } \mid c_{\text{high}})$

$$R(t) = \log \frac{P(\text{ } \mid c_{\text{low}})}{P(\text{  } \mid c_{\text{high}})}$$

$> \text{confidence value K} \rightarrow$ Decision: c_{low}

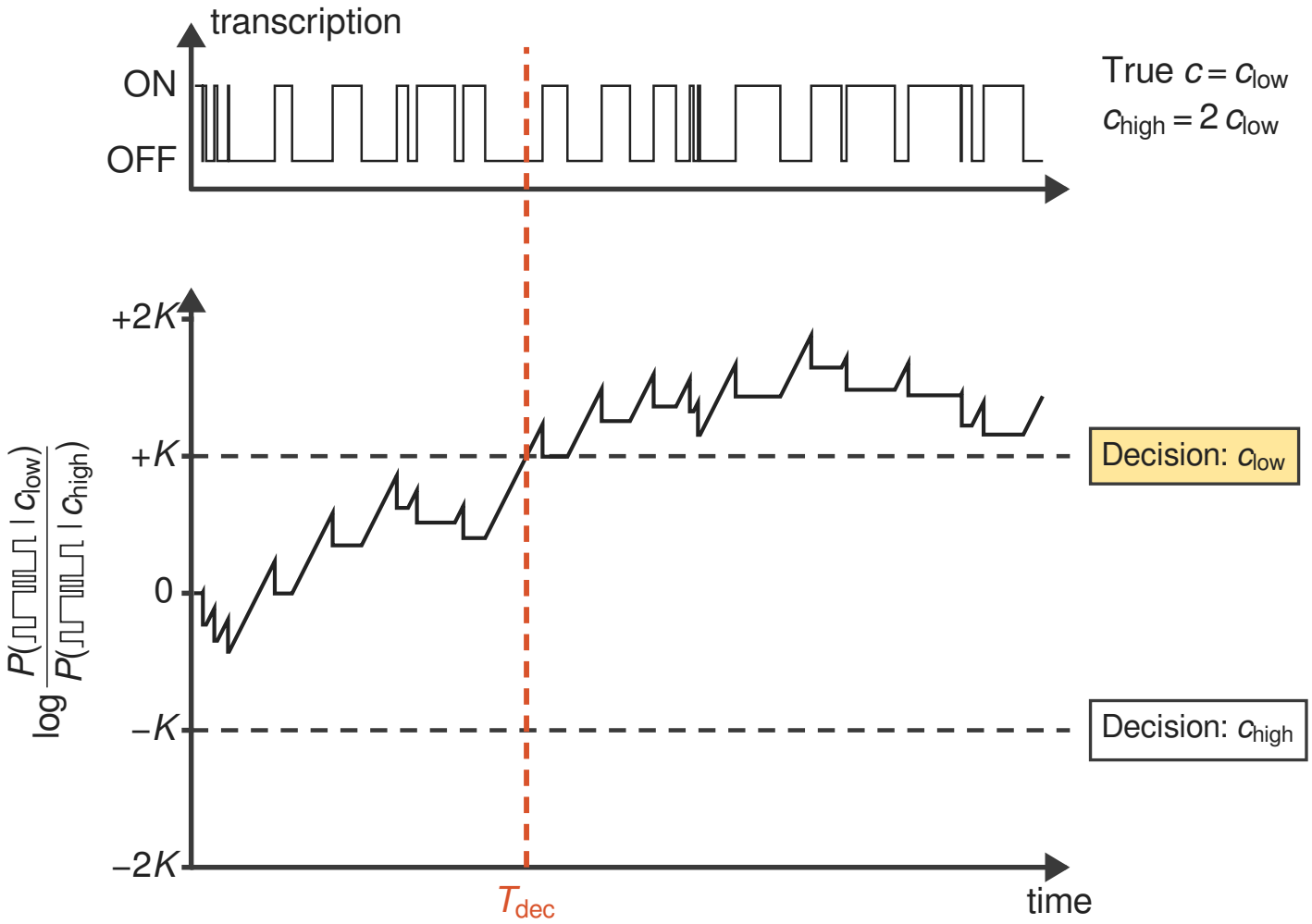
$< -\text{K} \rightarrow$ Decision: c_{high}

Sequential Probability Ratio Test (SPRT)

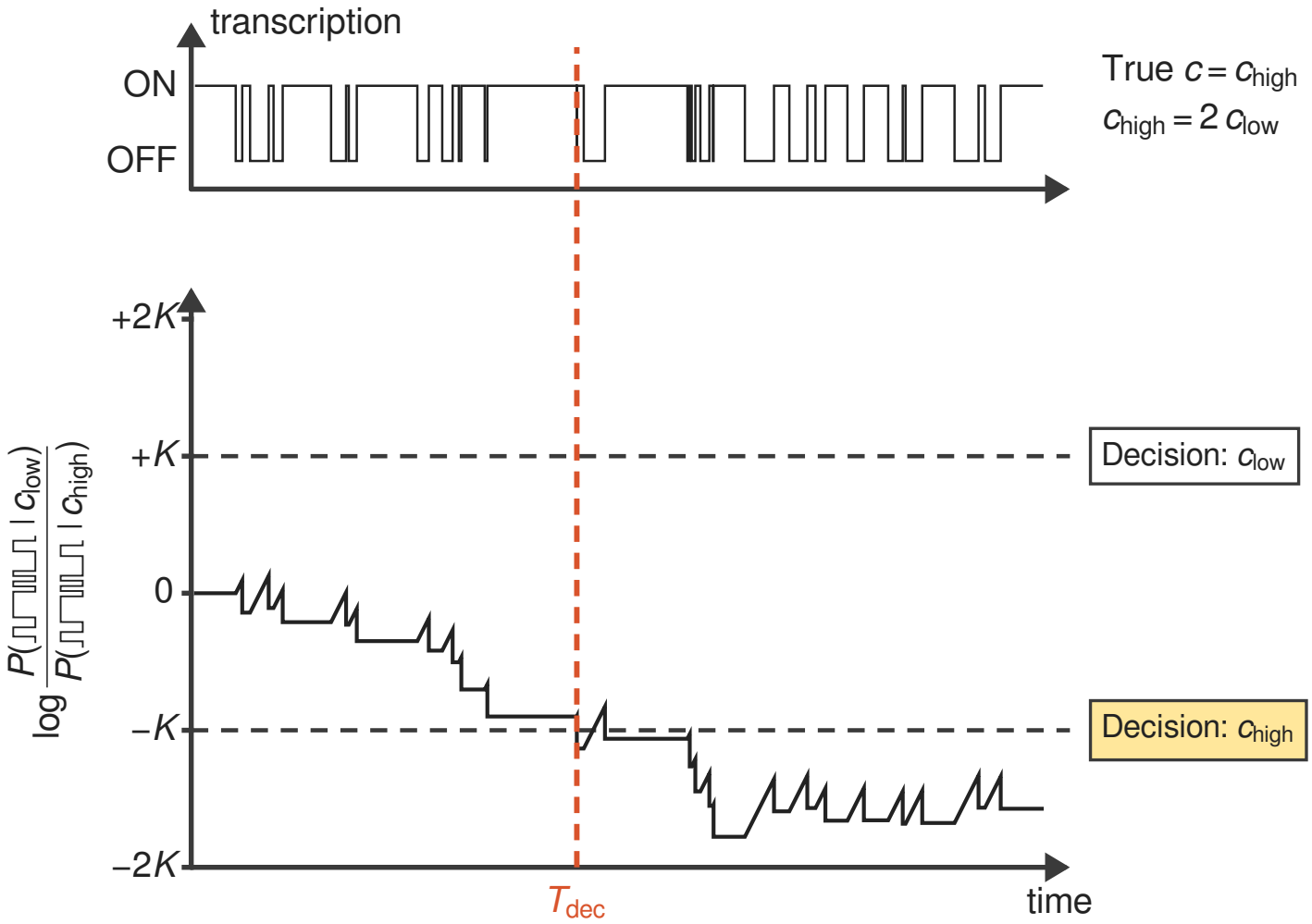
Wald 1949 (conception)
Siggia, Vergassola 2012 (application to biological receptors)
Desponds, Vergassola, Walczak 2022 (optimization)
Lammers, Flamholz, Garcia, 2024 (nonequilibrium receptors)

Philosophy:
Quantify the most information available to a cell
without worrying about how it is used

Sequential hypothesis testing



Sequential hypothesis testing



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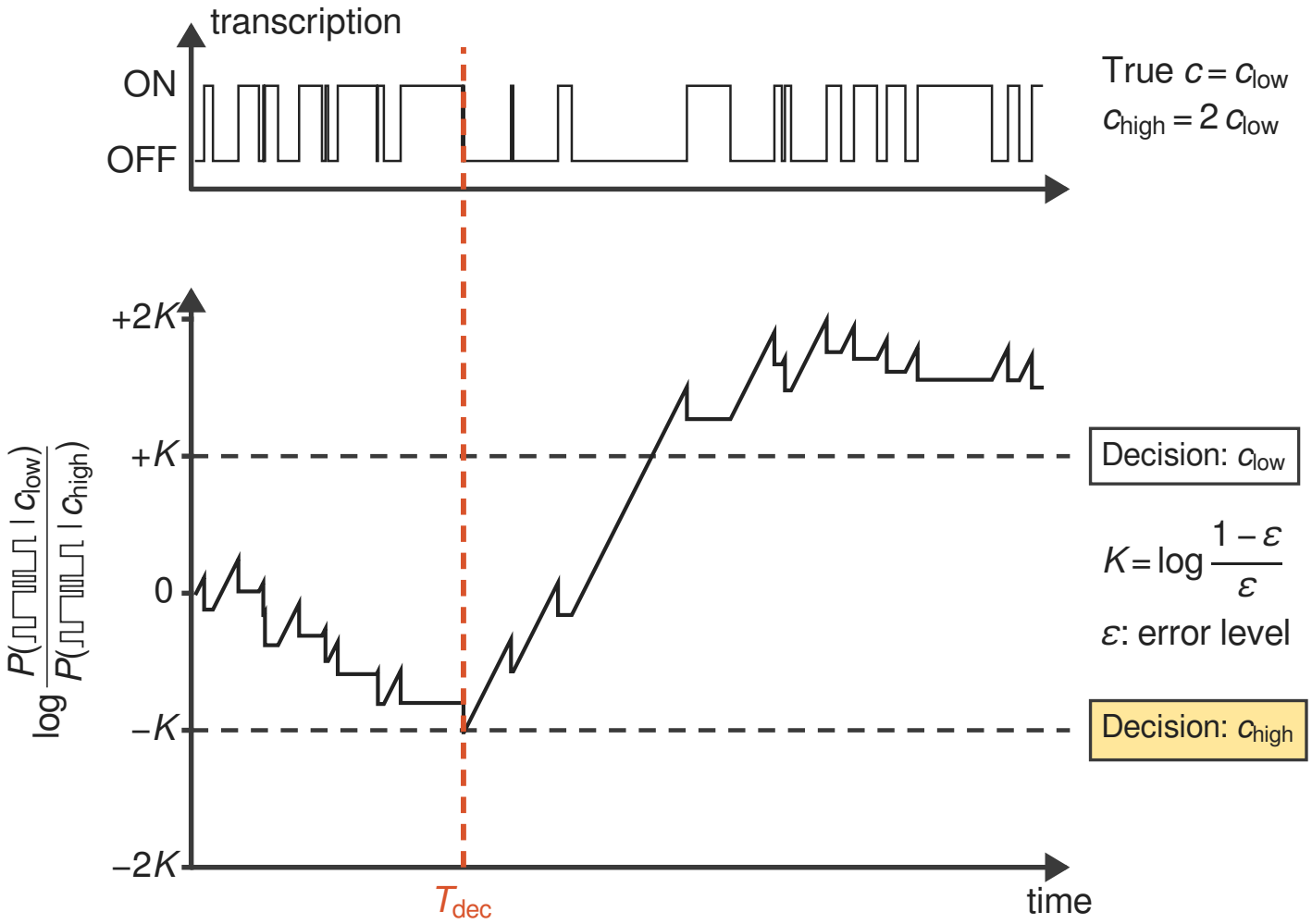
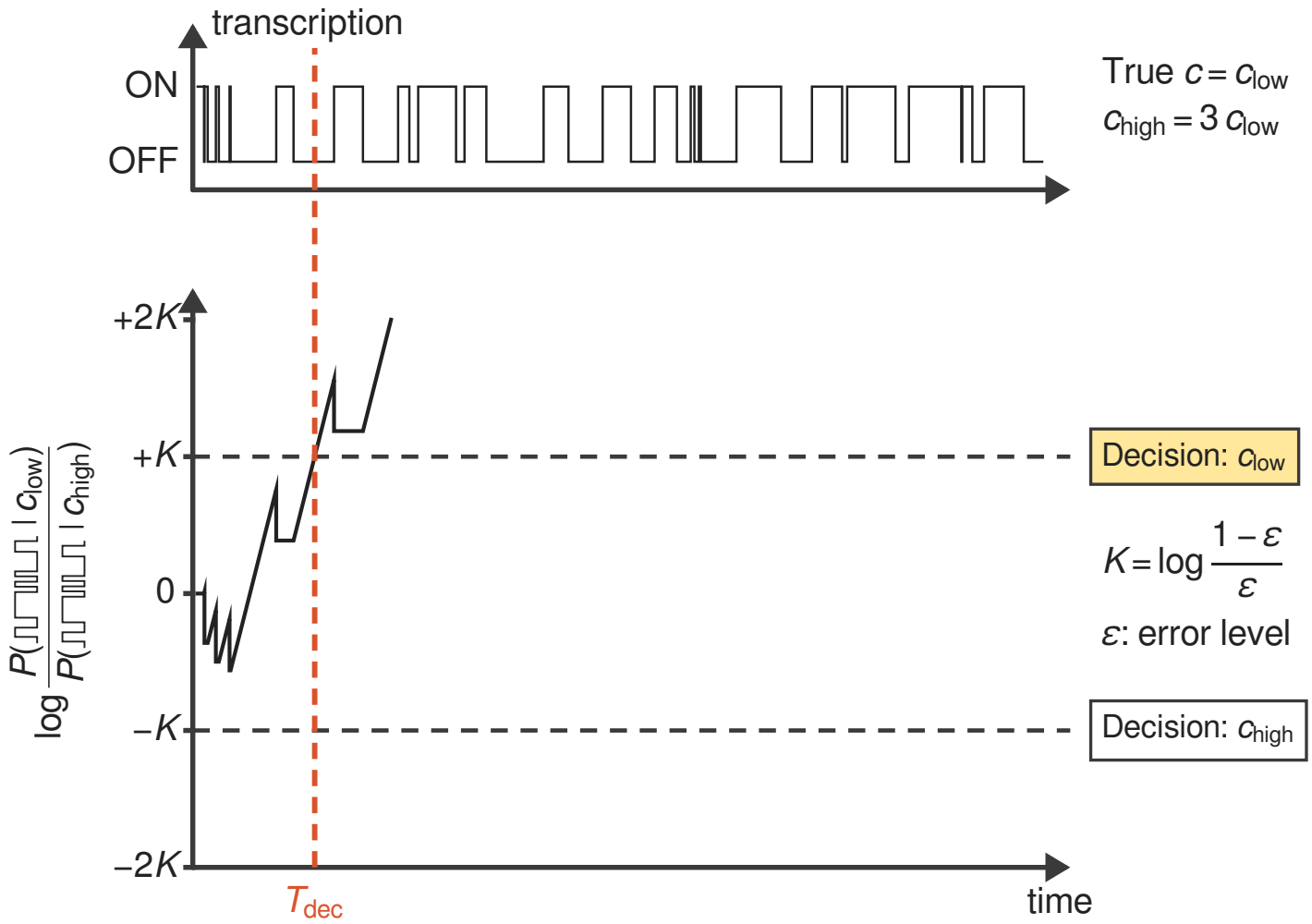


Sequential hypothesis testing



Fisher information rate and decision time

$$\mathcal{R}(t) = \log \frac{P(\text{⎍⎍⎾⎍⎍⎿⎍⎍}|c)}{P(\text{⎍⎍⎾⎍⎍⎿⎍⎍}\,|c+\delta c)}$$

Fisher information rate and decision time

$$\mathcal{R}(t) = \log \frac{P(\text{⎍⎍⎍⎍⎍}|c)}{P(\text{⎍⎍⎍⎍⎍}|c+\delta c)}$$

Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = V + \sqrt{2V}\eta \leftarrow \text{Gaussian noise } \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')$$

Drift = Diffusion

Fisher information rate and decision time

$$\mathcal{R}(t) = \log \frac{P(\text{⎍⎍⎍⎍⎍⎍⎍}|c)}{P(\text{⎍⎍⎍⎍⎍⎍⎍}|c+\delta c)}$$

Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = \underbrace{V}_{\text{Drift}} + \underbrace{\sqrt{2V}\eta}_{\text{Diffusion}} \leftarrow \begin{array}{c}\text{Gaussian noise} \\ \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')\end{array}$$

SPRT mean decision time (Siggia & Vergassola 2012)

$$\langle T_{\text{dec}} \rangle = \frac{K}{V} \tanh\left(\frac{K}{2}\right)$$

Fisher information rate and decision time

$$\mathcal{R}(t) = \log \frac{P(\text{⎍⎍⎍⎍⎍⎍⎍}|c)}{P(\text{⎍⎍⎍⎍⎍⎍⎍}|c+\delta c)}$$

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Drift (our result)

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

Holds for any model

Fisher information rate and decision time

$$\mathcal{R}(t) = \log \frac{P(\text{⎍⎍⎍⎍⎍}|c)}{P(\text{⎍⎍⎍⎍⎍}|c+\delta c)}$$

Drift-diffusion approximation when $\delta c/c \ll 1$

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Drift (our result)

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

Holds for any model

Fisher information rate

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(\text{⎍⎍⎍⎍⎍}|c)$$

The optimization problem

$$\text{Maximize } \dot{\mathcal{F}}(c)$$

subject to the constraints

$$k_b \leq 4aD \text{ and } k_{\text{u,on,off}} \leq k_{\max}$$

Diffusion Activation

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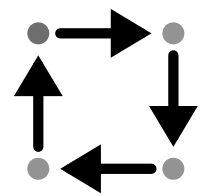
Diffusion Activation

In practice:

First pass: Monte Carlo simulation (approximate)

All solutions:



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Second pass: exact expression assuming this topology

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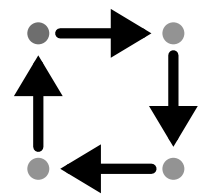

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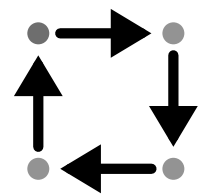
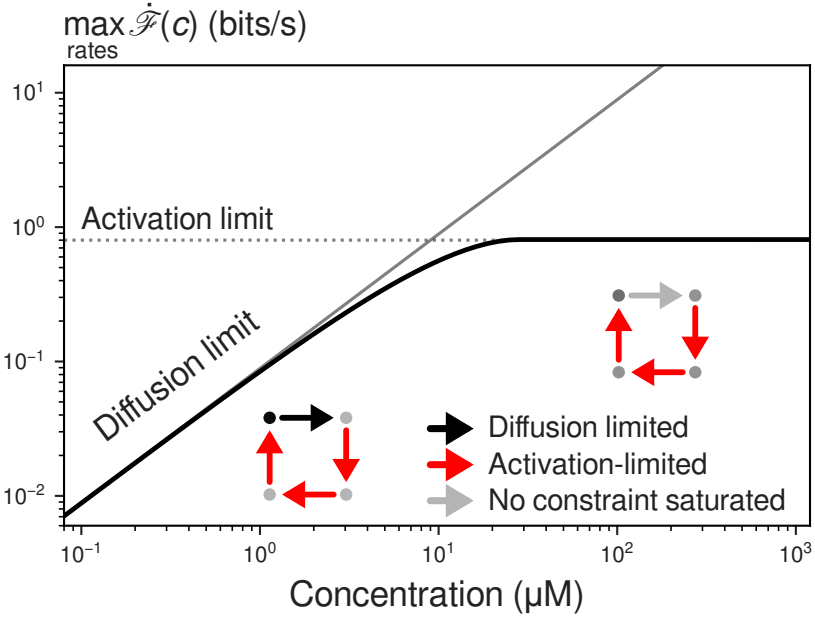


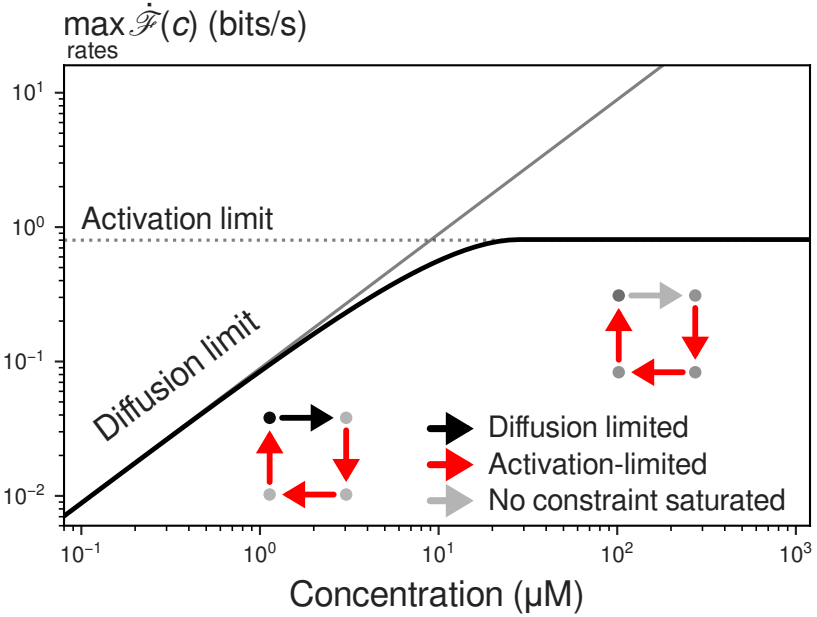
Control ignoring dynamics: use likelihood of mean mRNA

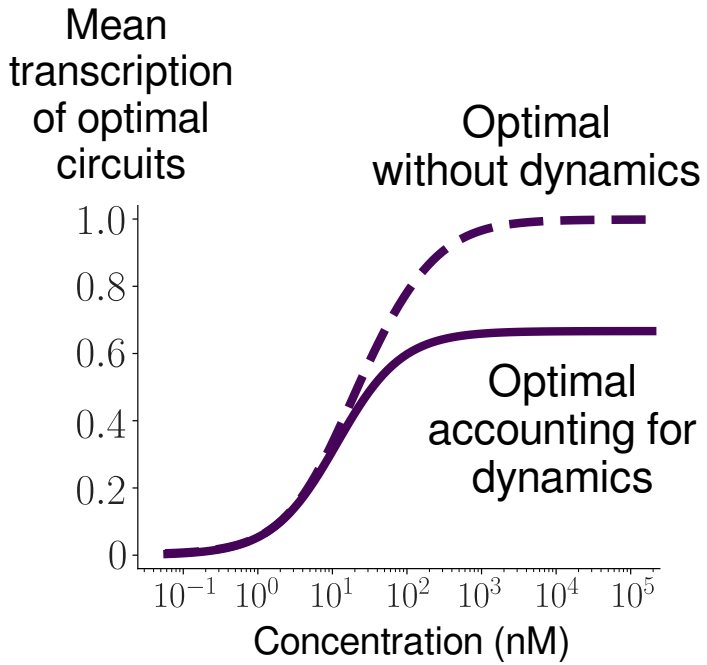
$P(\text{mRNA level } | c)$

Optimal promoters: two distinct regimes

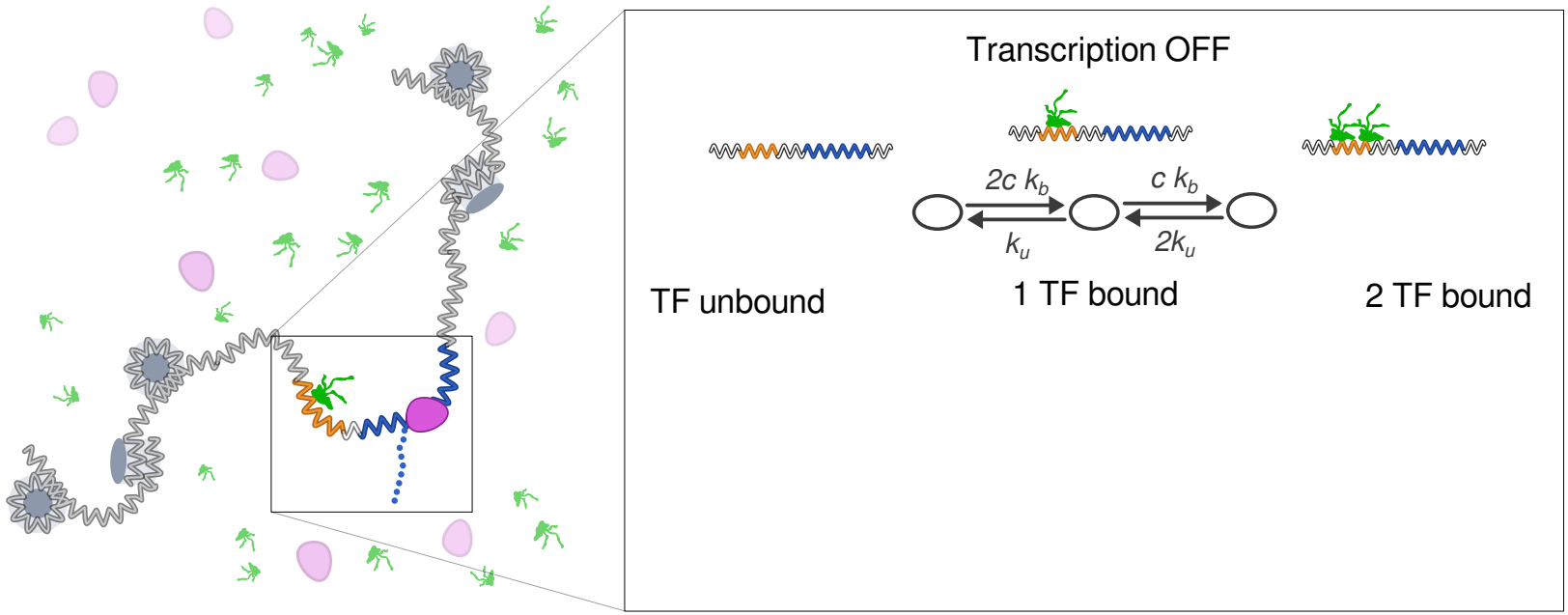


Optimal promoters: two distinct regimes



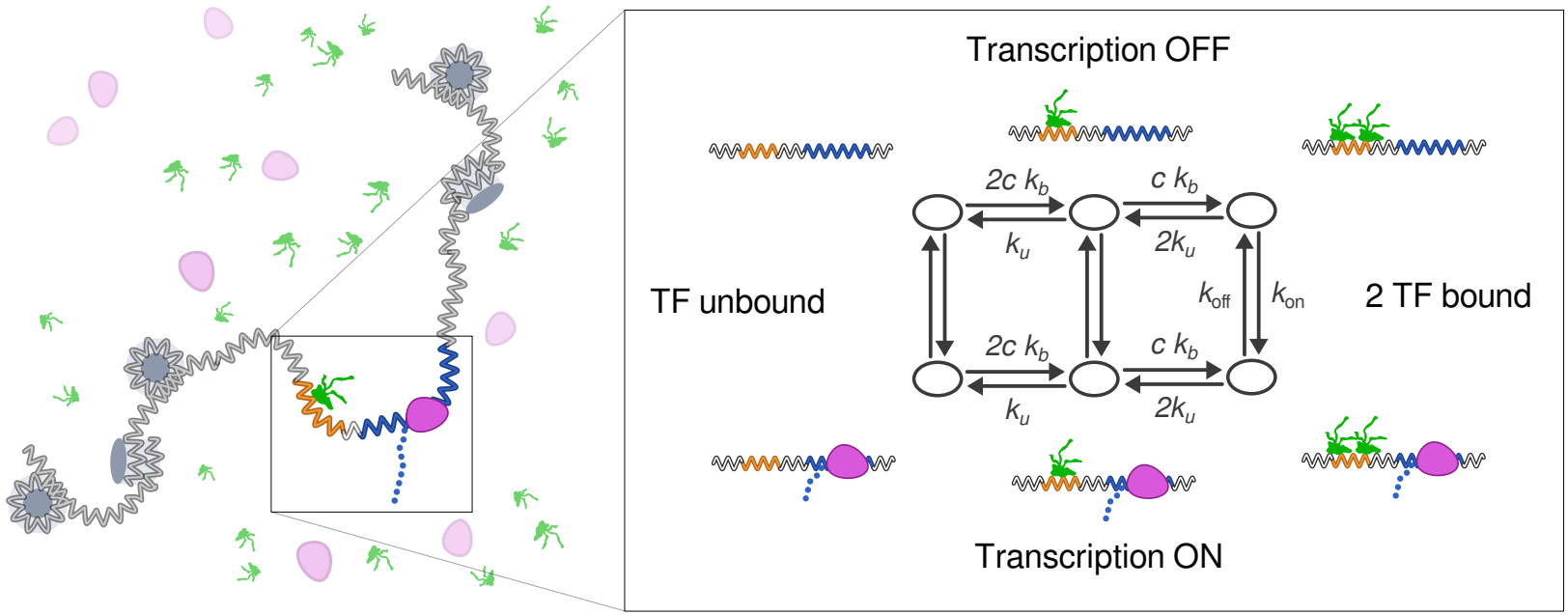


Cooperative binding



DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

Cooperative binding



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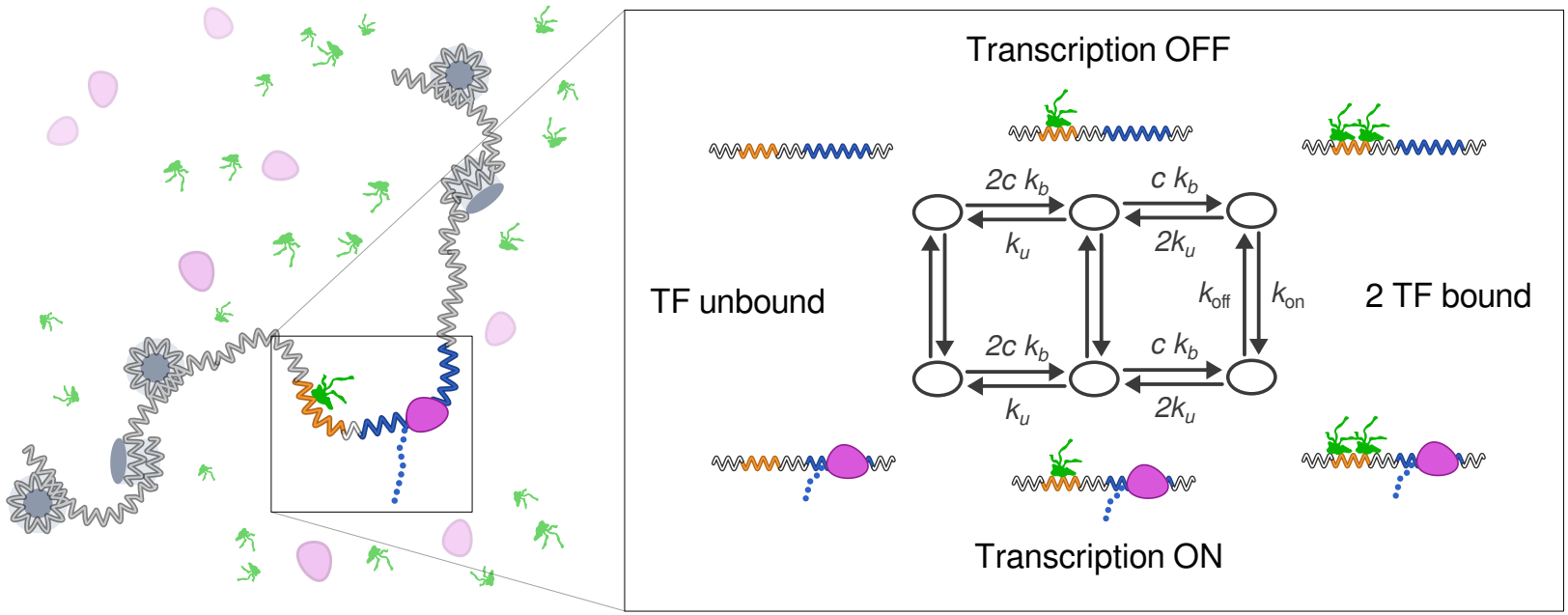
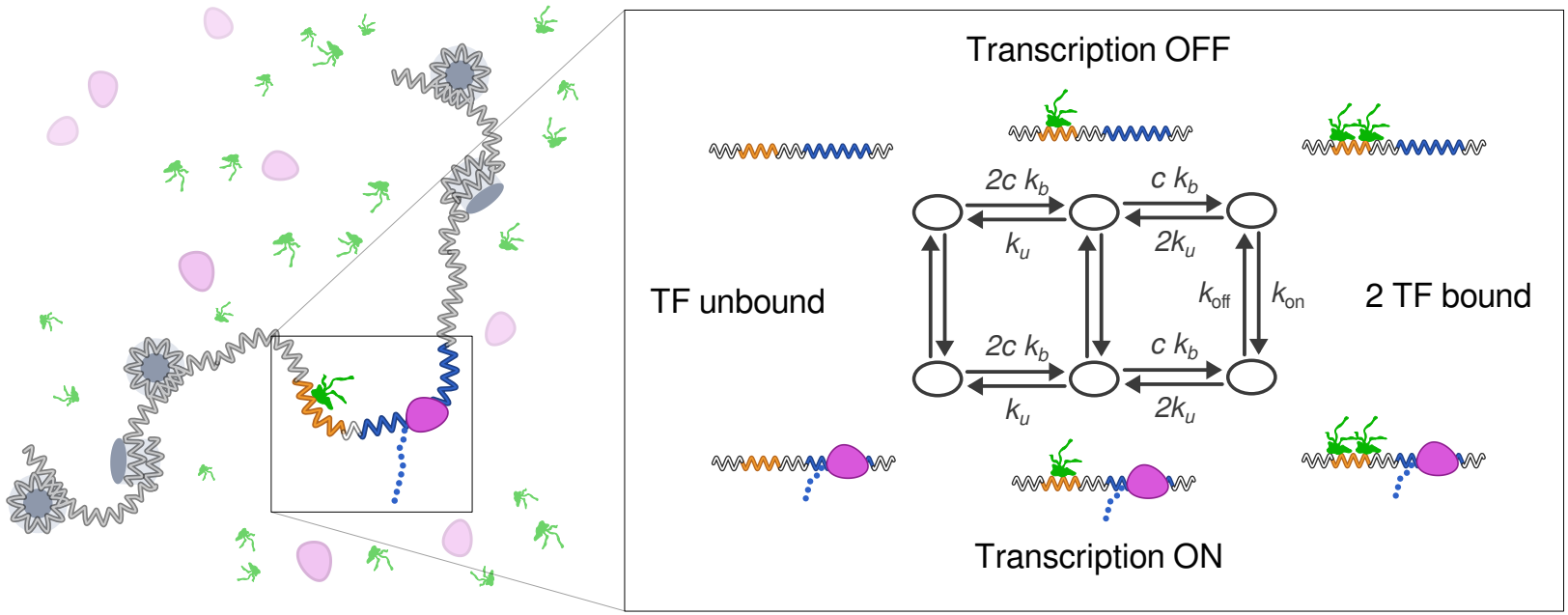
Cooperative binding



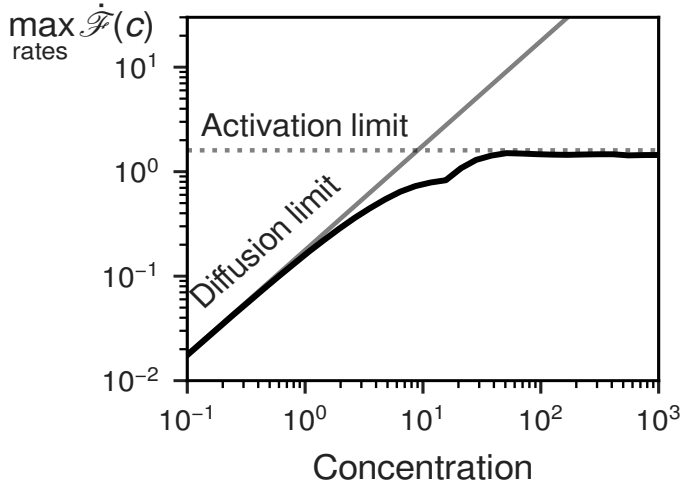
$\downarrow$ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints $\downarrow$

Cooperative binding



- DNA
- Promoter
- Gene
- Transcription factor
- RNA Polymerase
- mRNA

↓ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints ↓



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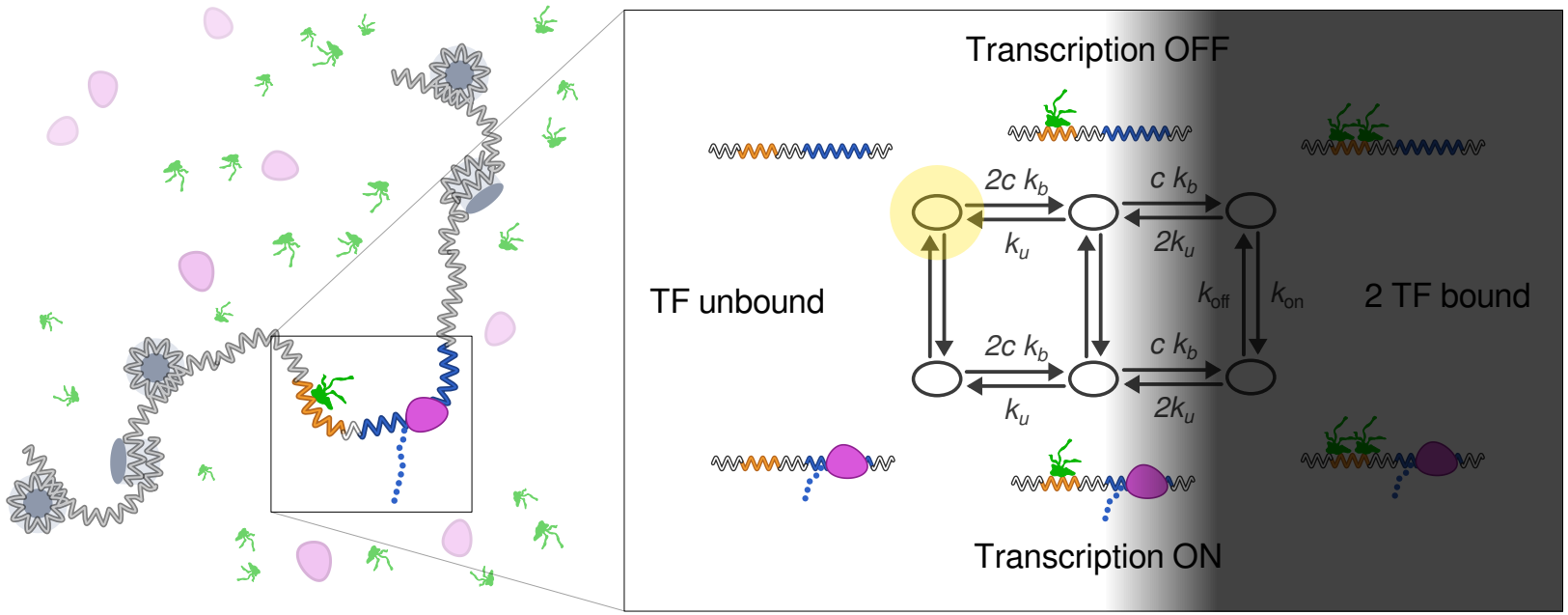

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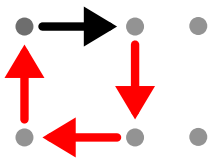
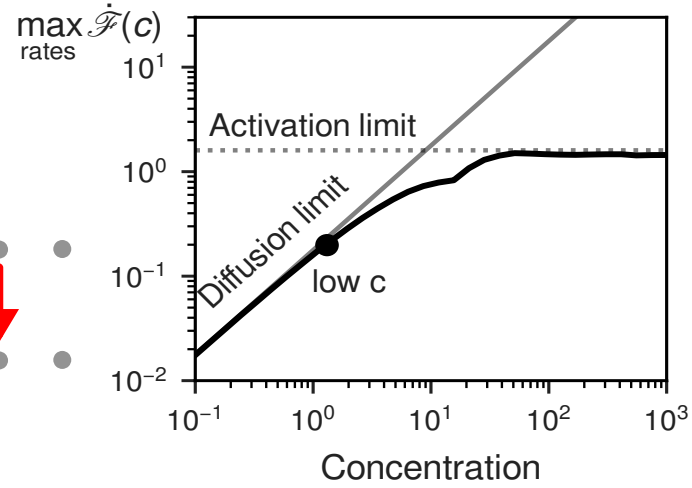
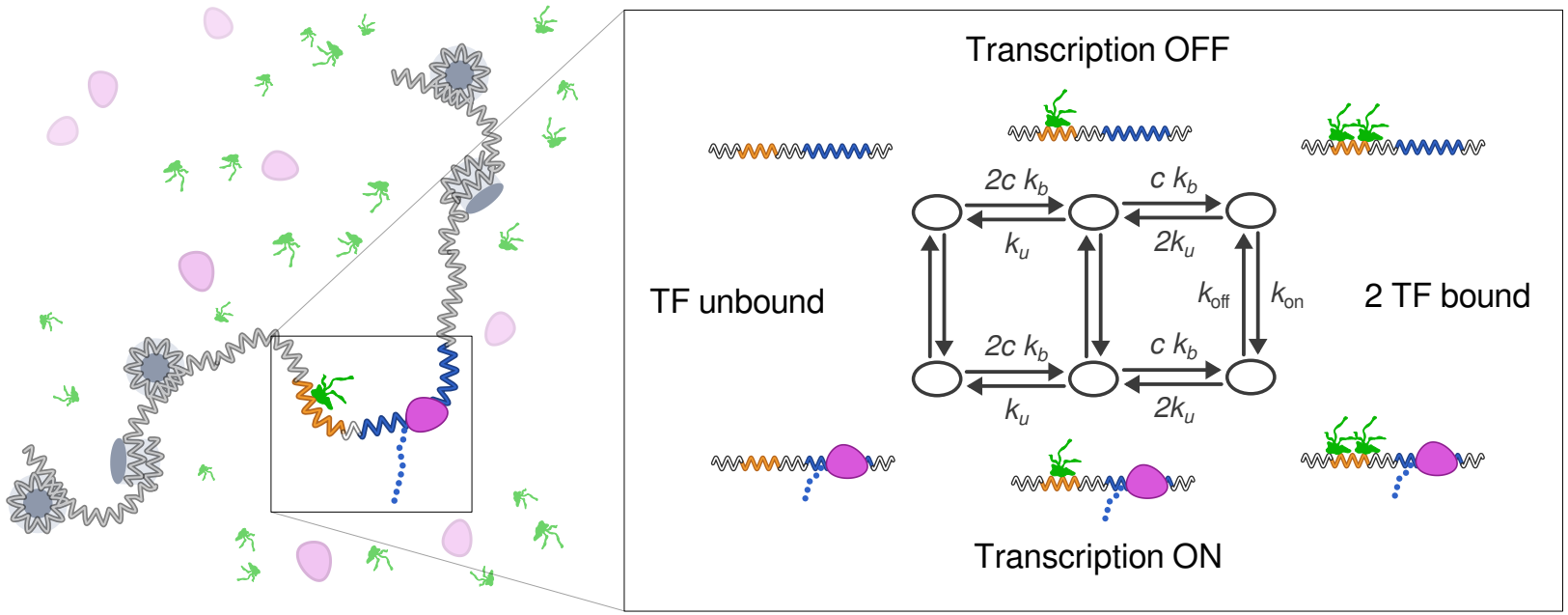


Diffusion limited
Activation-limited
No constraint saturated



Cooperative binding



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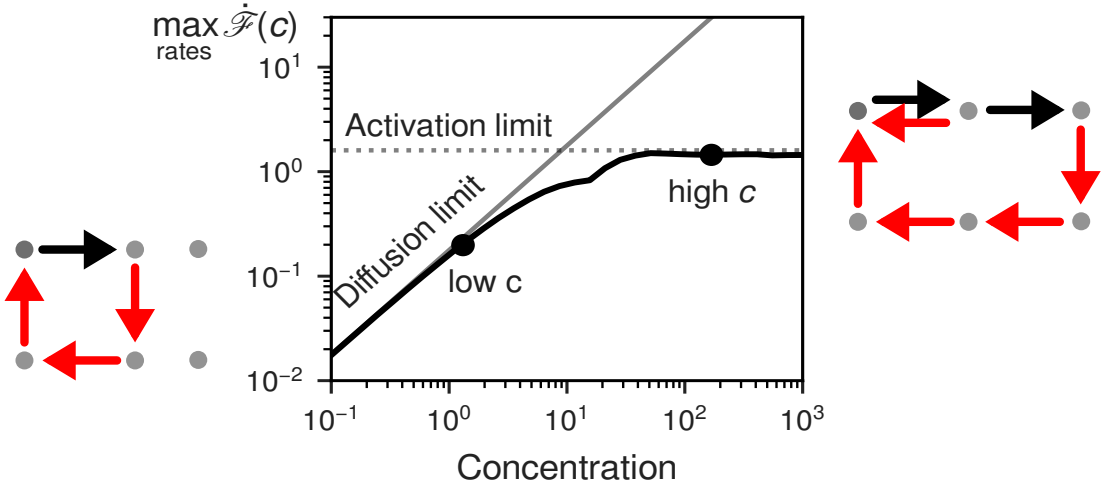
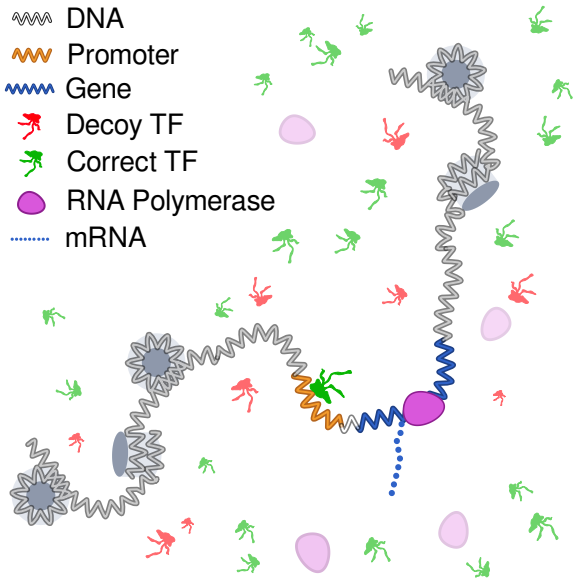


Proofreading



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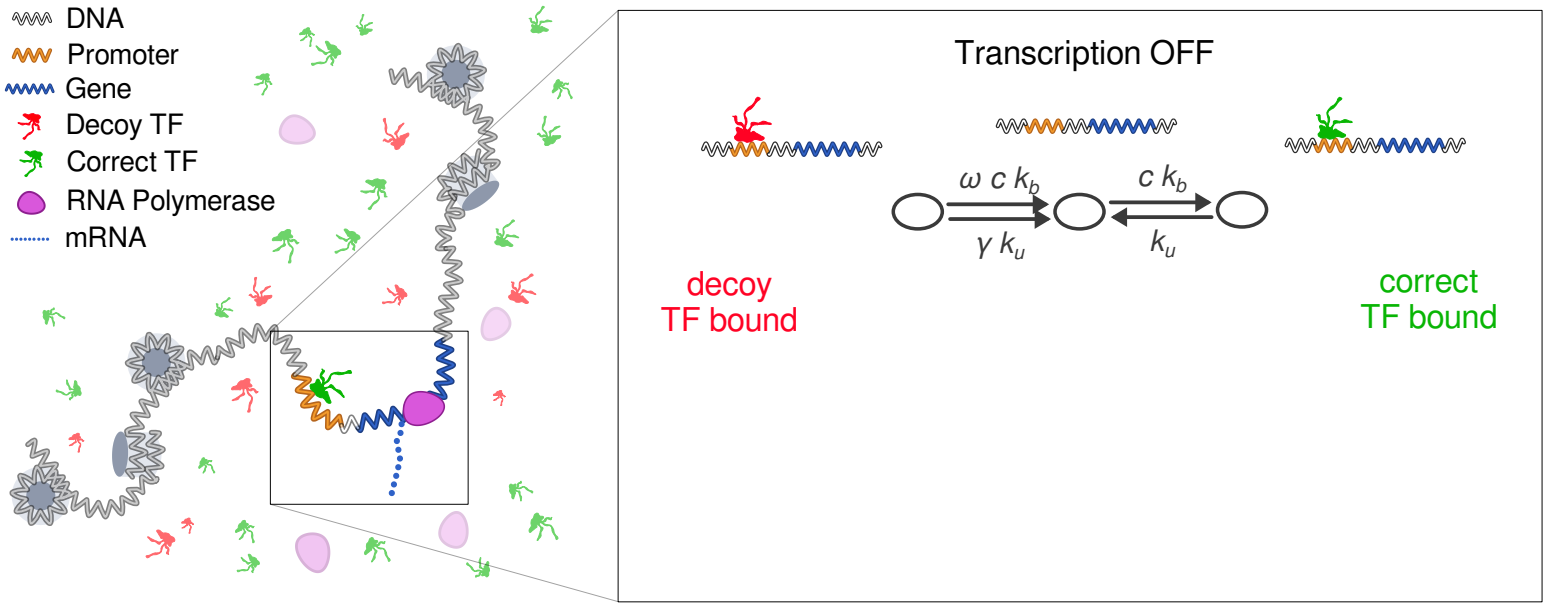

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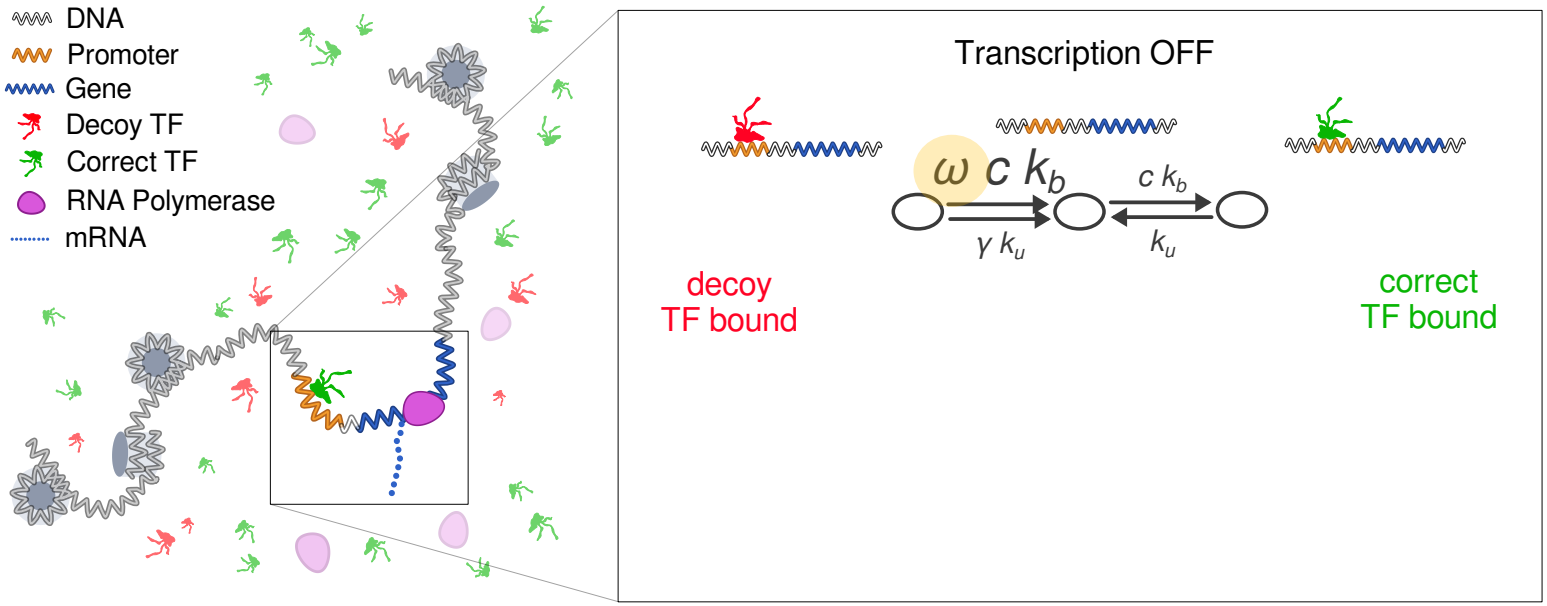
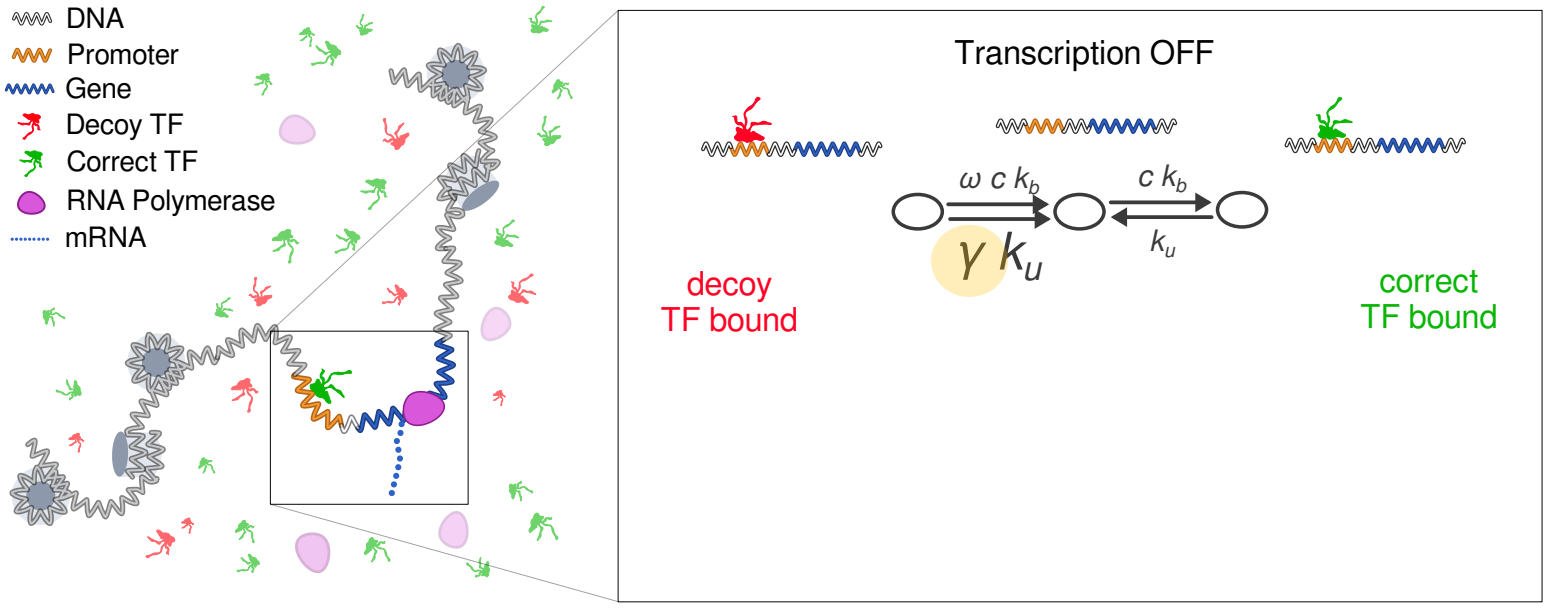


ω : [decoy TF] / [correct TF]

Proofreading



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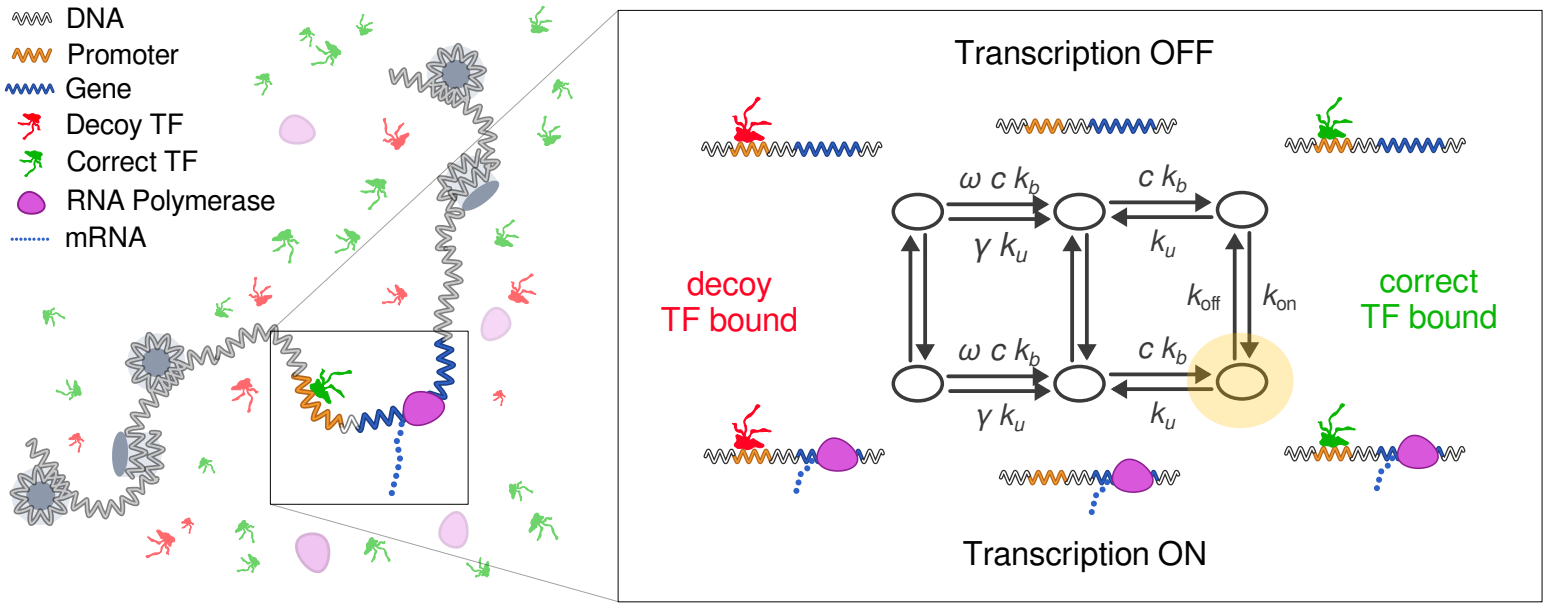
$$\gamma : \frac{\text{decoy unbinding rate}}{\text{correct unbinding rate}}$$

Proofreading



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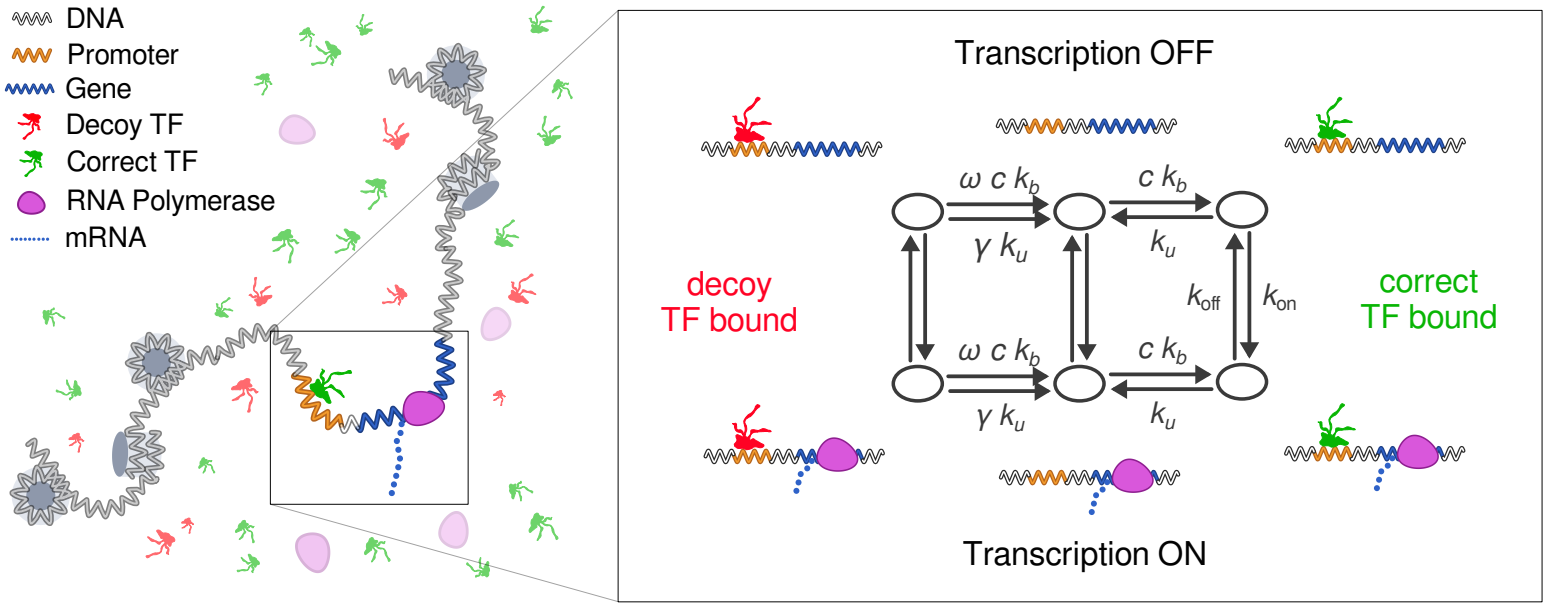
Selectivity f = P(transcribing | correct TF)

Proofreading



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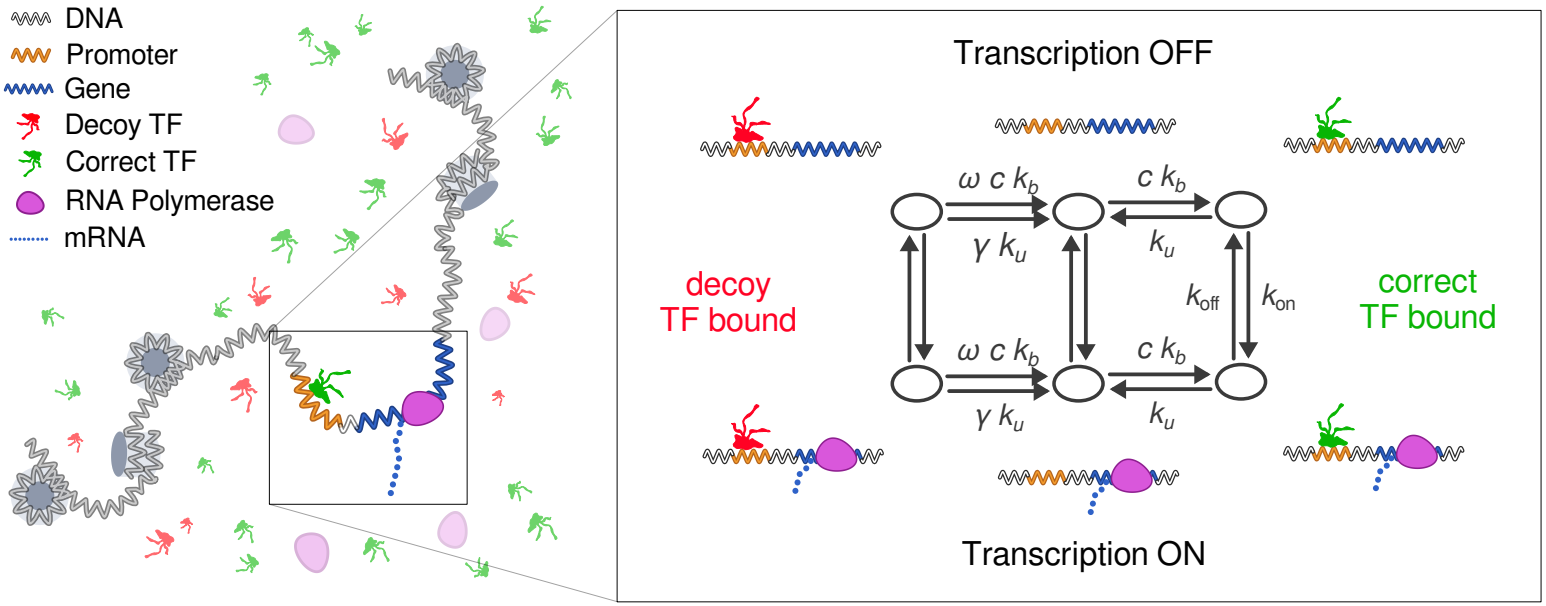
$$\max f = \gamma^2 / (\omega + \gamma^2)$$

$$\min f = \gamma / (\omega(\gamma+1)/2 + \gamma)$$

Proofreading



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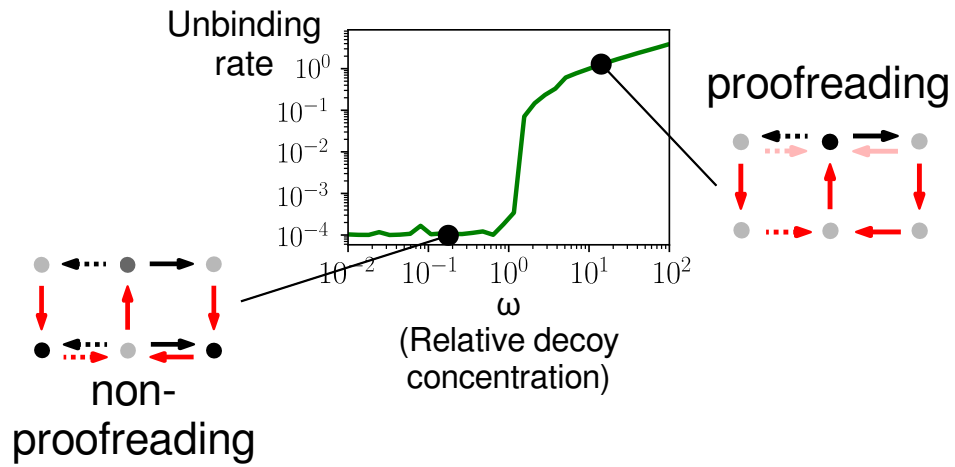
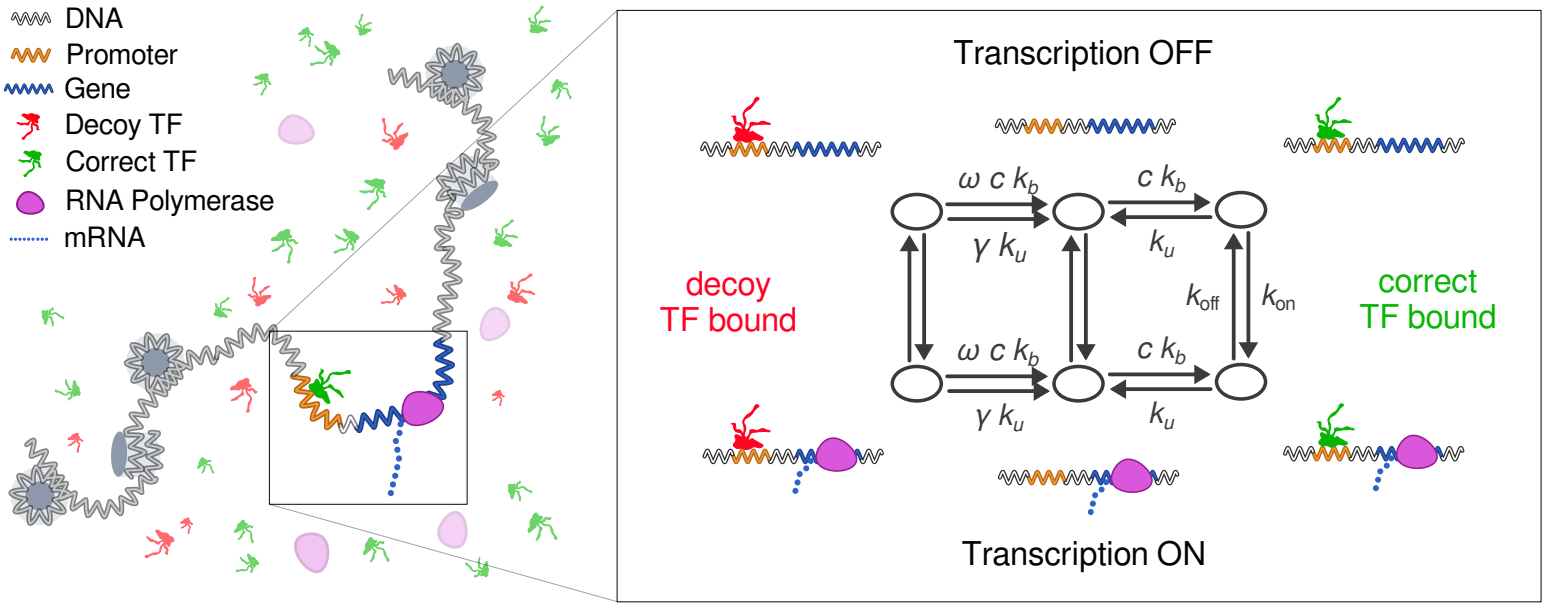


Proofreading



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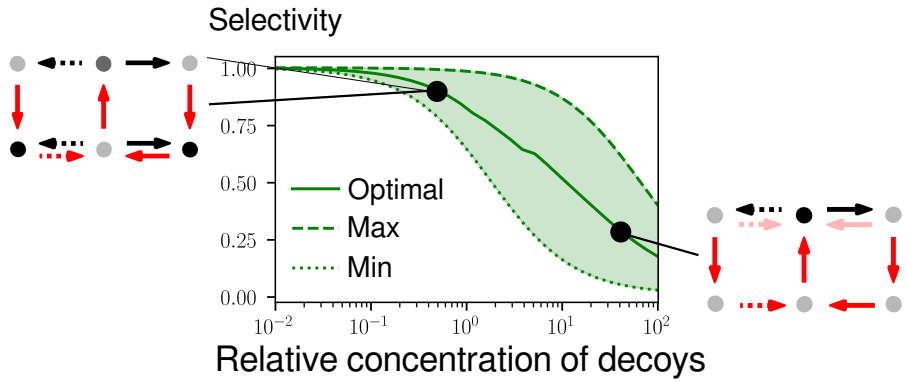
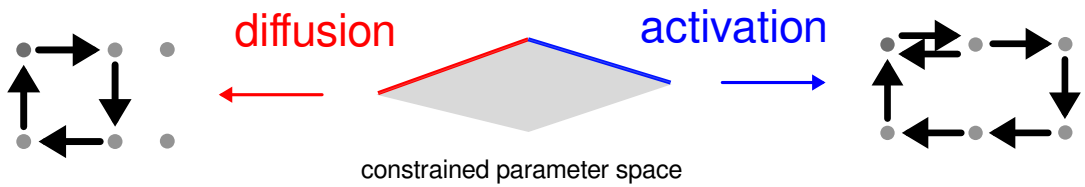


Takeaways

The limiting constraint shapes the optimal architecture



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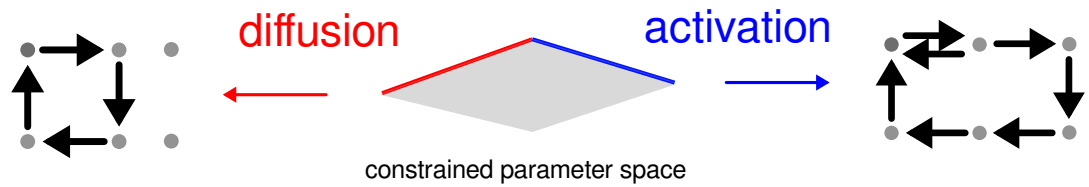
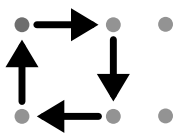
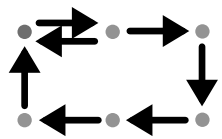


Solution set is disconnected: each cluster points to a different sensing strategy



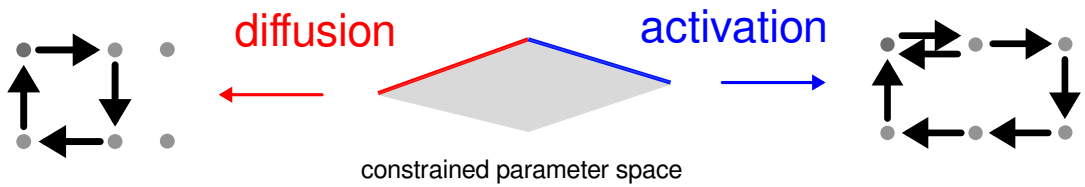
count rare
binding events



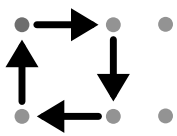
leverage
cooperative binding

Takeaways

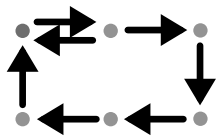
The limiting constraint shapes the optimal architecture



Solution set is disconnected: each cluster points to a different sensing strategy

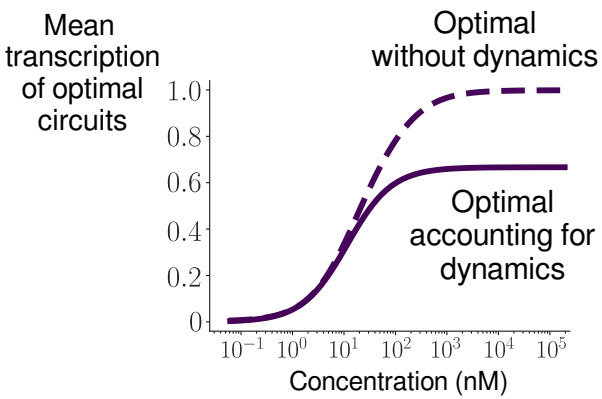


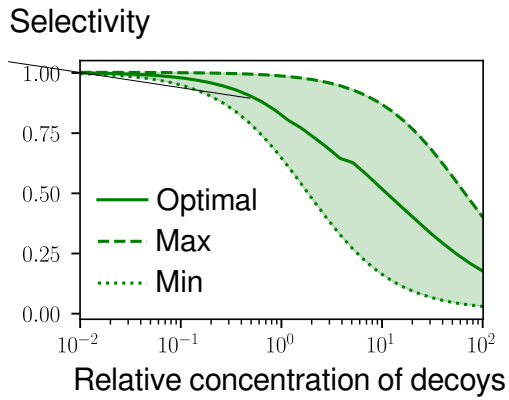
count rare
binding events



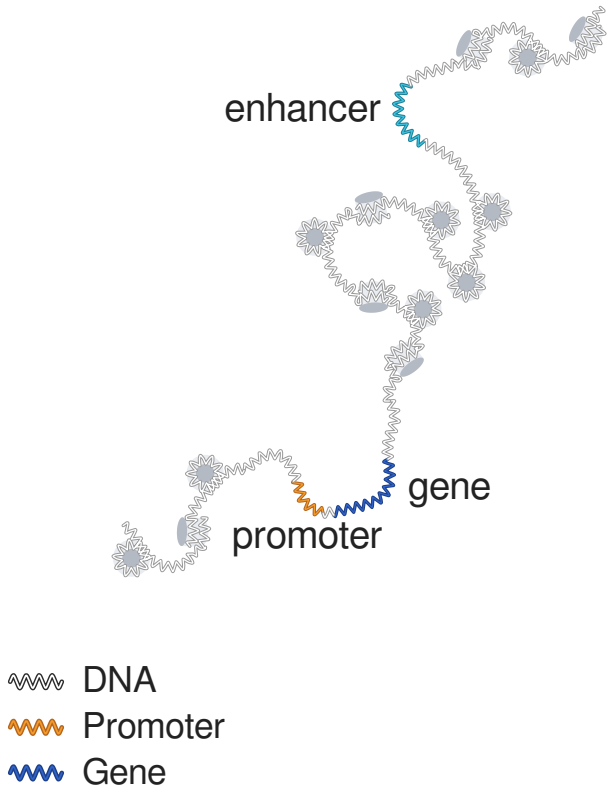
leverage
cooperative binding

Sharpness, selectivity do not lie at extrema

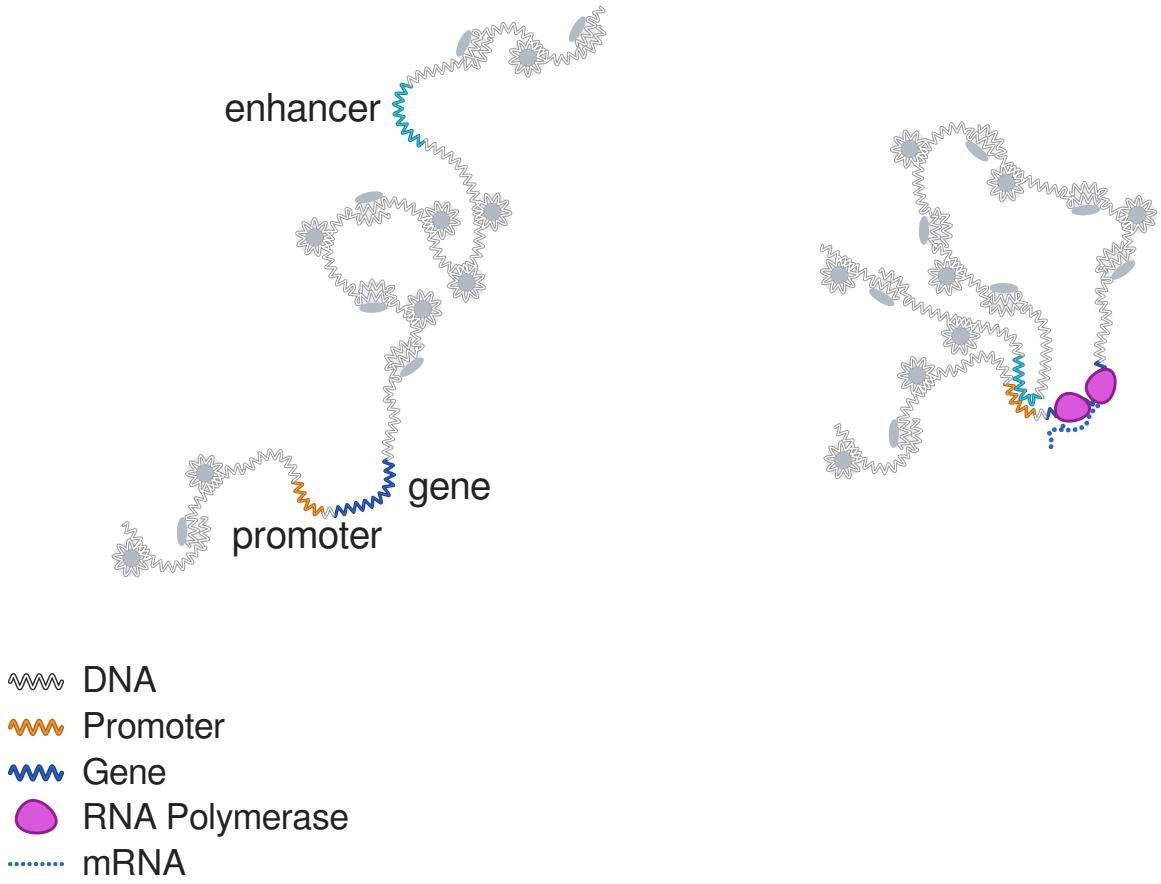




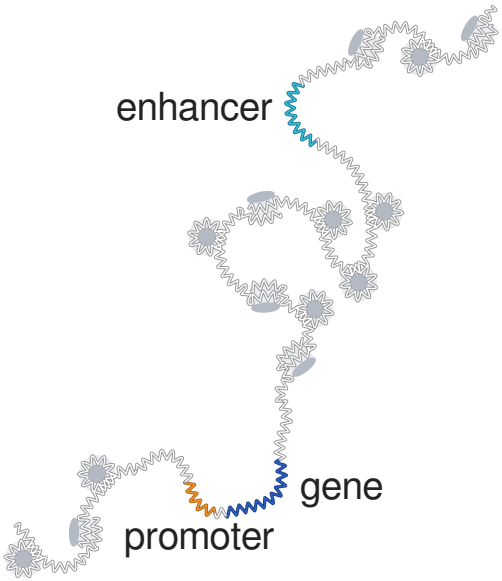
Transcriptional control by distal elements

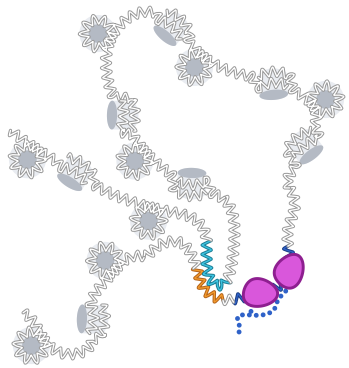


Transcriptional control by distal elements



Transcriptional control by distal elements

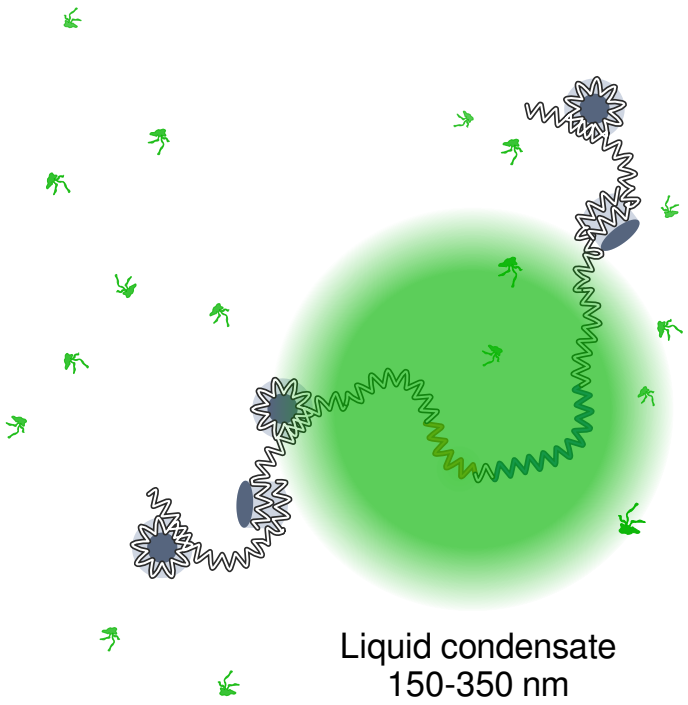




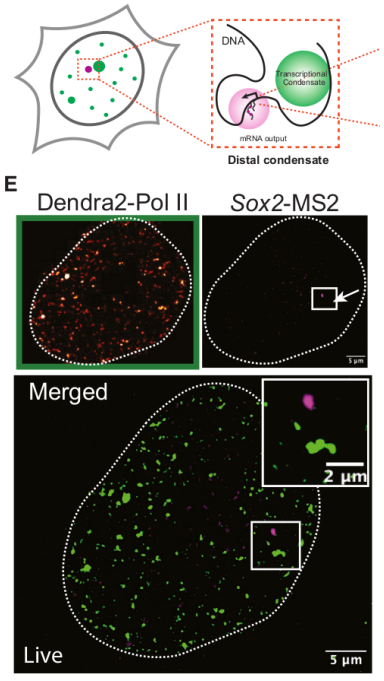
DNA
Promoter
Gene
RNA Polymerase
mRNA

Fukaya et al., *Cell,* 2016
Chen et al., *Nature Genetics,* 2018
Brückner et al., *Science,* 2023

Transcriptional control through liquid phase separation?



Liquid condensate
150-350 nm



Du et al.,*Cell* 2024

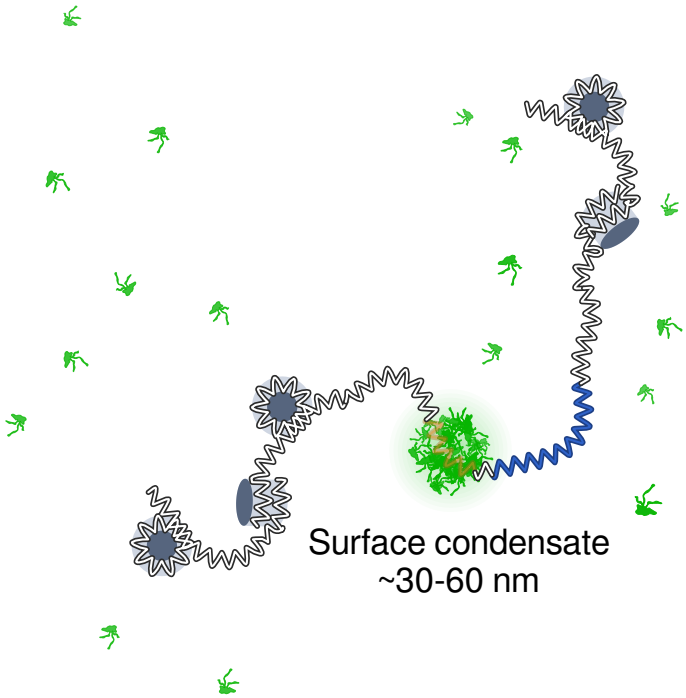
- DNA
- Promoter
- Gene
- Transcription factor

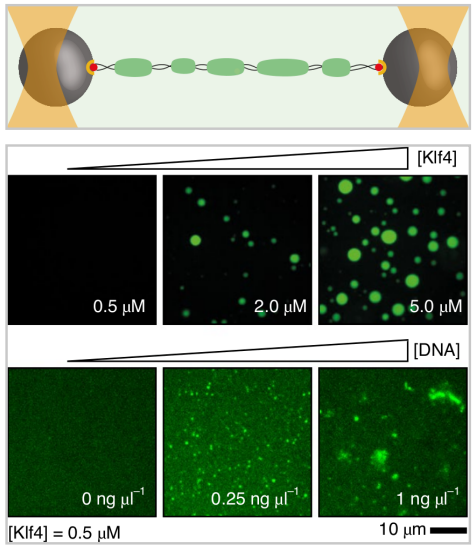
Hnisz et al., *Cell* 2017

Sabari et al., *Science* 2018

Munshi et al., *Science Advances* 2025

Proteins condense on the surface of DNA





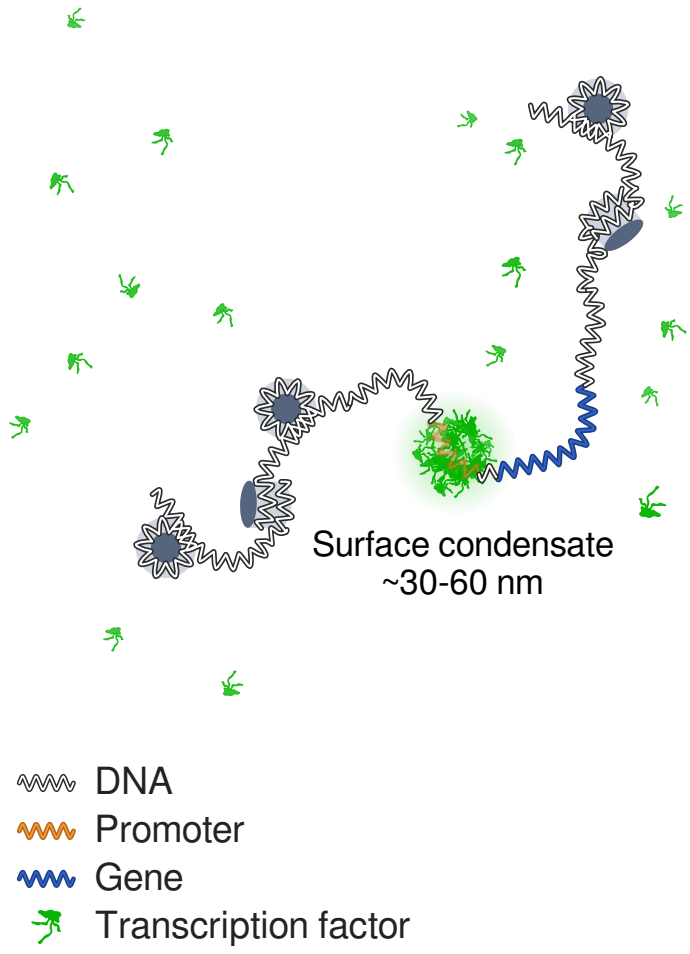
Morin et al., *Nature Physics* 2022

DNA
Promoter
Gene
Transcription factor

Pancholi et al., *MolecularSystems Biology* 2021

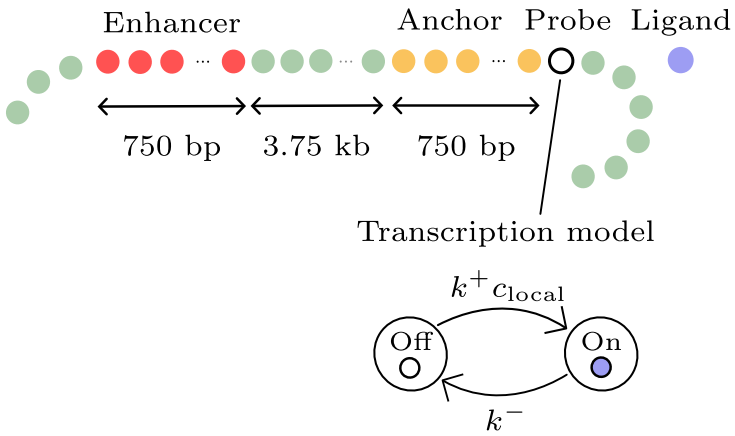
Rouches & Machta, *Biophysical Journal* 2025

Proteins condense on the surface of DNA

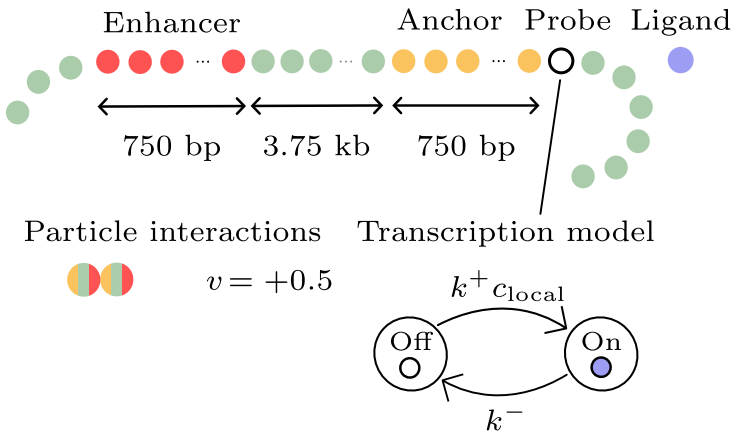


Influence on precision of gene expression?

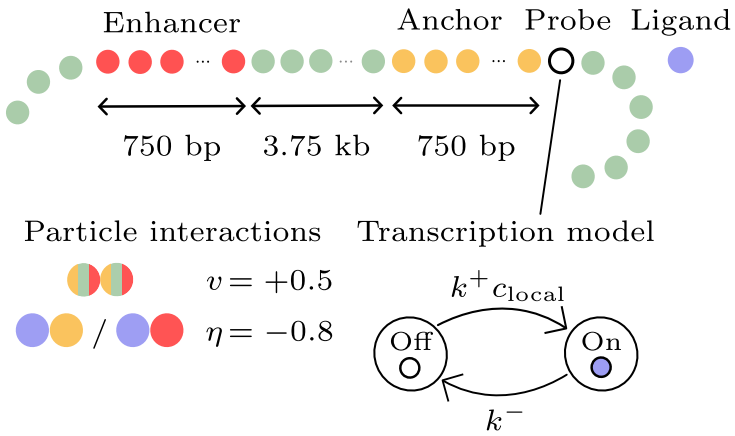
Coupling condensation and polymer dynamics



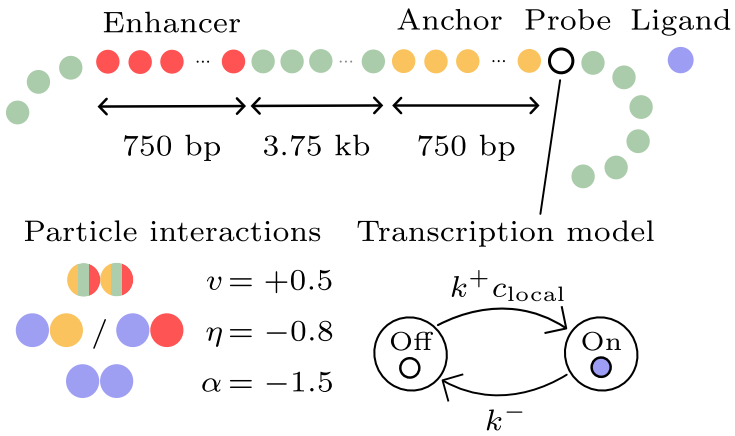
Coupling condensation and polymer dynamics



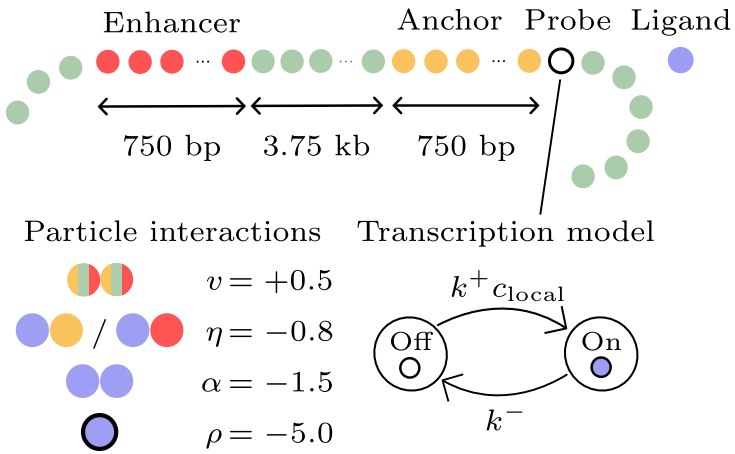
Coupling condensation and polymer dynamics



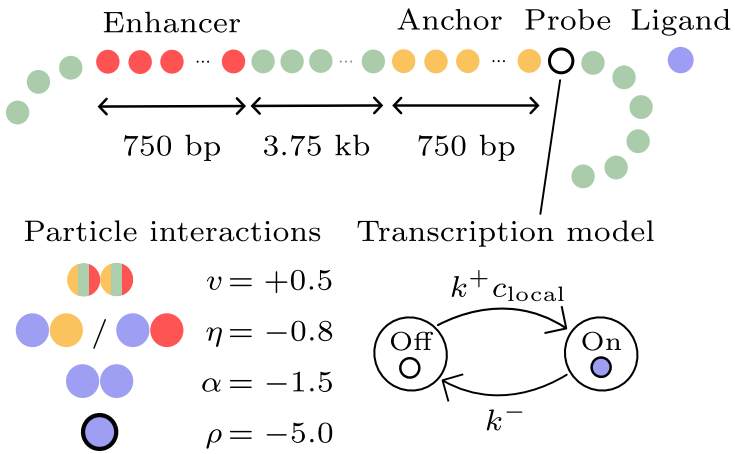
Coupling condensation and polymer dynamics

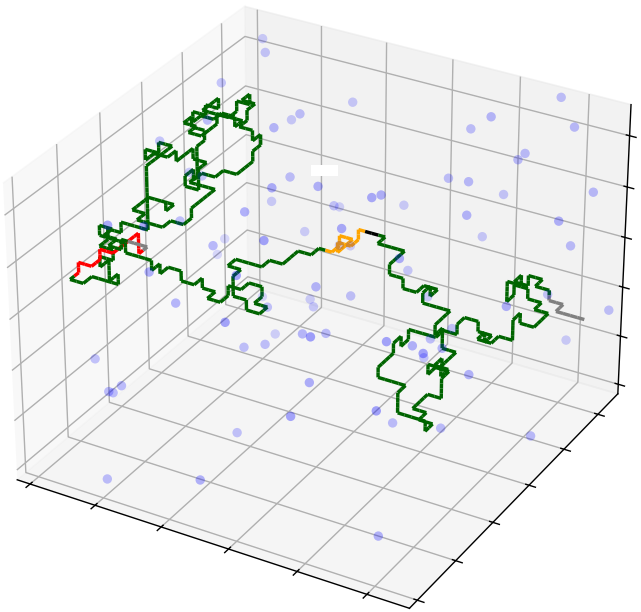


Coupling condensation and polymer dynamics

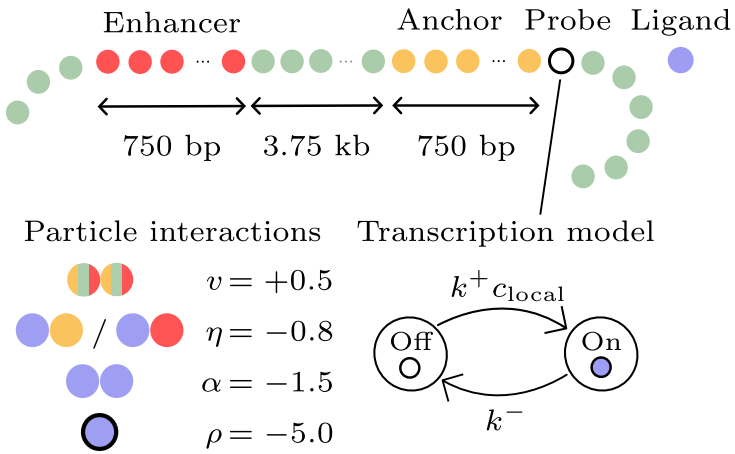


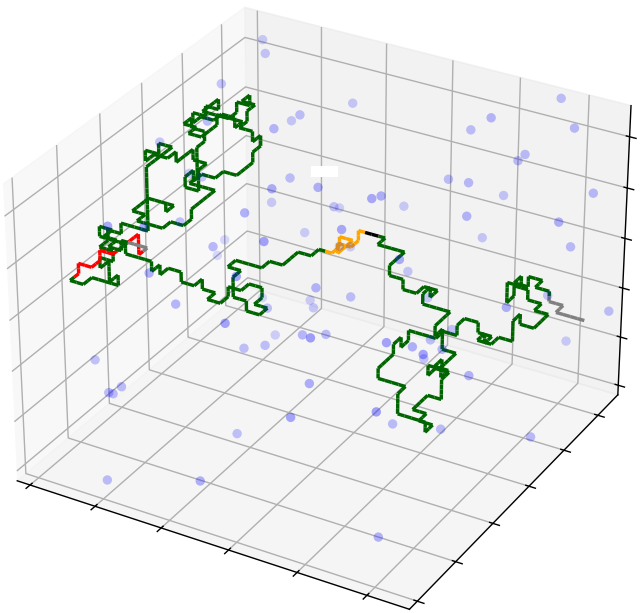
Coupling condensation and polymer dynamics

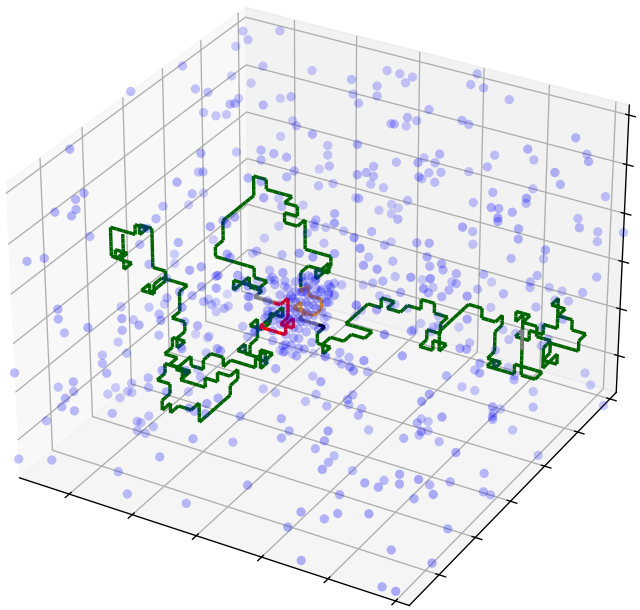




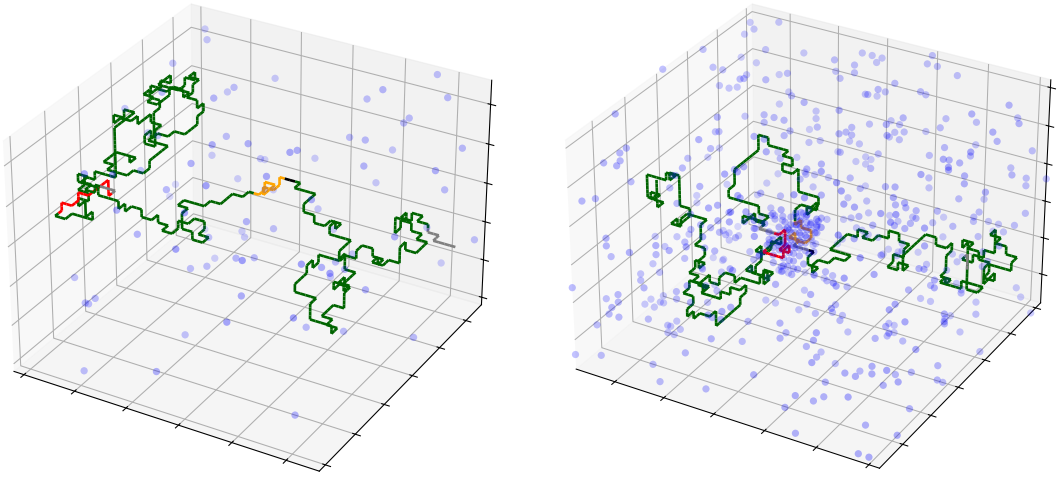
Coupling condensation and polymer dynamics

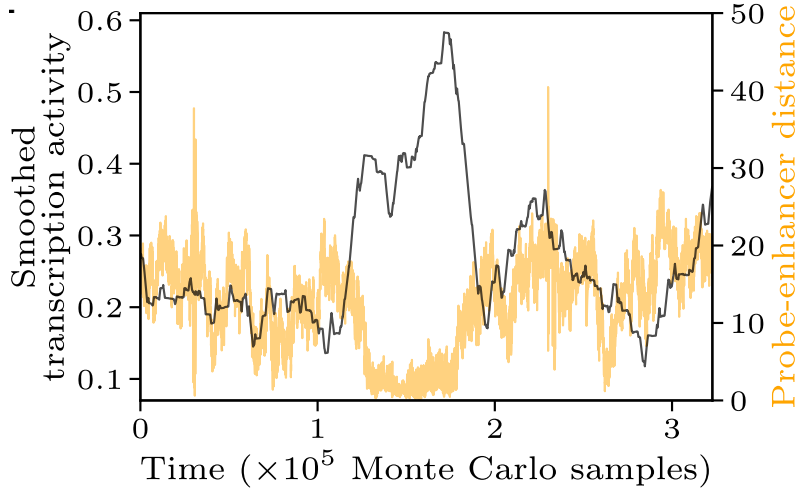




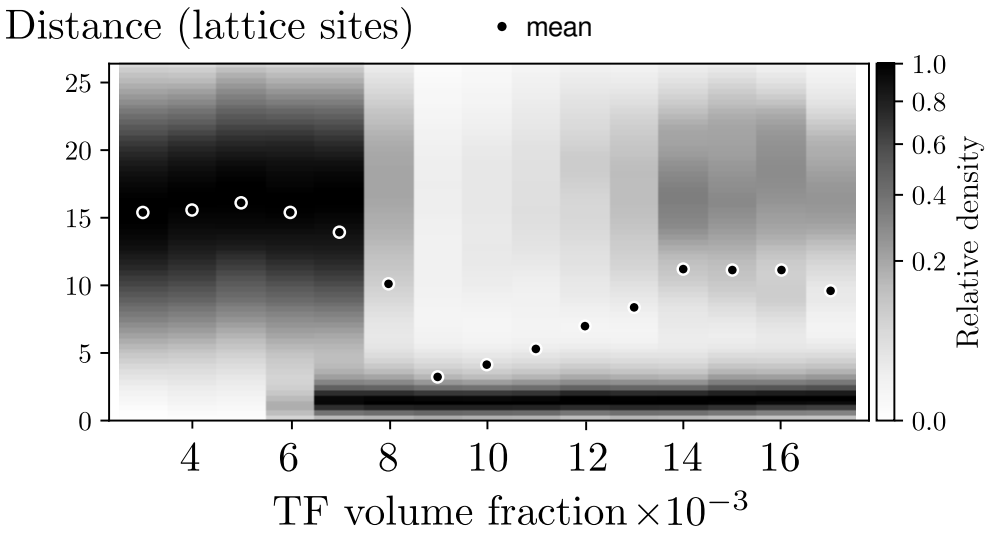


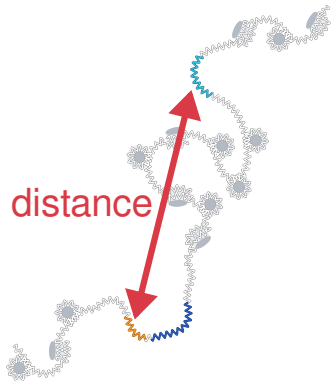
Coupling condensation and polymer dynamics



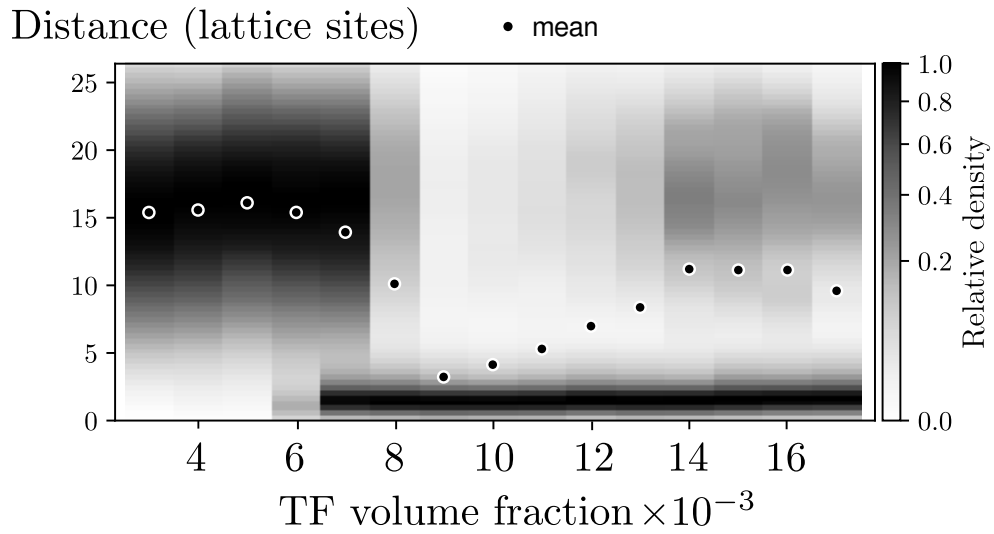


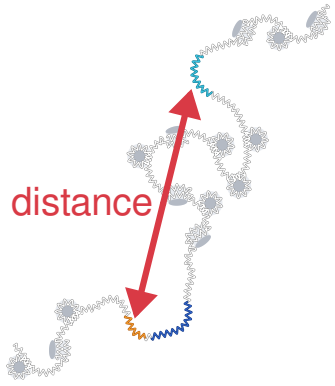
Polymer configuration depends sensitively on concentration

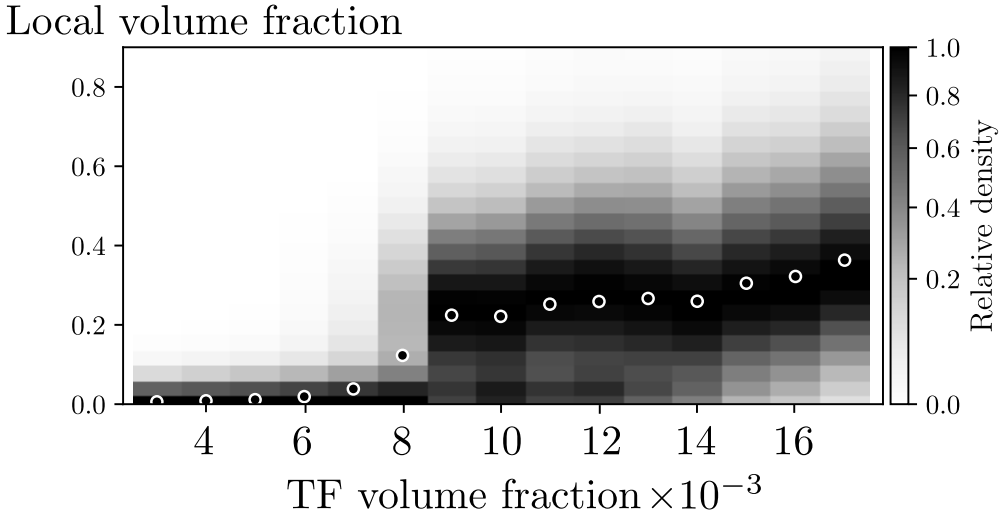


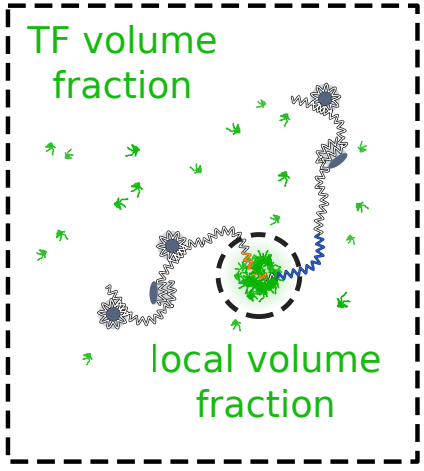


Polymer configuration depends sensitively on concentration

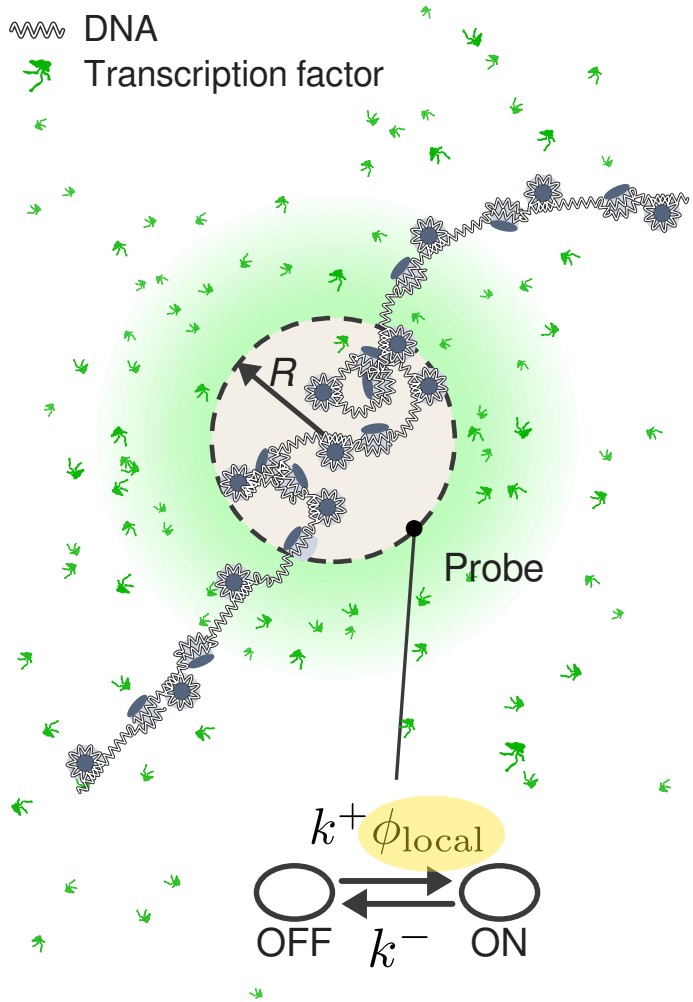








Continuous model of surface condensation



Flory-Huggins potential

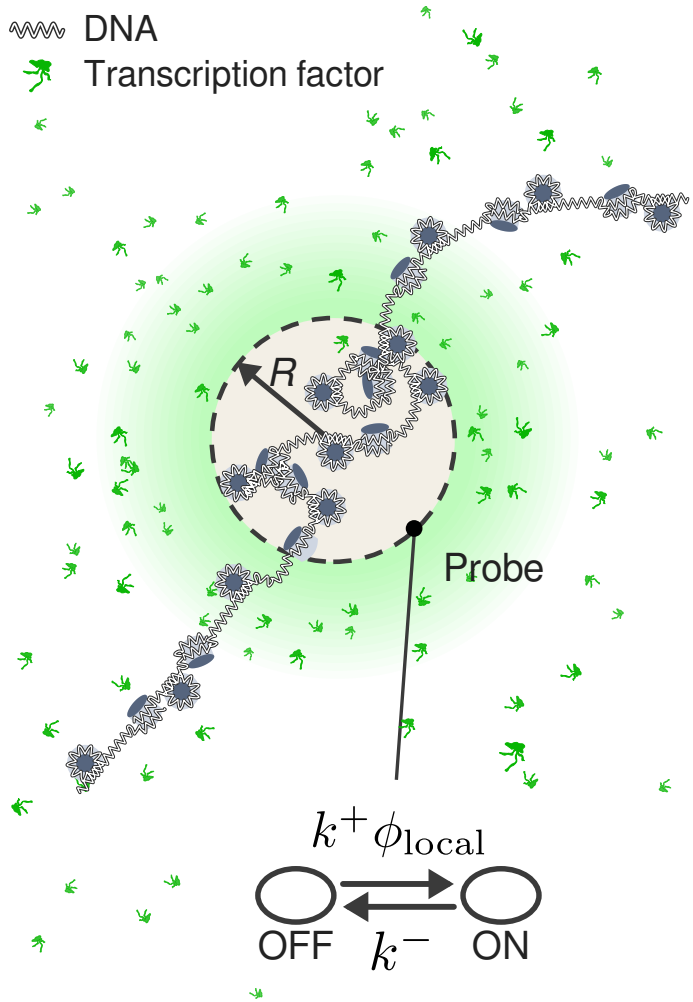
$$f(\phi) = \frac{\phi}{\nu}\ln\phi + (1-\phi)\ln(1-\phi) - \chi\phi^2 - \mu\phi$$

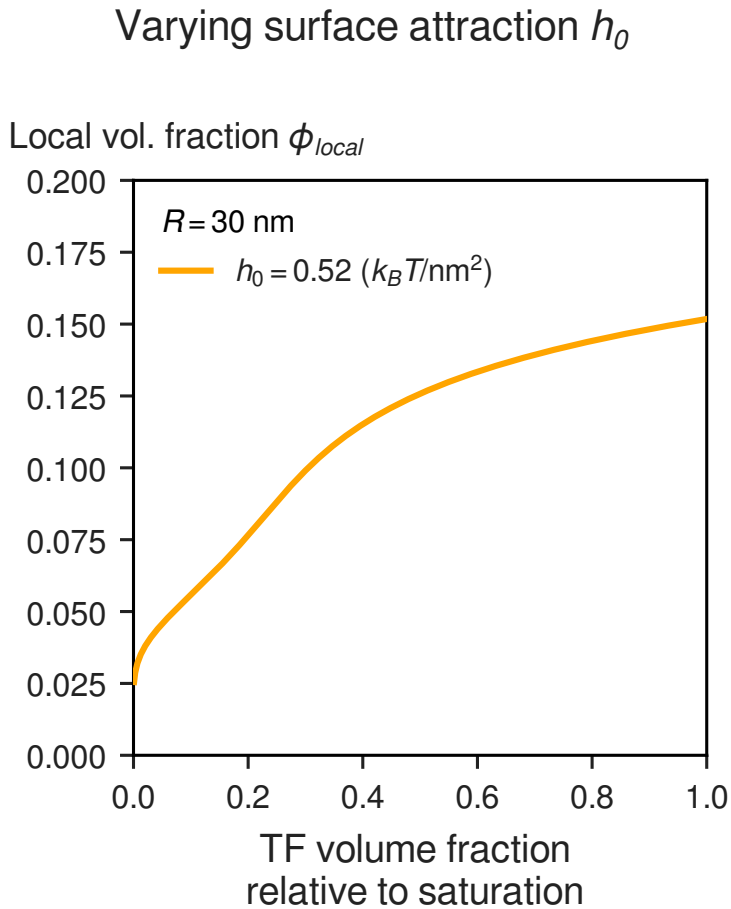
Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_S h_0\phi(\mathbf{x})dS$$

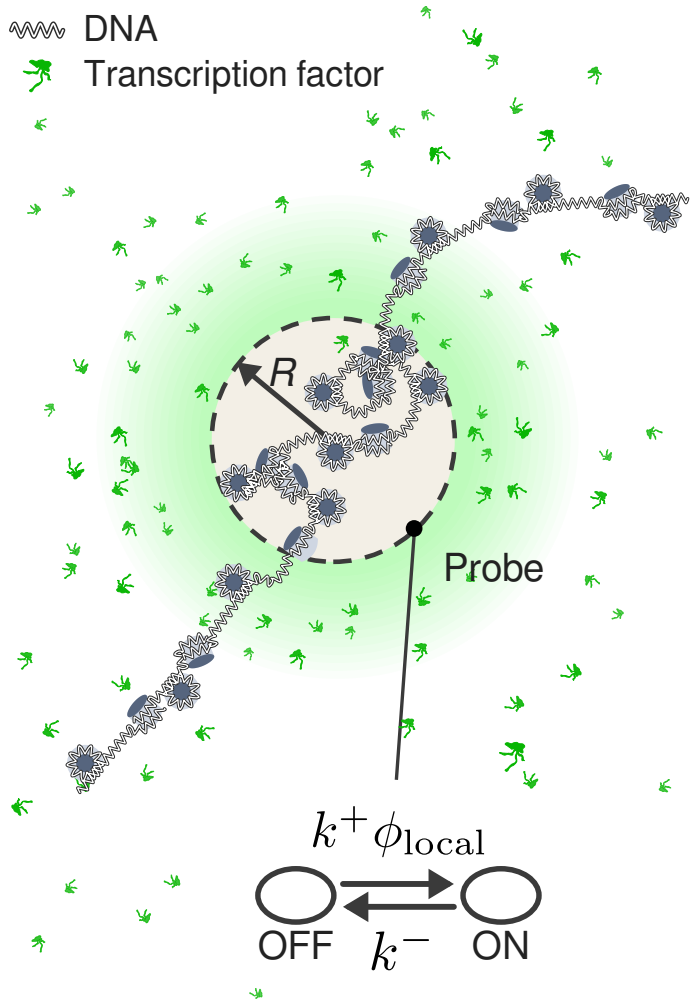
ϕ TF volume fraction
ν TF / solvent volume ratio
χ TF-TF attraction
μ chemical potential
h_0 surface attraction
κ Interface stiffness

The prewetting transition: varying surface attraction

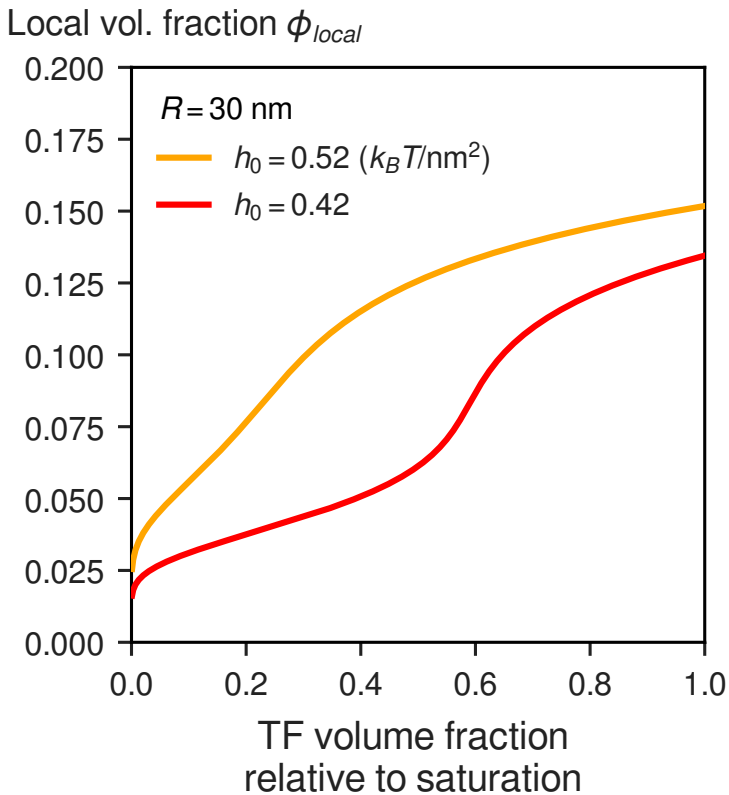




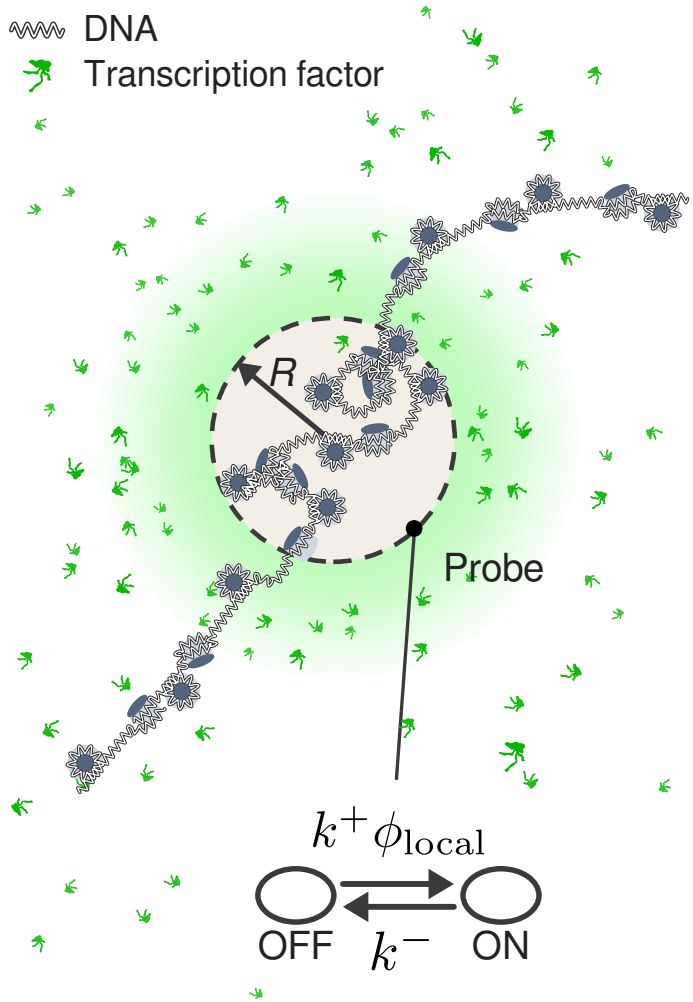
The prewetting transition: varying surface attraction



Varying surface attraction h_0



The prewetting transition: varying surface attraction



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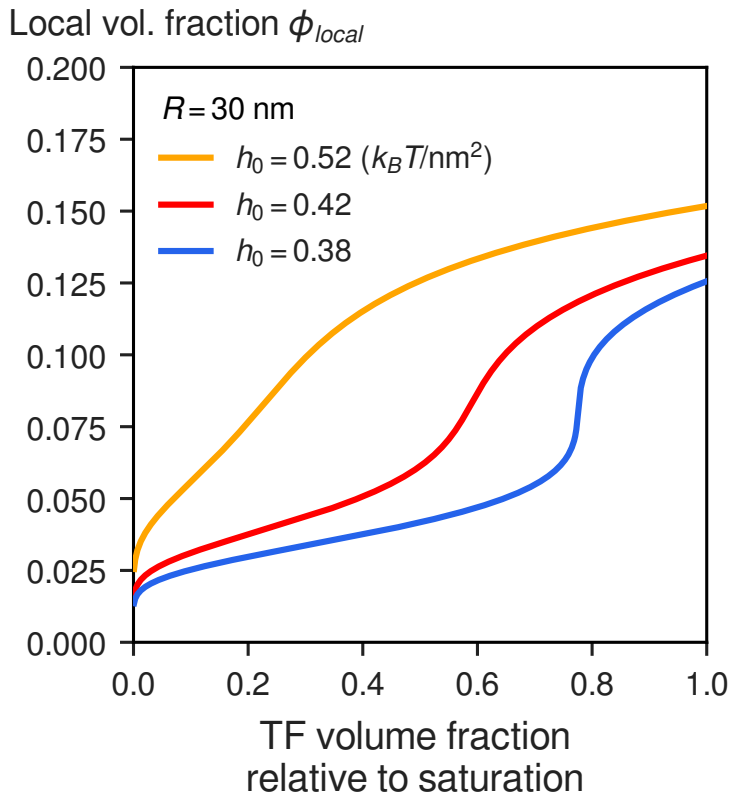
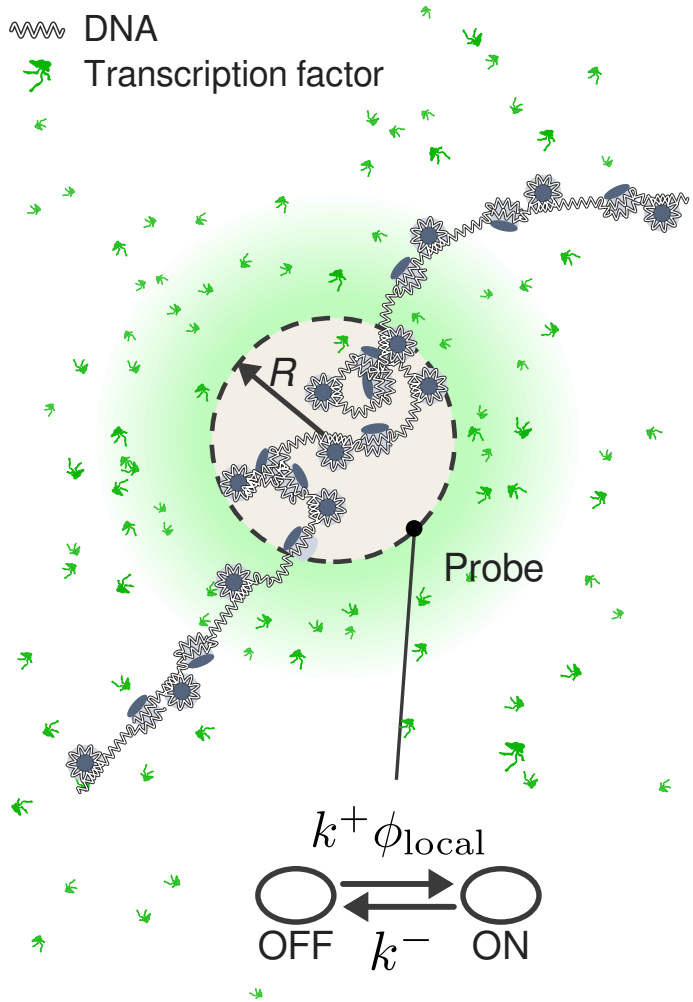
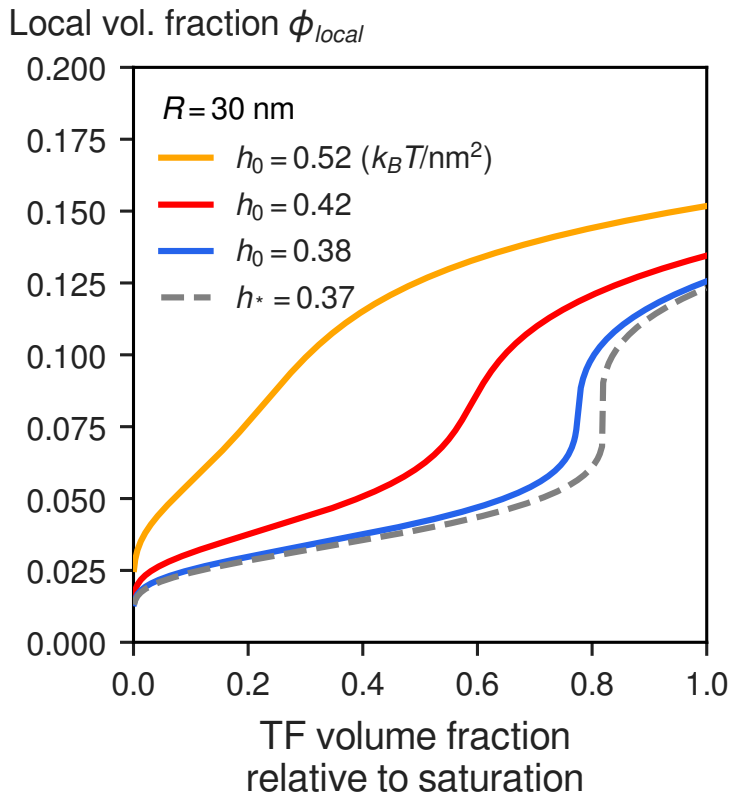


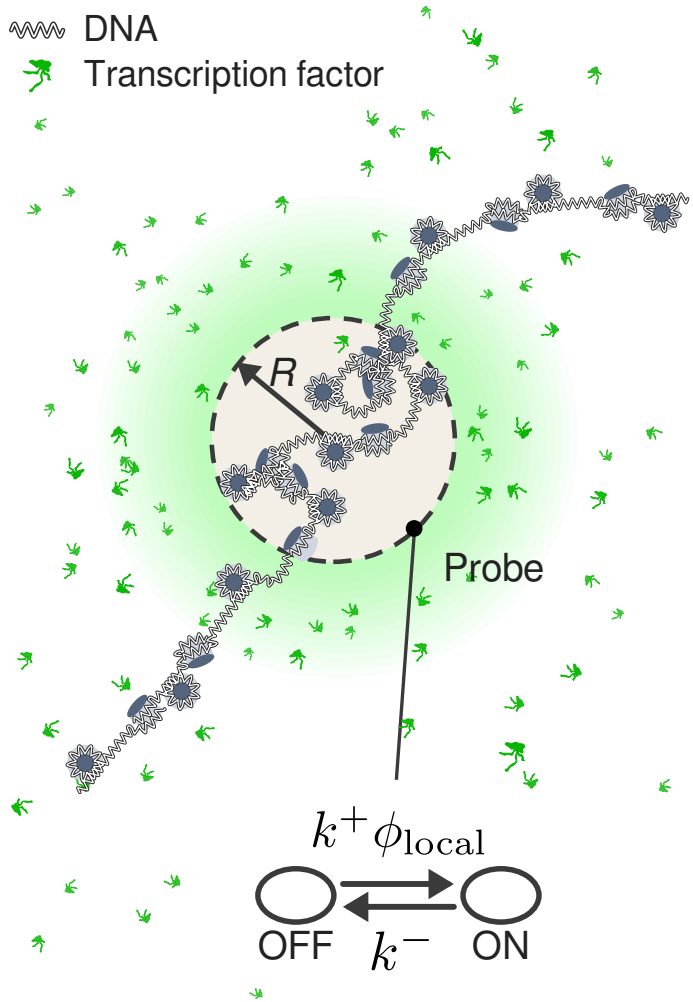
The prewetting transition: varying surface attraction



Varying surface attraction h_0



The prewetting transition: varying surface attraction



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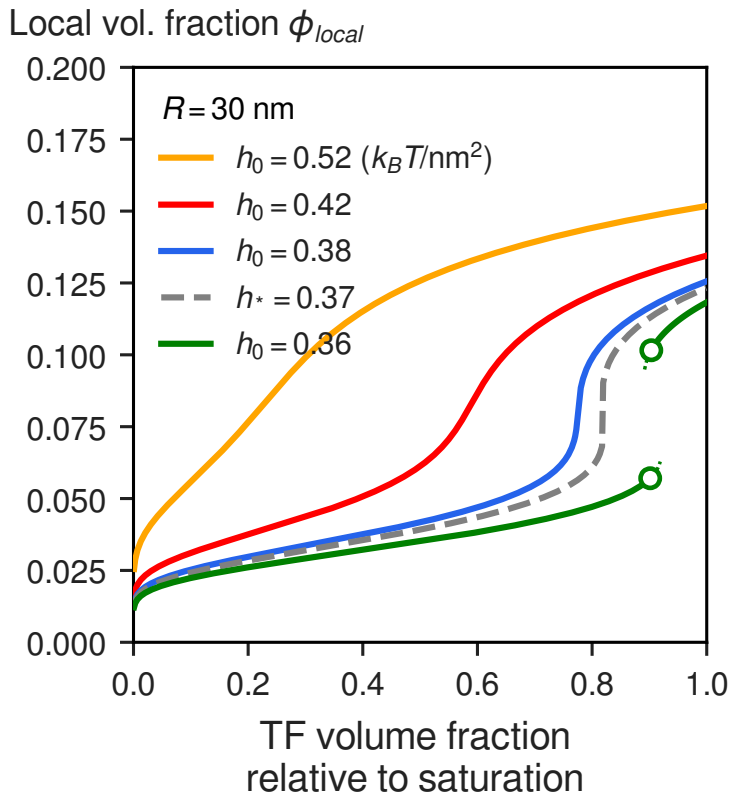
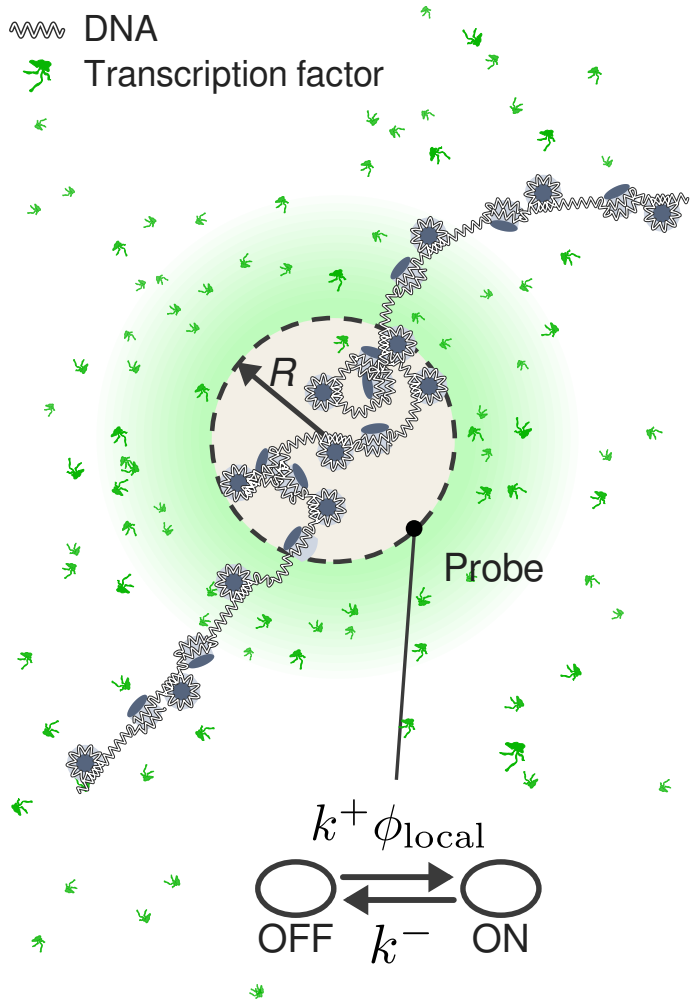


Quantifying sensitivity of local concentration



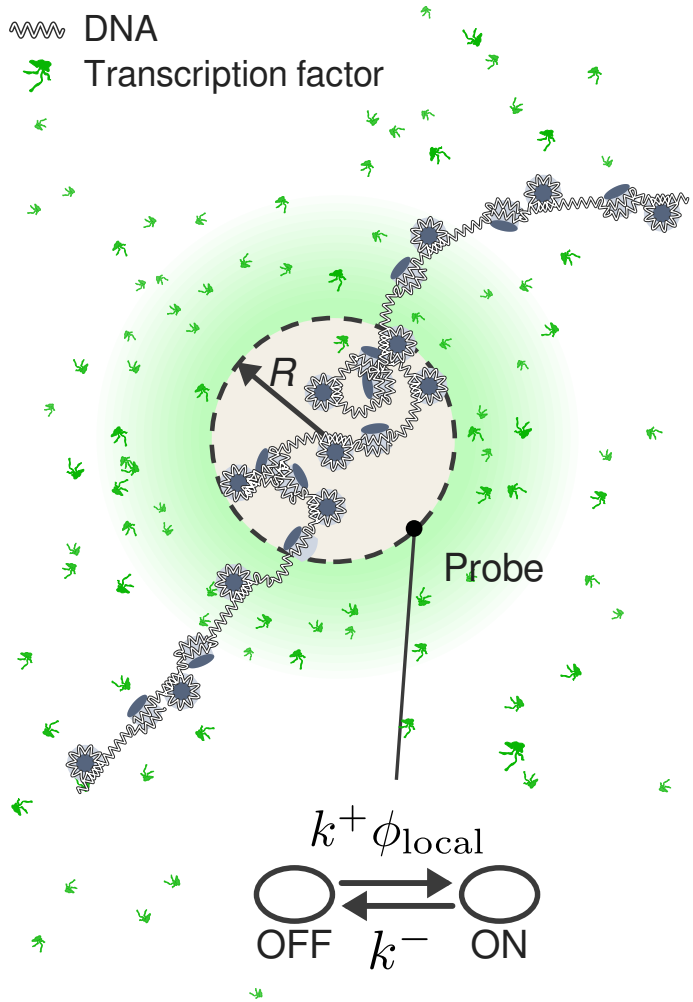
Free receptor

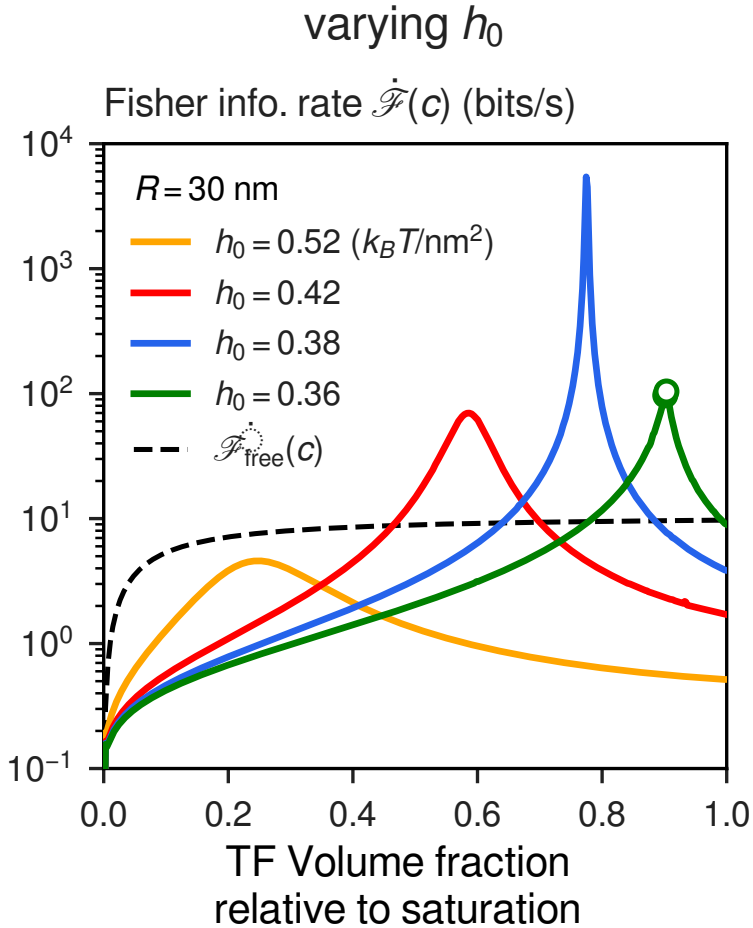
$$\dot{\mathcal{F}}_{\text{free}}(\phi) = \frac{1}{\tau_{\text{free}}(\phi)}$$

Receptor at the attractive surface

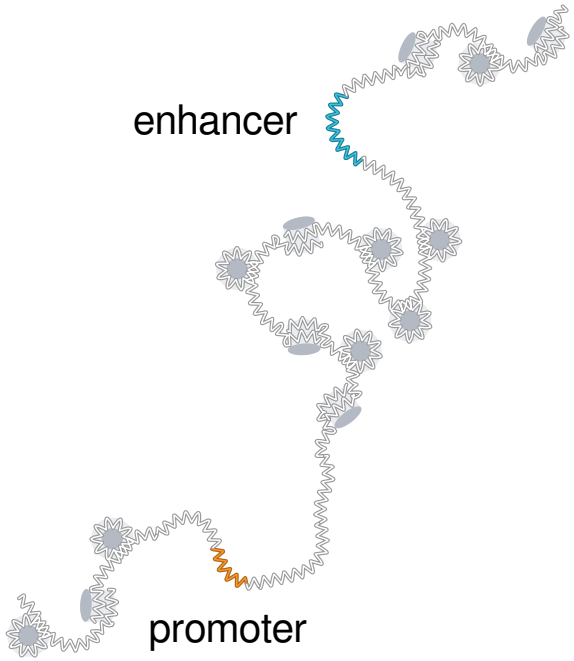
$$\dot{\mathcal{F}}(\phi) = \underbrace{\frac{1}{\tau_{\text{free}}(\phi_{\text{local}})}}_{\text{Probe}} \underbrace{\left(\frac{d\log\phi_{\text{local}}}{d\log\phi}\right)^2}_{\text{Surface}}$$

Quantifying sensitivity of local concentration

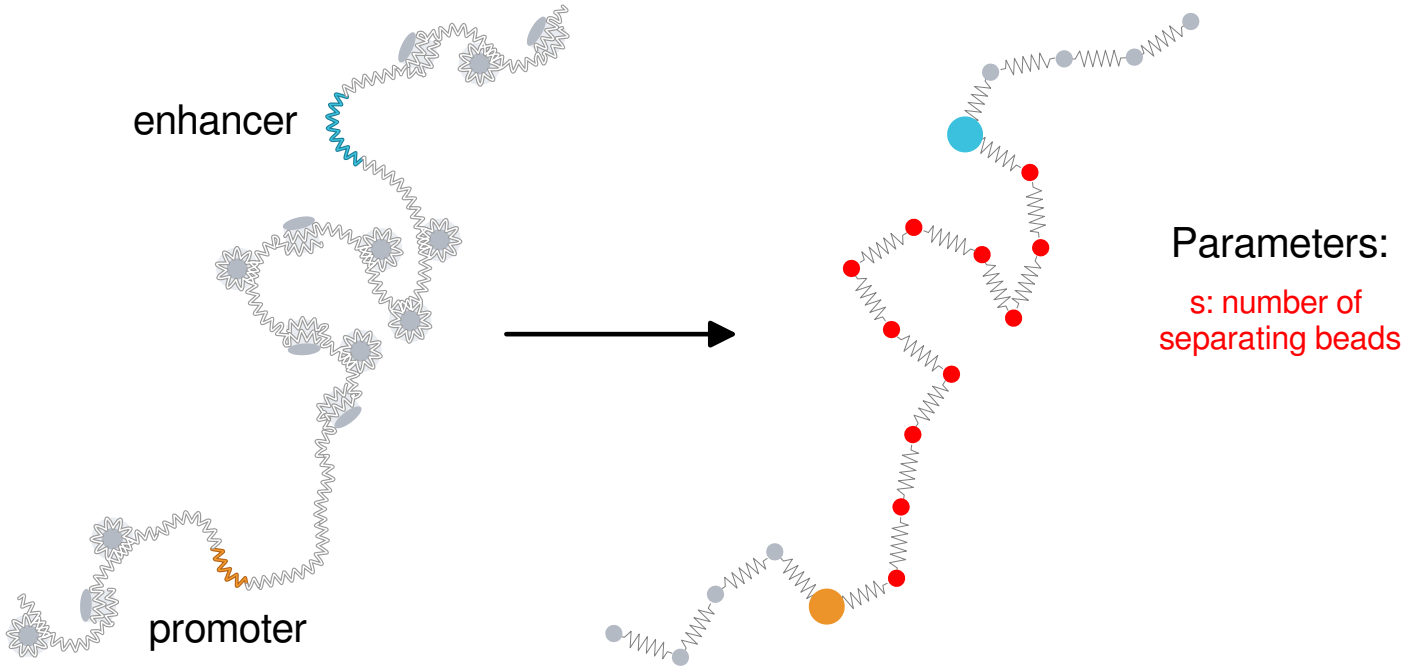




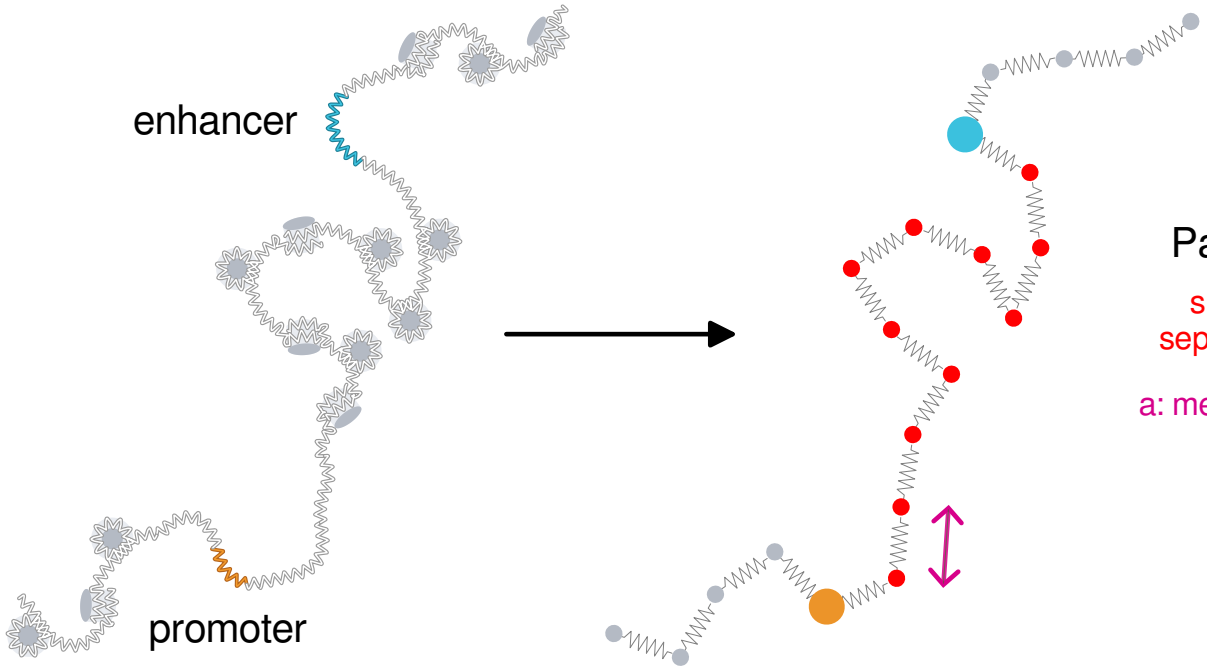
Rouse model of DNA



Rouse model of DNA



Rouse model of DNA

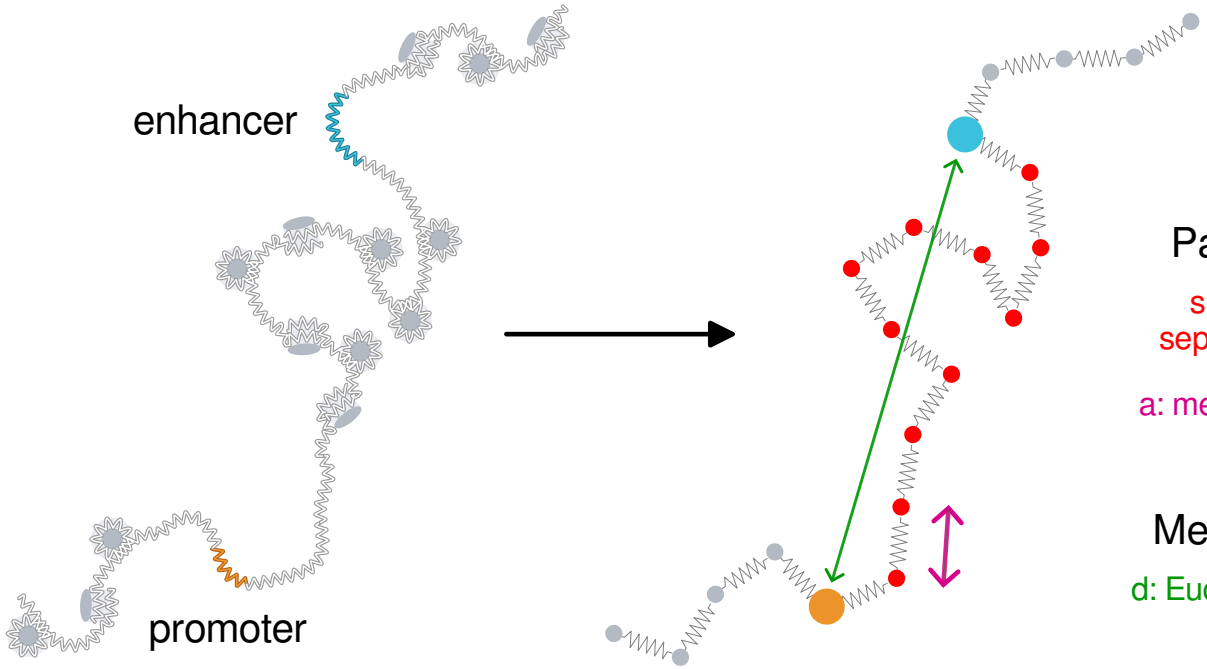


Parameters:

s: number of separating beads

a: mean bond length

Rouse model of DNA



Parameters:

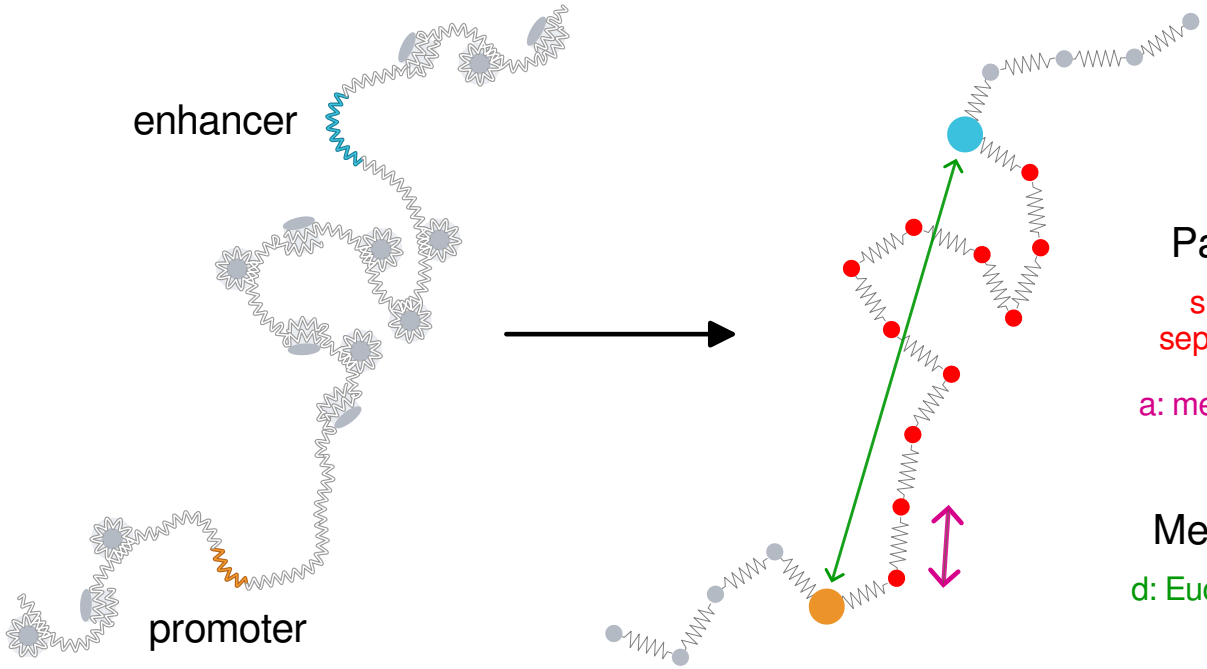
s: number of separating beads

a: mean bond length

Measurement:

d: Euclidean distance

Rouse model of DNA



Parameters:

s: number of separating beads

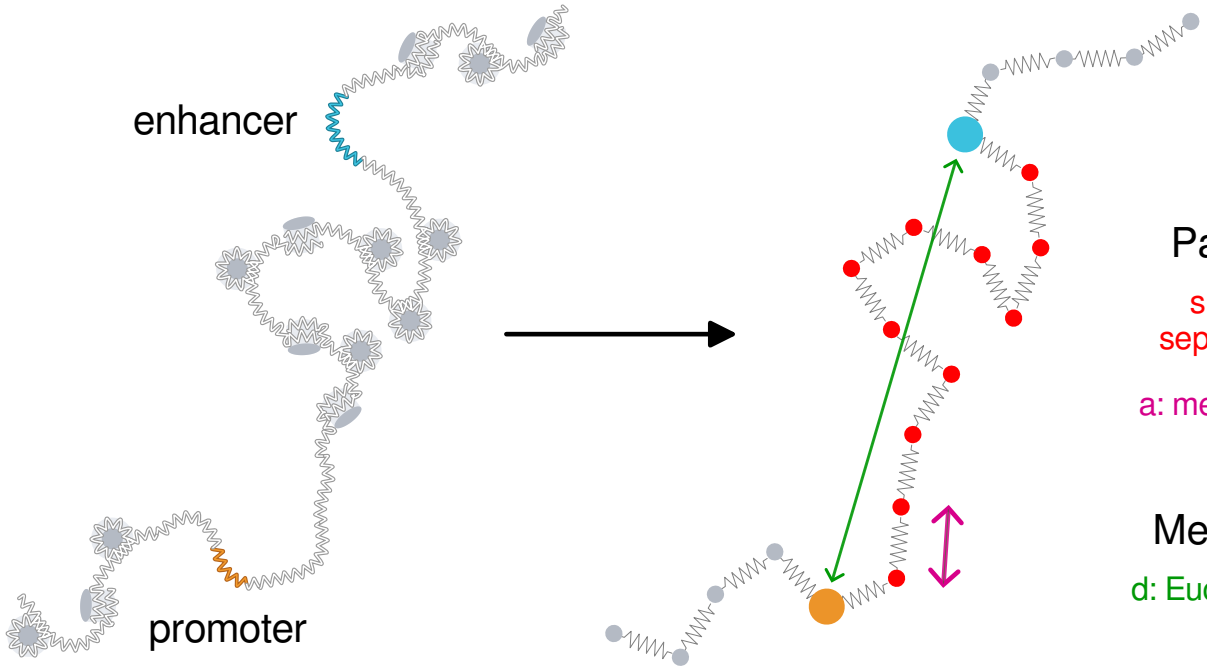
a: mean bond length

Measurement:

d: Euclidean distance

$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T}$$

Rouse model of DNA



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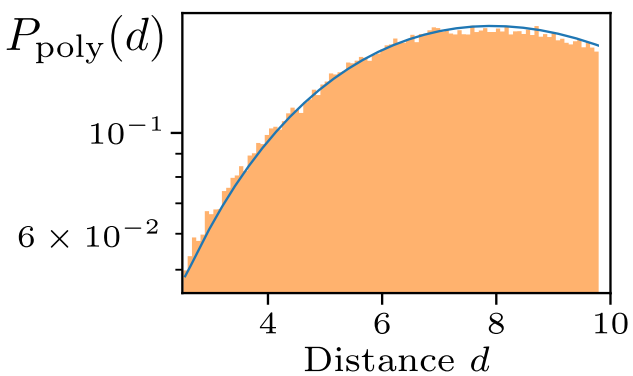
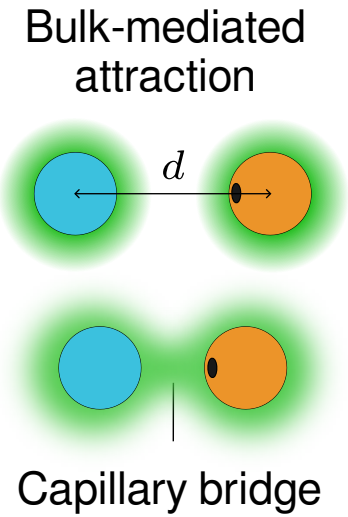
Measurement:

d: Euclidean distance

$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T}$$



Promoter-enhancer attraction



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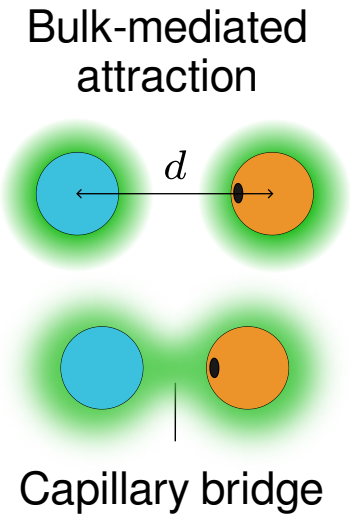

Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_{S_1} h_0\phi(\mathbf{x})dS - \int_{S_2} h_0\phi(\mathbf{x})dS$$

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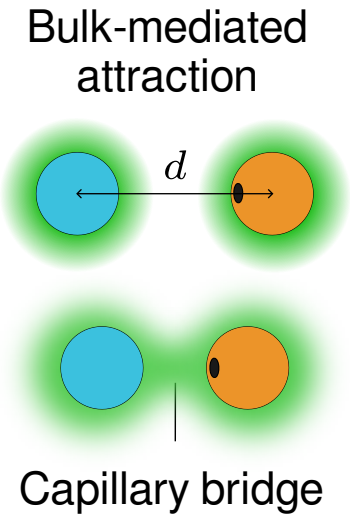

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$$U_{\text{attr}}(d) = \min_{\phi(\mathbf{x})}\{F\}$$

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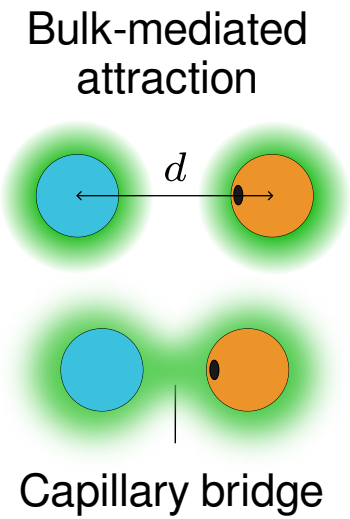

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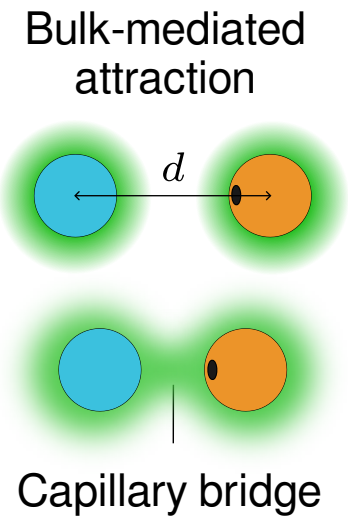

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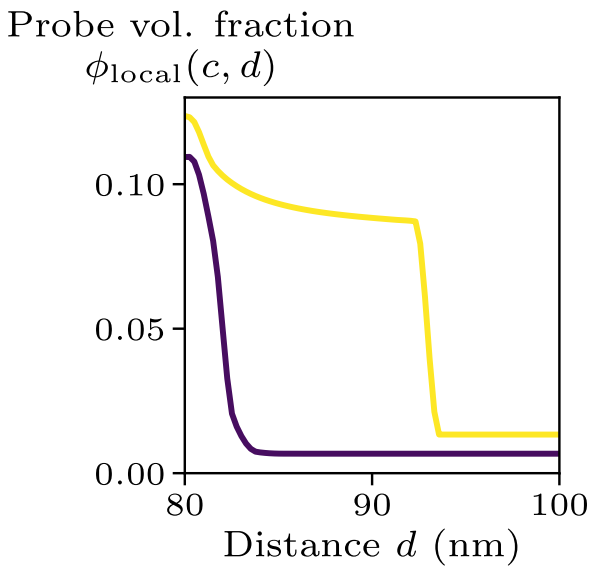

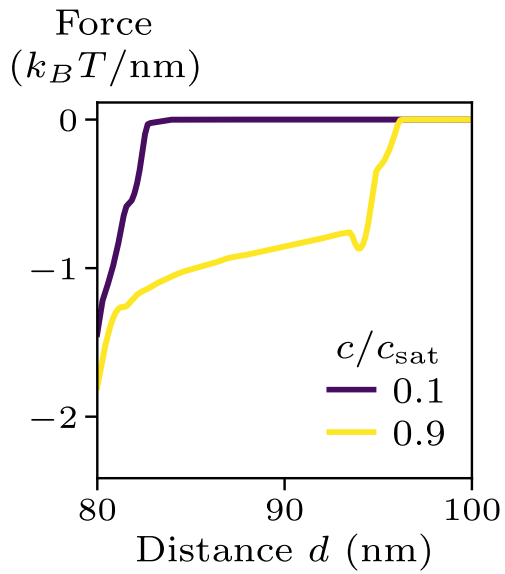

Polymer + capillary attraction

$$U_{\text{attr}}(d) + U_{\text{poly}}(d) = U_{\text{tot}}(d)$$

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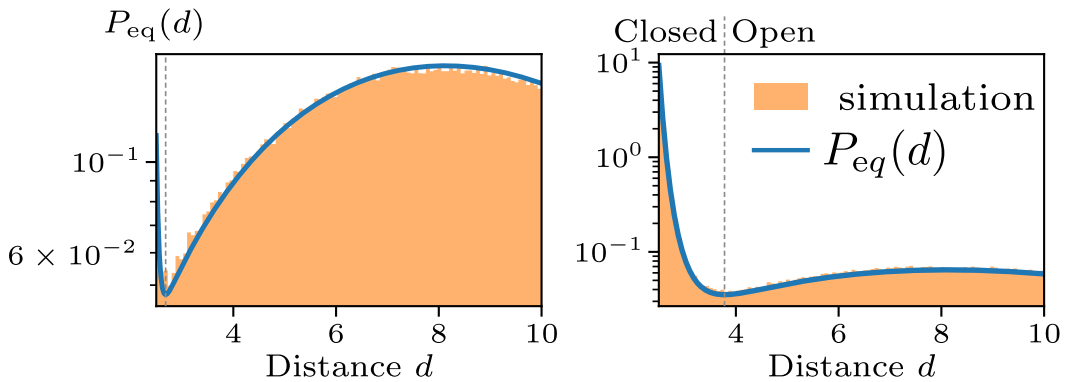

low concentration high concentration

Polymer dynamics

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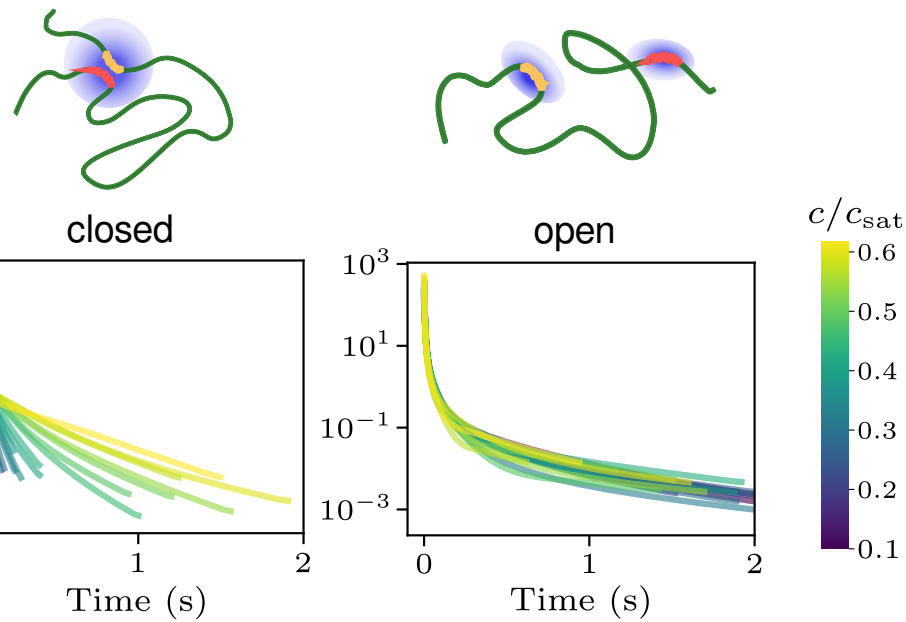

Sensitivity from polymer dynamics

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(\,\text{⎍⎍⎍⎍⎍}\,|c)$$

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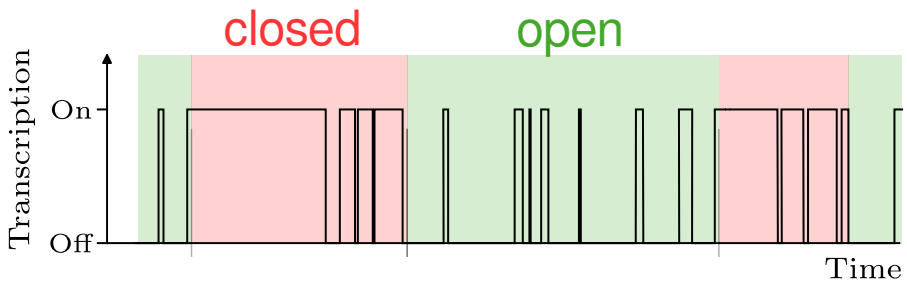

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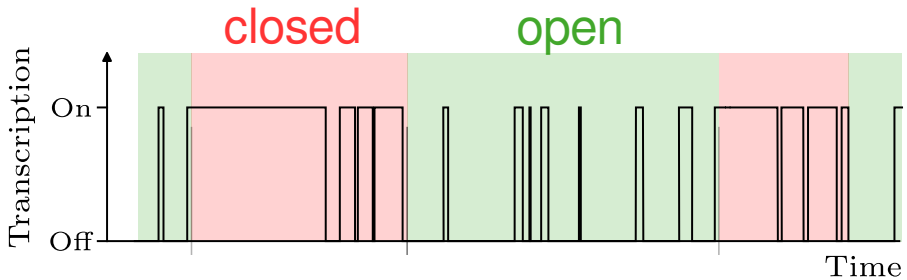

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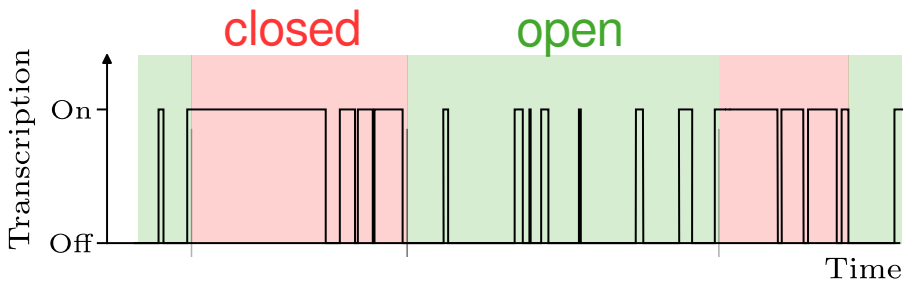

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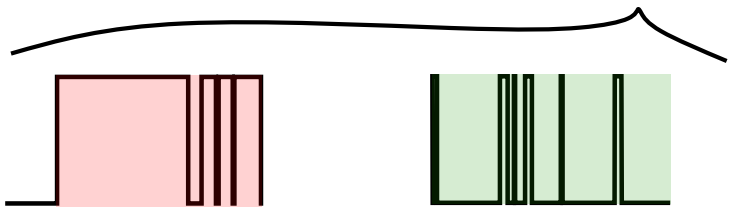

$$\dot{\mathcal{F}}(\phi) = \underbrace{\frac{1}{\tau_{\text{free}}(\phi_{\text{local}})}}_{\text{Probe}} \underbrace{\left(\frac{d\log\phi_{\text{local}}}{d\log\phi}\right)^2}_{\text{Surface}}$$

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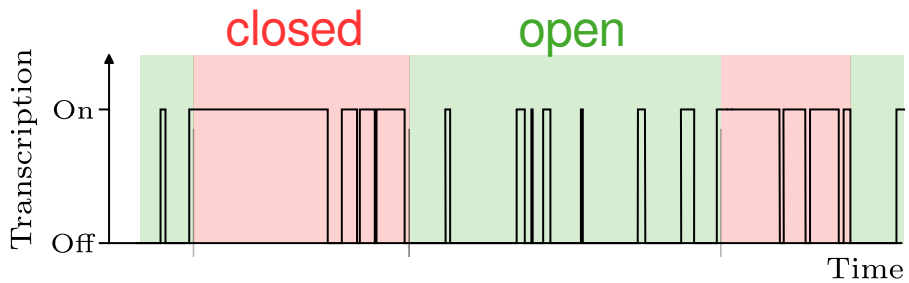

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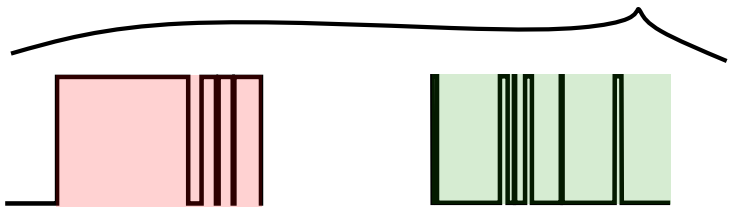

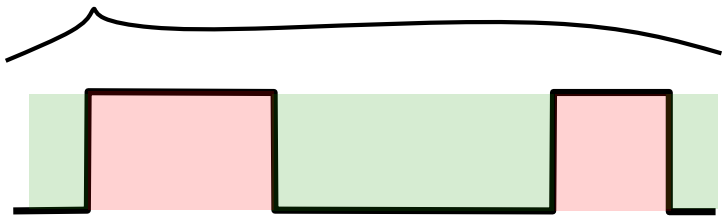

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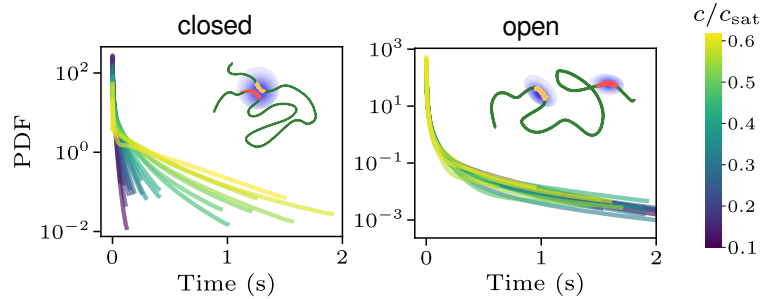

Total sensitivity

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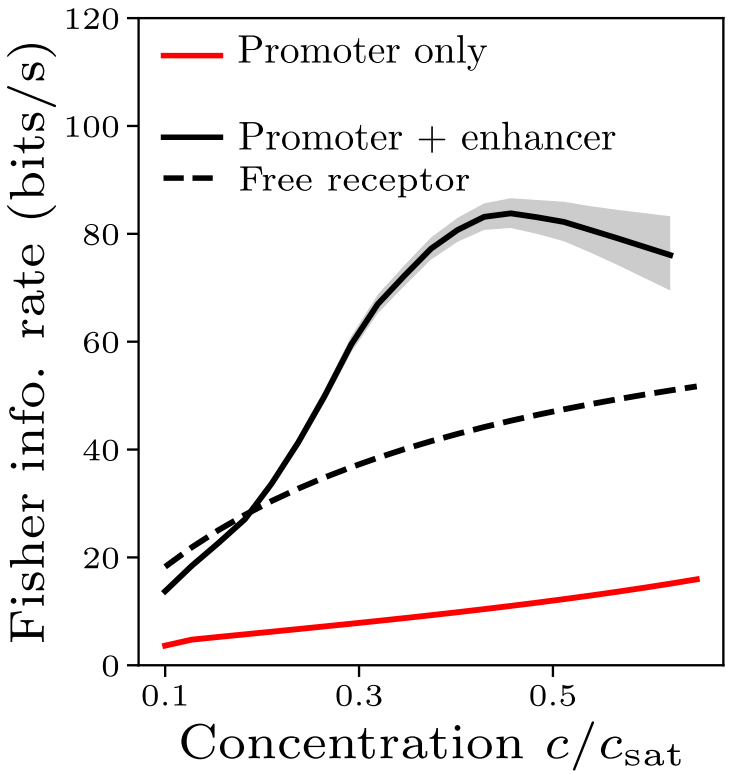

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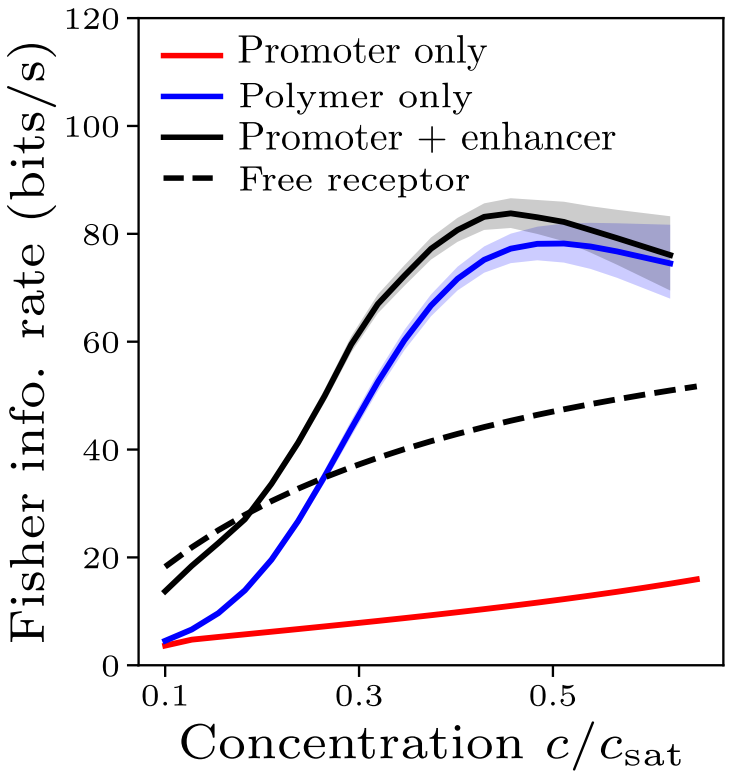

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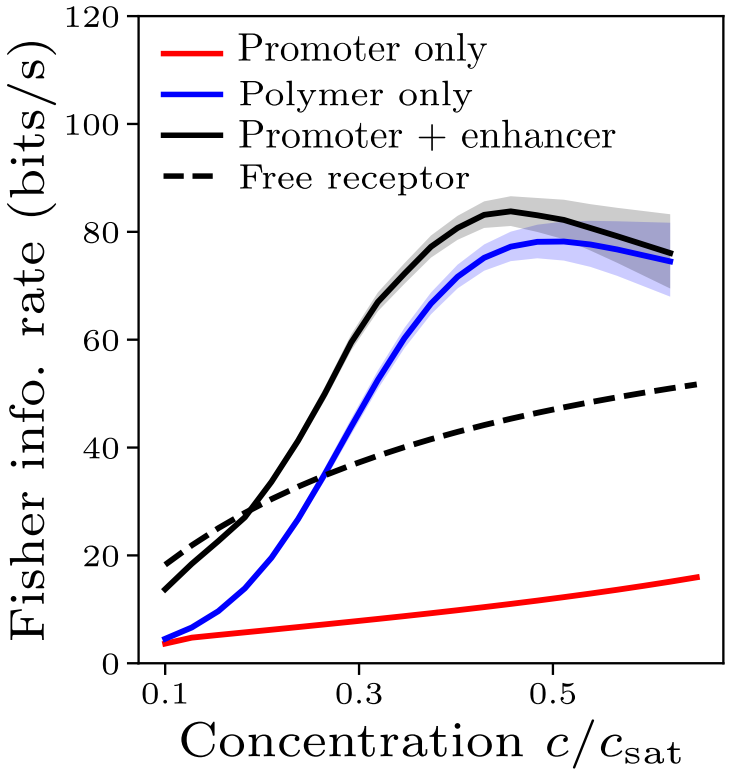

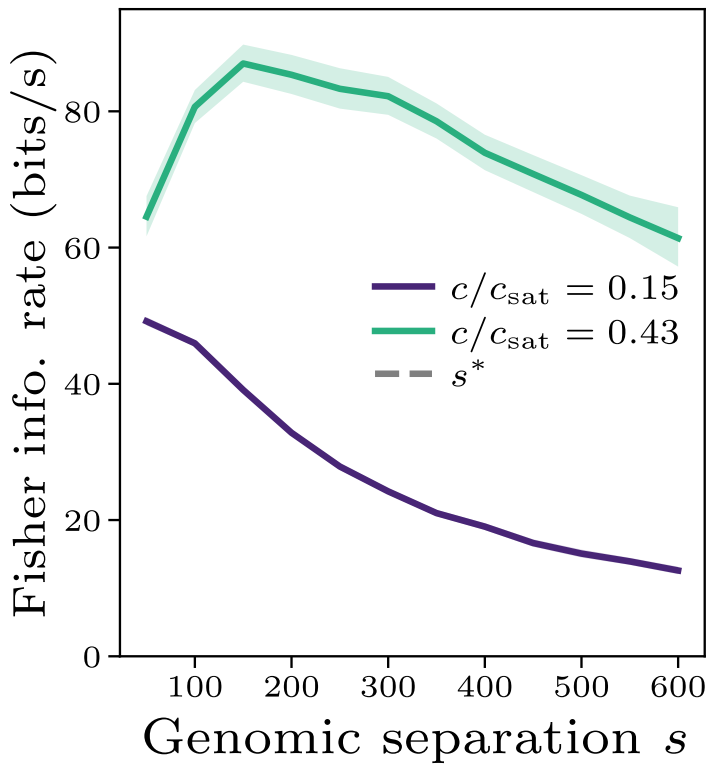

Takeaways

Surface condensation can boost concentration sensitivity

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Polymer dynamics could contribute to gene expression sensitivity

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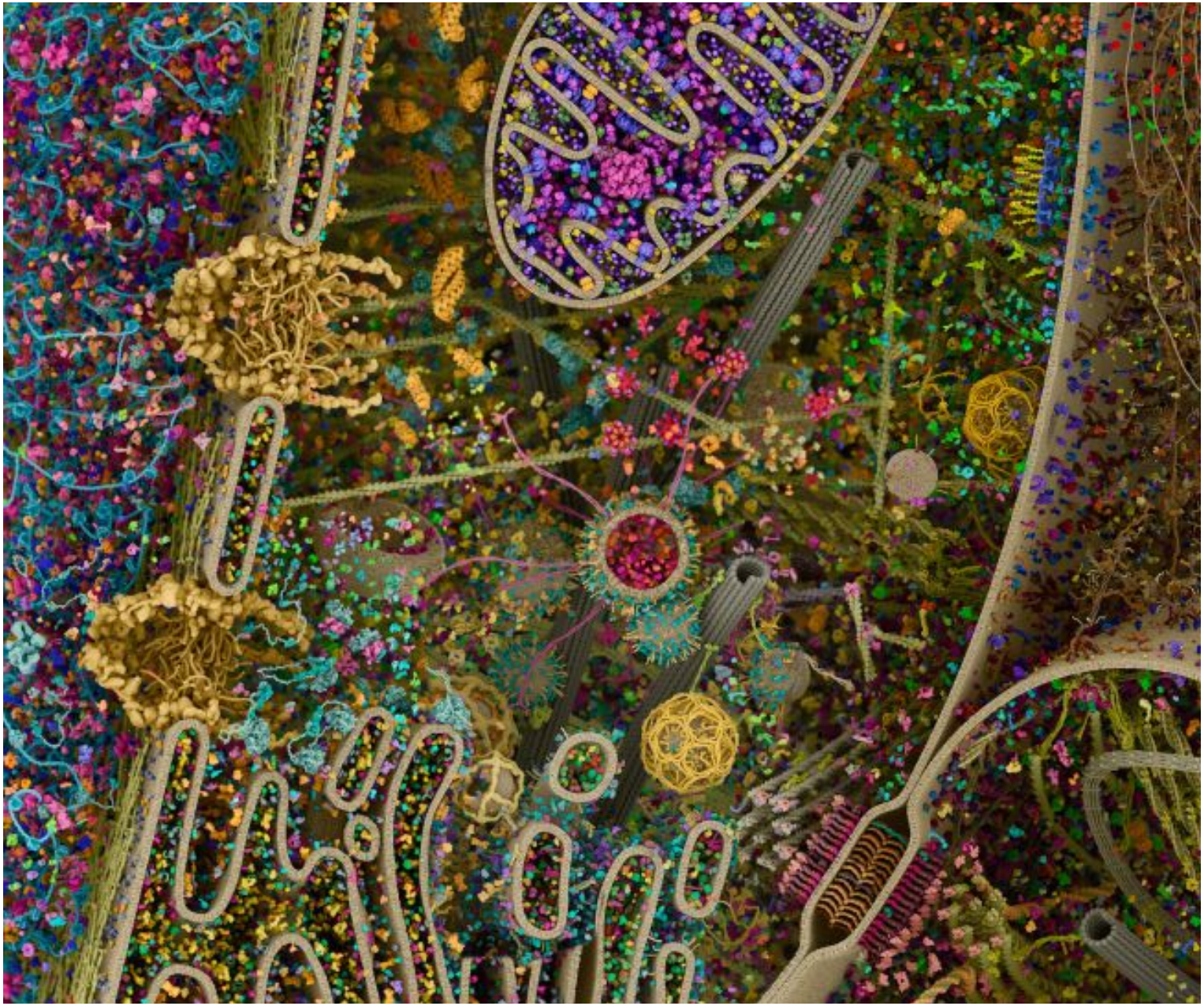
Polymer dynamics could contribute to gene expression sensitivity

Limitations of the model:

- Rouse model unrealistic for DNA
- Loop extrusion, TADs
- Membrane binding behaves differently than bulk

A Eukaryotic cell



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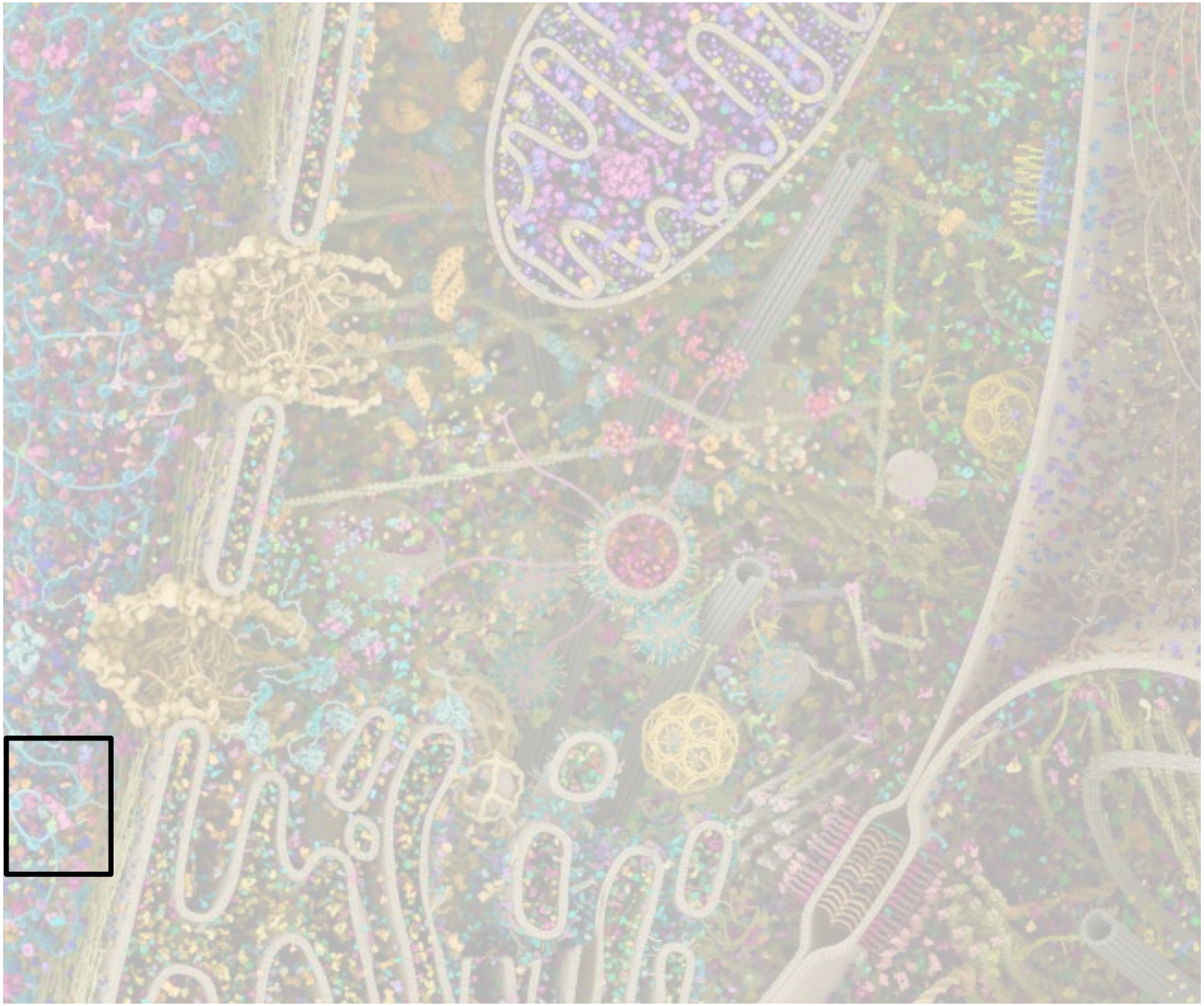

Remember: Tanscriptional regulation is very precise

Conclusion

Q. What is the most concentration-sensitive promoter architecture?

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sharp response?
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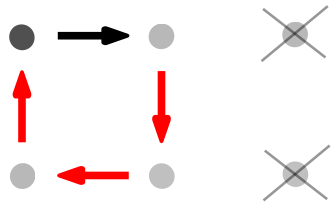
1. Cooperativity only optimal above critical concentration

"Keep binding sites free" (diffusion-limited)



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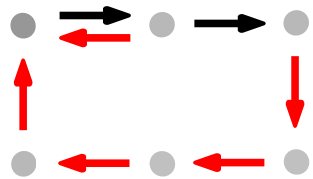

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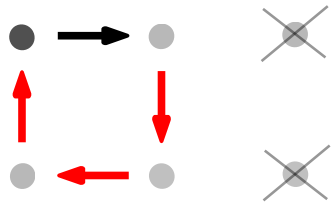

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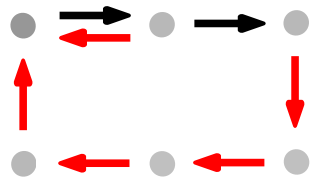

Proofreading only optimal above critical **decoy** concentration

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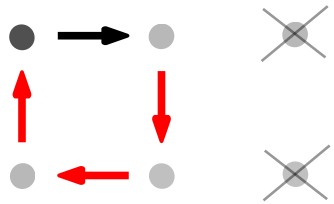

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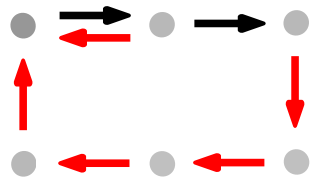

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2. Solutions sets are generically **discontinuous**

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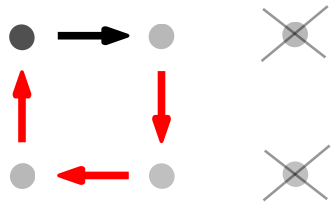

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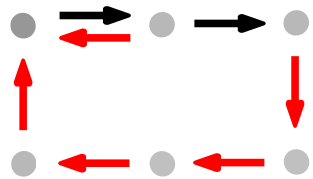
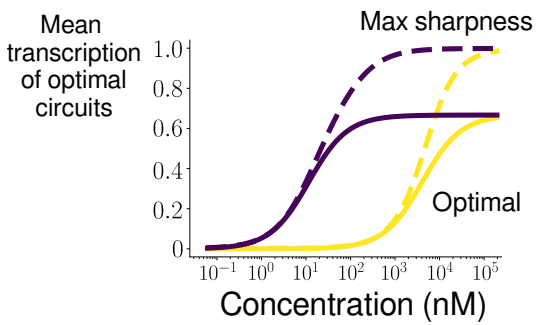
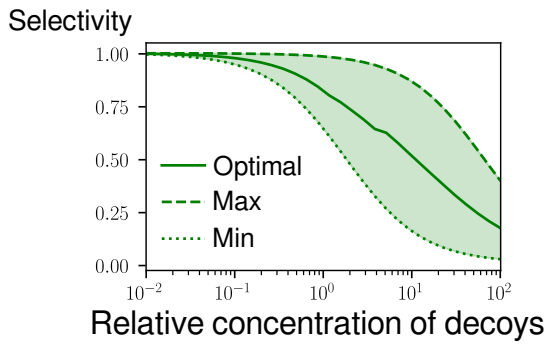


Proofreading only optimal above critical **decoy** concentration

2. Solutions sets are generically **discontinuous**

3. Optimal promoters neither maximize sharpness nor selectivity





Discussing optimality requires precise accounting of underlying constraints

Conclusion

Q. How does a promoter's environment influence concentration sensitivity?

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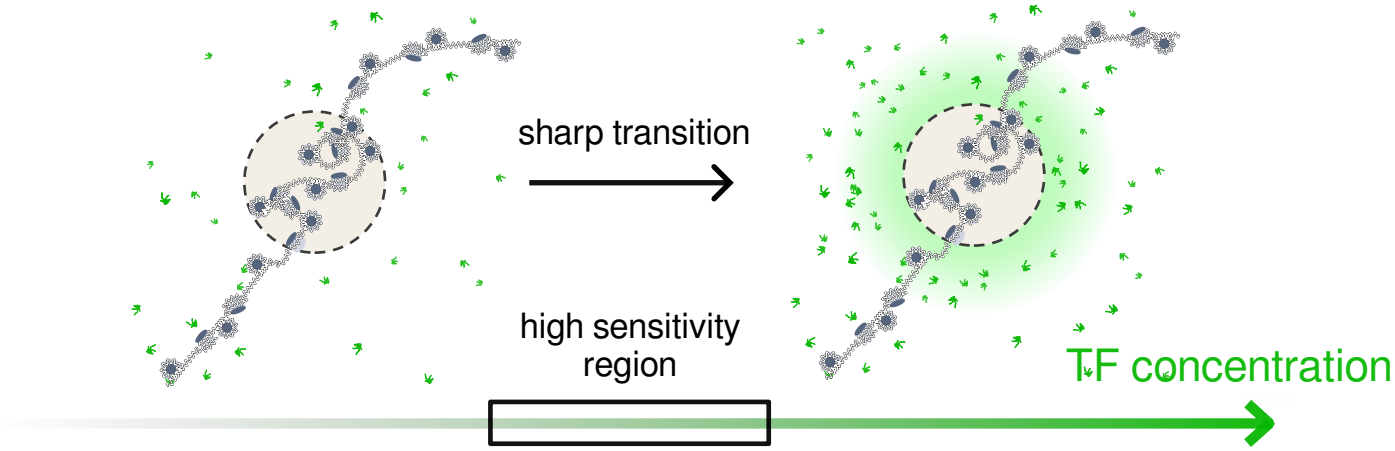
Surface condensation of transcription factors can amplify gene expression sensitivity



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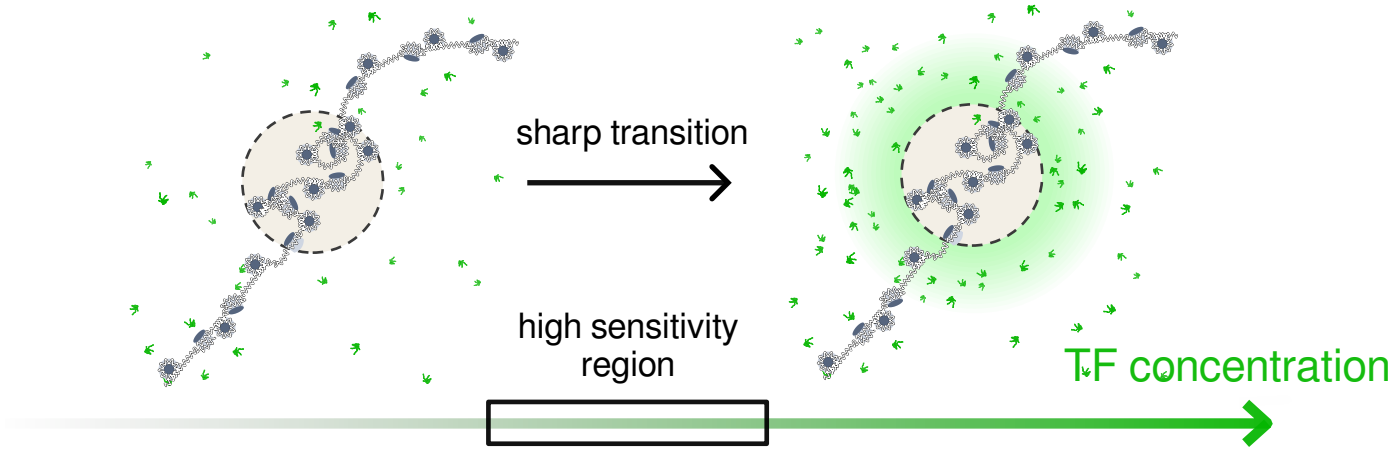
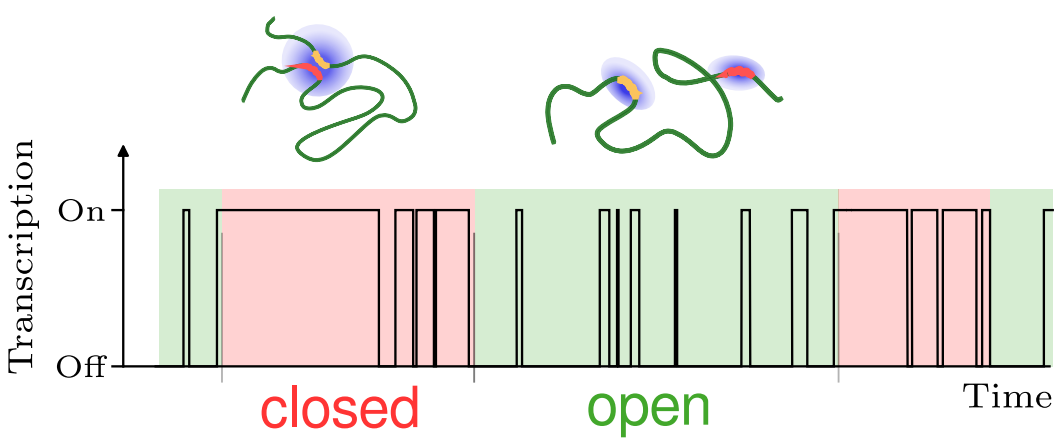
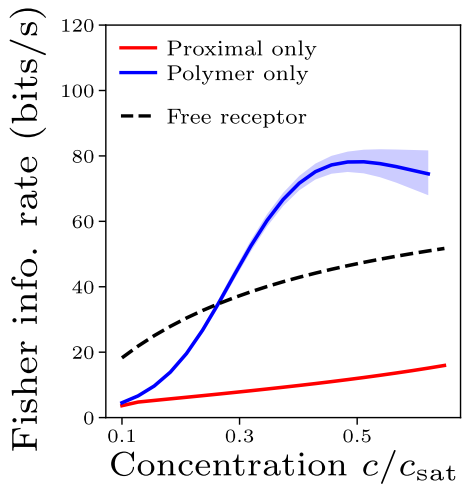


Polymer dynamics itself can encode a signal about concentration in transcription activity





Thanks.

StatBioPhys@ENS

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Aleksandra Walczak
Thierry Mora

Collaborators

Massimo Vergassola (1st project)
Jacqueline Janssen (2nd project)
Henry Alston (2nd project)

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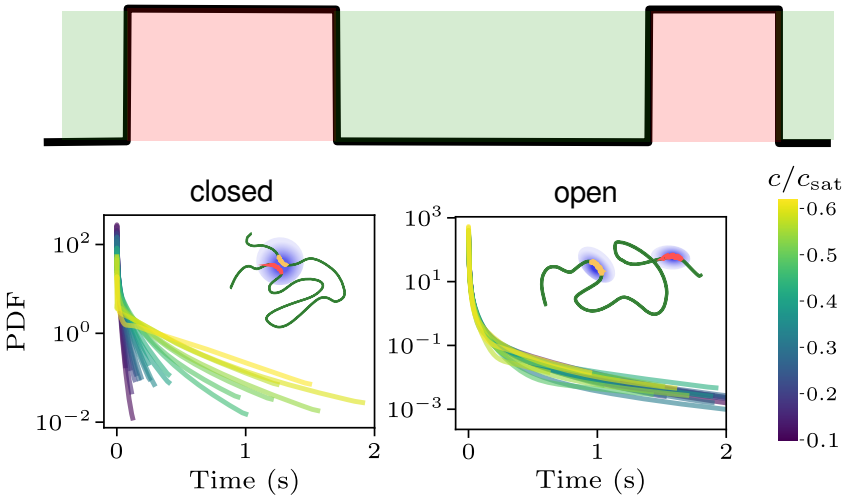
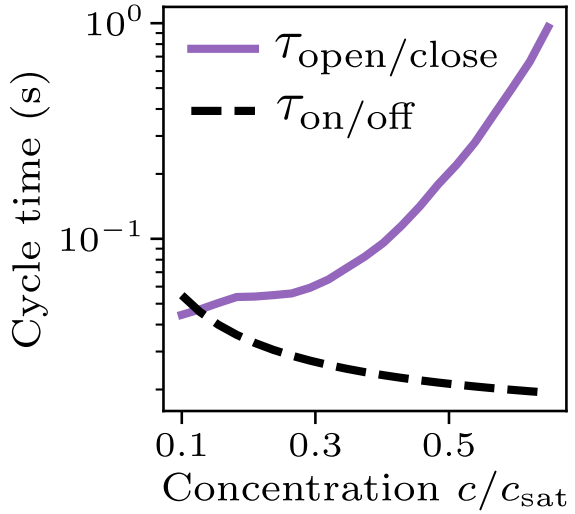
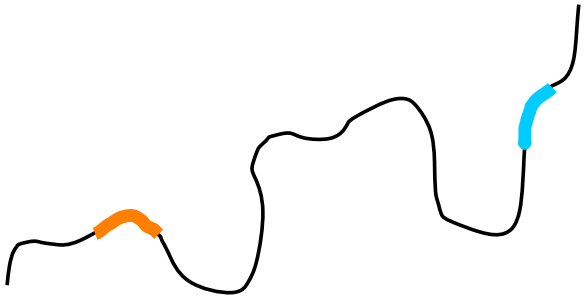
Mattia Greco
Anthia Prapa
Monika Rasz
Max Zayas
Johnathan Schrohe
Lander Bogers

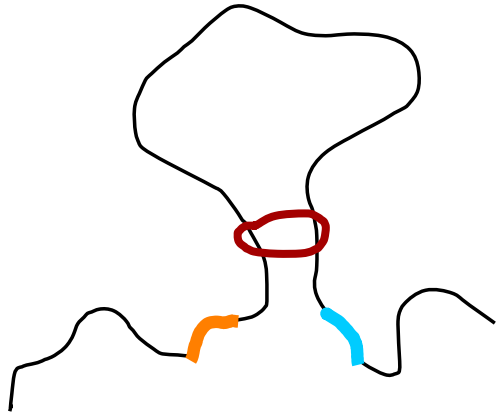
Tradeoff between cycle rate and per-cycle informativeness





$$\dot{\mathcal{F}}(\phi) = \frac{\mathcal{F}_{\mathrm{open}}(\phi) + \mathcal{F}_{\mathrm{closed}}(\phi)}{\tau_{\mathrm{open/close}}}$$





Loop extrusion decreases cycle time -> can only help sensitivity